Gap Analysis and Completion Specification
for the *Handbook of Formal Logic*

Campaign document — generated from the audit registers

10 August 2026 · registers at `spec/registers/`

Abstract

This document is the completion specification for transforming *Formal Logic: Language, Semantics, Deduction, and Metatheory* (v1.0, 9 August 2026) into a comprehensive CRC-style handbook of logic. It records every verified defect and gap from three exhaustive full-text audits of the 14,780-line manuscript, fixes the target architecture and editorial standards, and lays out the phased campaign. The registers in `spec/registers/` are the single source of truth; every table in this document is generated from them by `tools/csv2tex.py` and is regenerated, never hand-edited.

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1 Baseline and how to use this document

Source identification. The audited manuscript is `source/formal-logic-reference.tex` at git tag `v1.0-audit` (commit `676d170`), 14,780 lines, SHA-256 `81bf3c0ab12863d7d89e1b44648ccc1d63832dc6-8ab8e32b84aa9a0700cd6de1`. *Every line number in this document and in the registers refers to that revision.* Line references do not track the working tree; when a register row is executed, the executing commit is recorded in the row's status cell, and the frozen reference remains valid via the tag.

Audit provenance. Three independent full-text audits were performed on 10 August 2026, covering lines 1–4995 (Parts I–III), 4995–10622 (Parts IV–IX), and 10622–14780 (Parts X–XI and back matter). Each audit read its entire segment, re-derived displayed calculations and countermodels, verified the hypotheses of the major theorems against the literature, and ran mechanical checks over the whole file (labels, references, citations). The consolidated findings were reviewed and approved before this specification was drawn up.

Identifier scheme. Every finding carries a permanent identifier. Line numbers rot as soon as editing begins; identifiers plus the frozen tag do not.

Prefix	Register	Class
R-*nnn*	`defects.csv`	rigor defects (mathematical / proof content)
S-*nnn*	`defects.csv`	standards defects (status policy, notation, citation discipline)
L-*nnn*	`defects.csv`	LaTeX / production defects
D-P*n*-*nnn*	`depth-gaps.csv`	depth gaps within existing part n
N-*d*-*nnn*	`new-content.csv`	new-domain items, $d \in \{a, b, c, d\}$; N-x = recorded exclusions
A-*nnn*	`apparatus.csv`	apparatus items (index, registries, sheets, tables, figures, checks)

Dispositions. Each R/S/L row carries one of four dispositions, which is what prevents wasted work given the restructure: **FIX-SOURCE** (mathematical repair applied to the v1.0 text in Phase 2, because migrated text inherits it); **RESOLVE-BY-MIGRATION** (fixed for free when the text is rewritten into the new tree); **OBSOLETE** (mooted by the restructure; no action ever); **CARRY** (a requirement transferred to the new tree's specification).

Status vocabulary of this document. Every claim about the manuscript in this document is *verified-by-audit* unless marked *spot-check pending*. The handbook's own three-term result-status policy (complete proof / proof sketch / cited theorem) is quoted throughout and becomes compile-enforced in the new tree (§8).

2 Executive dashboard

The three fixed decisions. These were made by the author and define "done":

D1. Comprehensiveness spans four domains: (a) deep mathematical logic, (b) philosophical logic, (c) logic in computer science, (d) algebraic and categorical logic.

D2. Format: restructure toward a pure CRC-style handbook — topical lookup entries and tables; proofs compressed into a proof annex; the dependency-ordered v1.0 manuscript becomes verified source material and is not printed.

D3. First deliverable: this specification; the writing campaign runs against it in later sessions.

Headline findings. (i) The manuscript's mathematics is largely sound — every re-derived countermodel and calculation checked out — but roughly a dozen proofs have genuine defects, the worst being that the book does not prove $\mathrm{Con}(\mathsf{ZFC}) \Rightarrow \mathrm{Con}(\mathsf{ZFC} + \neg\mathrm{CH})$ (R-001). (ii) The book's signature status discipline is applied in only about a quarter of the text (48 of 60 status labels sit in lines 5000–7999). (iii) Whole fields required by decision D1 are absent (registers N-a. . . N-d). (iv) The reference apparatus is far below handbook grade: the index is ∼308 auto-generated theorem titles, there are no figures in 14,780 lines, rule sheets cover 3 of ∼10 calculi, and the notation register covers none of Part X. (v) The back matter's Exercise Feedback chapter is the exception — mechanically verified complete (112 = 112) — and its discipline is institutionalized by ruling 7.

Register totals: 88 defects, 141 depth gaps, 185 new-content rows (170 in-scope, 15 recorded exclusions), 201 apparatus items — 615 register rows in all.

Defects by class	
production	20
rigor	42
standards	26
total	88

Defects by priority	
P0	13
P1	54
P2	21
total	88

Defects by disposition	
CARRY	13
FIX-SOURCE	37
OBSOLETE	11
RESOLVE-BY-MIGRATION	27
total	88

Depth gaps by treatment	
E	56
P	6
S	68
T	11
total	141

Depth gaps by priority	
could	16
must	58
should	67
total	141

New content by domain	
a	71
b	35
c	40
d	24
x	15
total	185

New content by treatment	
E	40
P	12
S	75
T	43
total	170

Apparatus by category	
bibliography	3
check	11
environment	7
figure	40
index	6
navigation	7
rule-sheet	23
symbol-registry	13
table	91
total	201

3 Defect register

3.1 Priorities

P0 — mathematical falsity or unsoundness: the statement as printed is wrong, the proof as printed is invalid, or the source fails to compile. **P1** — violates the book's own declared policy (the three-term status policy, the one-symbol-one-role rule, the primary-source rule, the rule-display requirements). **P2** — polish. All P0 items and load-bearing P1 proof gaps are dispositioned FIX-SOURCE and repaired in Phase 2; fix designs for them follow in §3.3.

3.2 The register

Projection of `registers/defects.csv` (full columns — part, acceptance criterion, effort — live in the CSV).

ID	Pri	v1.0 lines	Finding	Required fix	Disp	Ph	St
R-001	P0	9062–9133	CH-independence treatment does not prove $\mathsf{Con}(\mathrm{ZFC}) \Rightarrow \mathsf{Con}(\mathrm{ZFC} + \neg\mathrm{CH})$: the sketch uses the countable-transitive-model presentation that the adjacent Warning (9107) says does not suffice, and the nonderivability step invokes soundness over two structures never produced, so the theorem statement (9066) overclaims.	Rewrite per the book's own Warning: either present relative consistency with honest cited status or give the correct syntactic route (forcing over finite fragments / ctm with the inaccessibility caveat), deleting the two-structures soundness justification.	FIX	2	fixed (4201fbf)
R-002	P0	7011–7031 vs 6567–6585, 7440–7454	Coding inconsistency: the text asserts iterated Cantor pairing is not the coding used for bounded verification while the fixed machine numbering is built from exactly that pairing, so $\mathsf{Comp}(e, x, s)$ lacks the bounded extraction from e it needs (the beta-function is applied only to histories), and the gap propagates to the Σ_1^0 status of Prov_T.	Adopt a single coding with a proved bounded-extraction lemma covering machine codes as well as histories, and re-derive the Σ_1^0 form of Comp and Prov_T from it.	FIX	2	fixed (563da95)
R-003	P0	7666–7699	Rosser sketch uses the object symbol $\leq$, never defined or registered anywhere (only $<$ at 6924), and silently uses two unstated Q-lemmas (bounded-case enumeration and the dichotomy $y \leq \bar{n} \vee \bar{n} \leq y$) that carry the proof's only nontrivial content.	Define and register object $\leq$ (or eliminate it in favor of $<$) and state and prove or cite both Q-lemmas as labeled results before the sketch uses them.	FIX	2	fixed (563da95)
R-004	P0	7154–7165	The jump is defined via halting although the numbering makes $\varphi_e(e)$ undefined on halt-without-valid-output, mismatching K, and the jump theorem is asserted in unlabeled prose without proof or citation while load-bearing for hierarchy strictness (7180–7189).	Align the halting predicate with the numbering's convergence notion (or adjust the numbering) and promote the jump theorem to a labeled result with status and proof or citation.	FIX	2	fixed (563da95)
R-005	P0	6564–6593	The enumeration theorem for the program numbering (every partial computable function occurs in it) is asserted in unlabeled prose without proof while load-bearing for Rice, the recursion theorem, and Post.	State the enumeration theorem as a labeled result with a proof (or precise citation) tied to the constructed numbering.	FIX	2	fixed (563da95)
R-006	P0	6952–6971	The closure-class definition of the arithmetical hierarchy (with bounded-quantifier closure) is never proved equivalent to the prefix-normal-form characterization; the required bounded-collection lemma is missing.	State and prove the bounded-collection lemma and derive the closure-class/prenex equivalence as a labeled theorem.	FIX	2	fixed (563da95)
R-007	P0	4420–4446	Prenex normal form theorem rests on an unstated and unproved first-order NNF existence theorem, unproved existence of rectified alpha-variants, and an unstated rectification-preservation invariant; the propositional NNF theorem (1899) does not transfer.	Prove first-order NNF existence and rectified-variant existence as labeled lemmas, state the preservation invariant, and run the prenex induction on them.	FIX	2	fixed (fe1f17d)
R-008	P0	4480–4507	Skolemization equisatisfiability is labeled complete but is a sketch: no induction hypothesis is formulated, the unregistered term quantifier game is used (no game semantics exists in the book), and AC is over-assumed for the choice-free direction.	Write the induction with an explicit hypothesis and cases, remove or define the game terminology, and separate the AC-using direction from the choice-free one.	FIX	2	fixed (fe1f17d)
R-009	P0	3934–3985	The Henkin–Lindenbaum maximality step silently reuses the omega-recursion argument of the propositional Lindenbaum lemma where a transfinite recursion with limit stages and a chain argument is needed (the nearby witness stage does invoke transfinite recursion properly).	Redo the maximality step by transfinite recursion with explicit successor and limit stages and the union/chain argument.	FIX	2	fixed (fe1f17d)

continued

ID	Pri	v1.0 lines	Finding	Required fix	Disp	Ph	St
R-010	P0	3006–3016	The substitution/alpha lemma applies structural induction to $\psi[z/y]$, which is not a subformula of the original formula, so the induction principle invoked does not apply.	Recast as strong induction on formula complexity via the substitution-complexity lemma, the fix the book itself uses at 2951–2958 and 3195–3199.	FIX	2	fixed (fe1f17d)
R-011	P0	1958–1965	The CNF half of the propositional normal-form theorem is procedural with no induction or termination argument and fails degenerate cases (valid or variable-free formulas) against the book's rigid CNF definition; the DNF half is exemplary.	Give the CNF half a genuine induction (or dualize the DNF argument) with the degenerate cases handled explicitly.	FIX	2	fixed (2465859)
R-012	P0	9513–9548, 9555, 9575	The cut-elimination sketch invokes a mix/multicut rule that is never displayed anywhere, states the termination measure for Cut rather than mix, and two corollaries with full proof environments rest on the sketch, violating the release checklist's own rule (14346).	Display and label the mix rule, restate the measure for the rule actually eliminated, and either complete the induction or downgrade the corollaries' status to match the sketch.	FIX	2	fixed (2465859)
R-013	P1	2020–2021, 2374, 2446–2449	ND repetition is permitted in prose but is not among the twelve displayed rules, and vacuous discharge is never licensed although the completeness proof needs it; the first-order analogue (3804–3806) handles both explicitly.	Add repetition to the displayed rule inventory (or derive it) and license vacuous discharge explicitly, matching the first-order treatment.	FIX	2	fixed (2465859)
R-014	P1	2488 vs 2295; 4165 vs 3674	LK completeness is stated only for single-formula succedents, strictly weaker than LK soundness for arbitrary sequents, and the book never closes the loop for arbitrary sequents.	State and prove LK completeness for arbitrary sequents, or scope both metatheorems to the same sequent class with an explicit remark.	FIX	2	fixed (fe1f17d)
R-015	P1	5994–6015	Model completeness of DLO is defined by quantification over all embeddings but the proof treats only substructure inclusions; the reduction step is missing.	Add the reduction from arbitrary embeddings to substructure inclusions or restate the definition to match the proof.	FIX	2	fixed (2465859)
R-016	P1	6068–6110	The back-and-forth corollary uses an extends-the-original-map clause that is available from the proof but absent from the theorem statement, which concludes only isomorphism.	Add the extension clause to the theorem statement.	FIX	2	fixed (2465859)
R-017	P1	678–702	Mixed-status theorem thm:infinite-cardinal-finite-coding: one label covers clauses cited to Jech, a clause unlocatable in the cited source, and a clause proved inline, against the status policy at 313–319.	Split into separately labeled results with per-clause status and locate or reprove the unlocatable clause.	FIX	2	fixed (4201fbf)
R-018	P1	1145–1152	The associativity-of-concatenation proof is a plan rather than a proof: the announced induction is not carried out.	Write the induction on string length in full.	FIX	2	fixed (89cfaa8)
R-019	P1	1239	The complexity-recursion example precedes the unique-readability theorem it depends on.	Reorder so unique readability precedes the example, or add an explicit forward-dependency remark.	MIGR	4	open
R-020	P1	1282	The path characterization of the transitive closure R^* is used without proof.	State and prove the path characterization as a labeled lemma or mark its status honestly.	FIX	2	fixed (89cfaa8)
R-021	P1	1819, 1823–1824	The controlled-prose translation table and its commentary on only and unless contain translation errors.	Correct the offending table rows and the accompanying only/unless commentary.	FIX	2	fixed (89cfaa8)
R-022	P1	2299–2317	The LK soundness wording slides between valid and satisfied.	Rewrite the soundness argument keeping the two registered notions distinct throughout.	FIX	2	fixed (58ef447)
R-023	P1	2547	Davis–Putnam elimination is unattributed and stated for finite clause sets only, with no remark relating the finite restriction to compactness.	Add the primary-source attribution and an explicit finiteness/compactness scope remark.	FIX	2	fixed (58ef447)
R-024	P1	3415–3418	Conservativity of definitional extensions is asserted in an example without proof although Beth definability later leans on it.	State conservativity of definitional extensions as a labeled result with proof or citation.	FIX	2	fixed (58ef447)
R-025	P1	3683–3685	The LK universal-quantifier-right soundness case is hand-waved.	Write the case in full with the eigenvariable condition doing explicit semantic work.	FIX	2	fixed (58ef447)

continued

ID	Pri	v1.0 lines	Finding	Required fix	Disp	Ph	St
R-026	P1	4199–4215	The compactness theorem statement is garbled (a dangling equivalently-clause mixes the model form with the consequence form) and the proof closes the loop redundantly.	Restate the equivalent forms unambiguously and remove or repair the redundant converse step.	FIX	2	fixed (58ef447)
R-027	P1	546–548, 4127	Broken forward promise: the front matter promises countable-language versions of the completeness theorems, but no countable or choice-free completeness theorem exists anywhere, and the AC hypothesis on completeness (4127) is thereby over-stated for the countable case.	Add a countable-language choice-free completeness theorem (or delete the promise) and scope the AC hypothesis accordingly.	FIX	2	fixed (58ef447)
R-028	P1	5456–5472	The type-space compactness proof omits the complete-type/Lindenbaum step while presented as complete.	Supply the missing step or downgrade the result's status.	FIX	2	fixed (b3eb76d)
R-029	P1	5573–5585	The saturation-implies-homogeneity proof omits its compactness appeal while presented as complete.	Supply the compactness appeal explicitly or downgrade the result's status.	FIX	2	fixed (b3eb76d)
R-030	P1	6119, 13692	The DLO $\aleph_0$-categoricity proof suppresses the four interval cases, which are supplied only by the exercise solution at 13692 – the appendix doing the theorem's work.	Promote the four-case analysis from the solution into the theorem's own proof during migration (annex).	MIGR	4	open
R-031	P1	10092	The injectivity step in the second-order categoricity proof is hand-waved.	Write the injectivity argument in full.	FIX	2	fixed (13c1cea)
R-032	P1	10142	PA_2 is used but never defined.	Define PA_2 (language, axioms, semantics) before first use in the migrated part.	MIGR	4	open
R-033	P1	10162–10168	Failure of downward Löwenheim–Skolem for full second-order semantics is justified by an undisplayed, unproved, uncited categoricity of the second-order reals.	State the second-order categoricity of the reals as a labeled result with status and citation, and cite it in the justification.	FIX	2	fixed (13c1cea)
R-034	P1	8389–8394	The ordinal trichotomy proof idea (foundation applied to the symmetric difference) is not a recognizable argument.	Replace with the standard argument via $\alpha \cap \beta$.	FIX	2	fixed (13c1cea)
R-035	P1	6340–6349	The definition of partial computable is malformed by a double iff.	Restate the definition as a single well-formed biconditional.	FIX	2	fixed (563da95)
R-036	P1	5690–5703	Nonprincipal-ultrafilter existence is unlabeled prose whose induction is not carried out.	Promote to a labeled result and carry out the chain/Zorn argument or cite it.	FIX	2	fixed (b3eb76d)
R-037	P1	11273	Finite-matrix tableau completeness wears a proof environment over a sketch (the induction is asserted, not performed) and uses the term saturated, never defined anywhere in the book.	Define saturation for the tableau system and either perform the induction or downgrade the result to sketch status.	FIX	2	fixed (13c1cea)
R-038	P1	11911	The DPLL correctness proof is one sentence over a two-specification claim, violating the book's own template at 11791 and style rule 12354.	Supply the two-part correctness argument or downgrade to sketch status under the migration's compile-enforced status scheme.	MIGR	4	open
R-039	P1	11179–11191	Common knowledge dangles: C_G receives language and semantics but no axioms or induction rule anywhere, the non-compactness/weak-completeness caveat stated for LTL (11090) is omitted for C_G, and an exercise then uses it.	Supply the C_G axiomatization (mix and induction rules) with the standard caveat as new-tree content, or rescope the exercise.	CARRY	4	open
R-040	P1	10806	The Gödel–Gentzen translation recursion is not literally well-defined: whether $\bot$ counts as atomic (priority of the $\bot$-clause versus the double-negation clause for atoms) is unstated.	Fix the clause priority explicitly in the definition.	FIX	2	fixed (13c1cea)
R-041	P1	10735	The prime-theory description is ambiguous between $\vdash A \lor B$ and $T \vdash A \lor B$.	Disambiguate the defining condition.	FIX	2	fixed (13c1cea)
R-042	P1	13049 vs 9513	The rule sheet declares the equality rules part of the complete active first-order sequent calculus while the cut-elimination sketch never treats equality rules, so two passages refer to different calculi under one name; the sheet's symmetry-derivability claim is correct but silently requires cut.	Name the calculi apart under the canonical-LK ruling (LK versus LK-with-equality), scope the cut-elimination claim to the calculus it covers, and state the cut-dependence of the symmetry derivation.	CARRY	4	open

continued

ID	Pri	v1.0 lines	Finding	Required fix	Disp	Ph	St
S-001	P1	313–319 vs 5007	The book's three-term status policy is scoped to Part IV onward by the convention at 5007, leaving 56 of 66 numbered results in Parts I–III silently presented as complete proofs with no status labels.	Extend the status convention book-wide and label every Part I–III result, via the new tree's compile-enforced status keys.	MIGR	4	open
S-002	P1	550, 3028, 4317, 4386	Notation collision against the front-matter policy (239–241): $\lvert\cdot\rvert$ denotes both cardinality (550) and structure domain (3028), yielding nested-bars expressions at 4317 and 4386.	Ratify and enforce the ruling: $\lvert\cdot\rvert$ is cardinality only and domains are written $\mathrm{dom}\, M$.	CARRY	4	open
S-003	P1	1468, 1770, 3256, 4371	Notation collision: $\equiv$ carries four roles (metalinguistic sign, propositional equivalence, first-order equivalence, elementary equivalence – the last registered mid-theorem at 4371).	Ratify and enforce the ruling: $\equiv$ is elementary equivalence only, formula equivalence is $\Leftrightarrow$, and the definitional sign is $:\equiv$.	CARRY	4	open
S-004	P1	4256–4264, 4324	Parameter notation $\varphi(\bar{x})$ lacks the side condition $\mathrm{FV}(\varphi) \subseteq \{\bar{x}\}$ and conflicts with the variable-indexing convention at 4324.	State the side condition in the notation's registration and reconcile it with the variable indexing.	CARRY	4	open
S-005	P1	704–724	The Ultrafilter Lemma, an existence theorem, is housed in a definition environment (hence unlabeled and unsourced) and contains an uncited independence claim (strictly weaker than AC over ZF).	Rehouse as a labeled theorem with status and add the Halpern–Lévy citation.	MIGR	4	open
S-006	P1	4016, 4225	Results are cited at whole-chapter level because the intended targets are unlabeled examples.	Label the target results and repoint the citations at result level.	MIGR	4	open
S-007	P1	4995–10622	Status coverage collapses after Parts IV–VI: 26 of 77 theorem/lemma/corollary environments in the segment carry no status (all of Part VII zero, Part VIII two).	Apply the compile-enforced status keys of the new tree to every result in the segment.	MIGR	4	open
S-008	P1	5007–7710 vs 7710–10622	Three inconsistent status mechanisms are in use (Status line with proofplan environment, bare paragraph Proof sketch, paragraph Cited theorem), split by page range – the same defect the production audit records as two sketch mechanisms (proofplan 10 times before 7710, paragraph 20 times after).	Adopt exactly one compile-enforced status mechanism book-wide.	MIGR	4	open
S-009	P1	8970, 9839, 10224, 10493	The four most important imported results – the forcing theorem, Gentzen's consistency theorem, and both Henkin completeness theorems – are unnumbered, unlabeled paragraphs with no TOC or index presence.	Promote all four to numbered, labeled, statused theorem environments.	MIGR	4	open
S-010	P1	8733, 9065	Misclassification: results labeled sketch are by the book's own three-way definition cited theorems.	Relabel with the correct status.	MIGR	4	open
S-011	P1	6137–6145	Uncountable categoricity of ACF_p is packaged as an uncited example (Steinitz).	Promote to a labeled statused result with the Steinitz citation.	MIGR	4	open
S-012	P1	2248–2695, 3440–3780 vs 9135–9948	Name clash: LK denotes two different calculi – Parts II/III's presentation with primitive negation rules and contexted initial sequents (reproduced by the back-matter rule sheets) versus Part VIII's variant with defined negation and context-free identity – with no reconciliation, and the cut-elimination sketch is by its own admission incomplete for the former (9258–9260).	Ratify the canonical-LK ruling: the Parts II/III presentation is canonical and the Part VIII variant is documented as a named variant with a labeled equivalence theorem.	CARRY	4	open
S-013	P1	9492	Analytic derivation appears as a chapter title but is never defined.	Define the term at first use or retitle the chapter.	MIGR	4	open
S-014	P1	10622–14780	Status policy is nearly absent in Parts X–XI and the back matter: 3 Status lines in 4,159 lines, with 48 of the book's 60 status labels sitting in lines 5000–7999.	Apply the compile-enforced status keys to every result in the segment.	MIGR	4	open
S-015	P1	10926	The modal theorem titled Frame correspondence and completeness contains no correspondence claim – the book's own warned-against failure mode (12357).	Retitle the theorem or add the correspondence statement it advertises.	MIGR	4	open

continued

ID	Pri	v1.0 lines	Finding	Required fix	Disp	Ph	St
S-016	P1	10744	The IPL completeness sketch omits its inline citation (the controlling sources) where the modal, LTL, and epistemic sketches all cite inline.	Add the inline citation.	MIGR	4	open
S-017	P1	10625–11683, 12568, 14354	Primary-source rule violated: Glivenko 1929, Gödel 1933, and Gentzen 1936 are absent from the bibliography (their results cited only to Troelstra volumes) and the many-valued chapter has zero citations (no Kleene at K3, no Priest 1979 at LP), against the book's own rules at 12568 and 14354.	Add the primary sources, cite them at point of use, and enforce the no-zero-citation-chapter check as a standing control.	CARRY	4	open
S-018	P1	11628, 12024	Rule displays violate the book's own naming requirement (12518): the four exponential rules appear as bare unnamed fractions, and the resolution rule's soundness-critical side conditions (renaming apart, mgu) live only in prose.	Name and label every rule and display its side conditions as part of the rule, sourced from the calculus registry.	CARRY	4	open
S-019	P1	12117, 12126, 12053	Three imported theorems live in prose with no status or label: SAT NP-completeness, FOL validity computably enumerable and undecidable (with no citation at all), and refutational completeness.	Promote each to a labeled, statused, cited statement.	MIGR	4	open
S-020	P1	10628, 11566	Same-glyph collisions within the segment: A as formula metavariable and monoid carrier in the same chapter, v as world versus registered valuation, u, v, w registered as strings but used as worlds, M as matrix versus structure, and V and D carrying three and four roles.	Apply the glyph-role rulings and the symbol-registry lint of the new tree.	CARRY	4	open
S-021	P1	10677 vs 14071	The Kripke forcing relation is typeset with two different macros in the body (10677) and the Exercise Feedback chapter (14071), and the second macro is unregistered.	Ratify one registered forcing macro and use it everywhere.	CARRY	4	open
S-022	P1	14324, 14326, 14343, 14354	Release-checklist self-test fails: the book fails four items of its own editorial checklist (undefined saturated, glyph roles, inductions without objects and cases, primary sources) while passing 14339 and 14366–14367.	Make the release checklist a mechanically checked phase gate and re-run it at release.	CARRY	6	open
S-023	P2	8784, 8858, 8907	The five-condition definition template is applied only sporadically in the set-theory chapters (poset filter and genericity, names and extension, semantic forcing), leaving sibling definitions unstructured.	Apply the new tree's uniform definition/entry template to every definition during migration.	MIGR	4	open
S-024	P2	1–4995	Chapters 10 and 12 contain no exercises while every other chapter in the segment carries them (40 exercises in Parts I–III).	None: exercises are retired from print to verification assets by the format decision, so per-chapter exercise coverage is moot.	OBS	4	open
S-025	P2	13072–13220	The qualifying-examination chapter poses ten problems but supplies solutions for only five; the final five are listed without solutions.	None: qualifying problems are retired from print to verification assets by the format decision.	OBS	4	open
L-001	P0	9683	Compile breaker: amsmath parses the bracketed discharge annotation $[A]$ immediately after a matrix opening as the matrix's optional alignment argument, producing an illegal-preamble-token error.	Brace-protect the bracketed annotation (applied in Phase 0, commit aa34611).	FIX	0	fixed
L-002	P2	76–85	Auto-index wrapper defects: accent commands in indexed titles are escaping hazards under makeindex (the quote character in umlauted names will be mangled) and the wrapper emits its index entry after the sectioning head, risking page drift.	Retire the wrapper; the new tree indexes inline from birth with sort keys.	OBS	4	open
L-003	P2	preamble	Dead macros (BV, Theory, eqdef, HOL, coNP, CNF, DNF, bracket, paren, id – the complexity-class macros suggesting planned-never-written content – plus the later-retired proofplan environment).	Drop them from the new preamble and add a defined-but-unused macro lint.	OBS	4	open

continued

ID	Pri	v1.0 lines	Finding	Required fix	Disp	Ph	St
L-004	P2	2082–2104, 3537–3561	Array rows switch inconsistently between the cr primitive and the double-backslash line terminator.	Normalize the line terminator during migration retypesetting.	MIGR	4	open
L-005	P2	preamble	The rule-typesetting macro produces mixed small-caps metrics and overwide multi-premise displays (disjunction elimination, existential elimination).	Retire the macro in favor of registry-generated rule typesetting.	OBS	4	open
L-006	P1	2082–2104, 3537–3561 vs 12823–13071	Deduction rules are typeset in raw arrays – unciteable – and duplicated by the back-matter rule sheets, creating divergence risk.	Introduce the single calculus registry that body, entry cards, and rule sheets all input, making divergence impossible.	CARRY	4	open
L-007	P2	363, 11684	Two part headings are unlabeled while the other nine are labeled.	Label every sectioning unit in the new tree.	MIGR	4	open
L-008	P2	preamble	The listings package is configured but unused in Parts I–III.	Let configuration follow use in the new preamble.	OBS	4	open
L-009	P2	global	The shared theorem counter makes citation numbers jump (Lemma 12.7 style); whether this is intended is unverified.	Decide and document the numbering scheme in the new tree's spec, deliberately.	MIGR	4	open
L-010	P2	1–14780	The book has no figures at all in 14,780 lines while tikz and three libraries are loaded: no syntax trees, derivation trees, Kripke diagrams, or hierarchy pictures.	Execute the apparatus program's figure quota (roughly 30–40 figures) per part during migration.	CARRY	4	open
L-011	P2	preamble	Unused packages (mathrsfs, bm, multirow) and the unused color LogicRed.	Drop them from the new preamble.	OBS	4	open
L-012	P2	8678, 7069, 10049	Non-starred set-builder delimiters around tall bodies yield undersized braces.	Use auto-sized (starred) delimiters during migration retypesetting.	MIGR	4	open
L-013	P2	7069	Prose is set inside set-builder notation as math.	Move the prose out of math or set it with text commands.	MIGR	4	open
L-014	P2	8970, 9839, 10224, 10493 and 8 uses in X–XI	paragraph sectioning commands do semantic work (sketch markers, cited theorems, internal proof boundaries) at least eight times: results at paragraph level break numbering, TOC, and referenceability and use the wrong spacing model.	Replace with numbered environments (statused results, a numbered proofboundary environment) in the new tree.	MIGR	4	open
L-015	P2	76–85	The index wrapper passes math and accent commands into the idx file with no sort keys.	Retire with the wrapper; inline index entries carry explicit sort keys.	OBS	4	open
L-016	P2	10682+	Negated forcing is typeset as a not-slash composition instead of the dedicated negated symbol, and no negated-forcing macro exists, against the book's own semantic-macro doctrine.	Register a negated-forcing macro and use it at every occurrence.	MIGR	4	open
L-017	P2	12465	A label inside an lstlisting environment is not a real label and cannot be referenced.	Move the label outside the listing or use the listings label key.	MIGR	4	open
L-018	P2	12678–12774 and other longtables	Longtables lack endfirsthead, endfoot, and captions, so the 34-row register breaks across pages with no continuation marker.	Regenerate register tables from the CSV registers with proper longtable head and foot in the new apparatus.	OBS	4	open
L-019	P2	12678–12821	Register table column widths are near-overfull.	Size regenerated tables by template with compile warnings treated as errors.	OBS	4	open
L-020	P2	13056–13058	Typo in the modal rule sheet: the calculus is said to contain substitution instances of classical classically valid propositional formulas (doubled word).	None in print: rule sheets are regenerated from the calculus registry, which retires the typo with the sheet.	OBS	4	open
S-026	P1	13749–13759	Exercise solutions use marked quantifiers with $>$ ($\exists x > b$, $\forall x > b$) registered nowhere in the book; same family as R-003's $\leq$ defect (found during adversarial verification of the G2 fix designs)	Register $>$-marked bounded quantifiers alongside $<$ and $\leq$, or rewrite the displays using registered comparisons	MIGR	4	open

3.3 Fix designs for the Phase-2 repairs

Each design below is a work order: it states the defect, the exact mathematical repair (statements, proof routes, landing sites), and acceptance criteria. The designs were drafted against the v1.0 text and adversarially verified.

R-001 — Honest CH independence (v1.0 lines 9062–9133; chapter 8665–9133)

Defect. The theorem at 9065 claims "the constructions using L and forcing establish" $\mathsf{Con}(\mathsf{ZFC}) \Rightarrow \mathsf{Con}(\mathsf{ZFC} + \neg\mathsf{CH})$, but the only forcing presented is over a countable transitive model, which Warning 9107–9117 says "assumes more than the bare arithmetical statement $\mathsf{Con}(\mathsf{ZFC})$" and must be discharged; it never is. The sketch (9092–9094) derives nonderivability by soundness from "the two structures satisfying the displayed theories," but neither is a set-sized structure: L is a proper class and M is hypothesized (so too the completeness appeal at 8759–8760).

Repair. **(1) 8733–8762.** Keep "L is transitive and contains every ordinal" (*Status: proof sketch*). Delete the Con displays (8739–8750) and the completeness sentence (8758–8760). Replace the satisfaction display by a separately stated scheme (*Status: cited theorem*; Jech 2003, Ch. 13, pp. 175–200 — reflection, definable well-order, condensation, as ZF-theorems about the definable L): $\mathsf{ZF} \vdash \sigma^L$ for every axiom σ of $\mathsf{ZFC} + \mathsf{GCH} + (V{=}L)$, with $(\cdot)^L$ the convention of 8710–8731. **(2) 8764–8782.** Retitle the section "Forcing over a countable transitive set: a pedagogical device." After the convention insert a Remark (*Status: proof sketch*; omitted verifications: the translation convention, the absoluteness computation, and the interpreted derivability-condition bookkeeping). $\mathsf{Con}(\mathsf{ZFC})$ (as at 7808) is a sentence of arithmetic, and the manuscript fixes no translation of arithmetic into the membership language, so the Remark must first state one: numbers as finite ordinals under the canonical interpretation, writing $\sigma^{\in}$ for the membership-language translation of an arithmetic sentence σ. Route: if such an M exists, soundness for the set-sized $(M, \in)$ gives metatheoretic consistency of ZFC; correctness of the fixed proof predicate (Warning at 7818–7824) makes $\mathsf{Con}(\mathsf{ZFC})$ true in the standard structure; absoluteness of arithmetic over transitive models containing ω (Jech, Ch. 12, pp. 155–174) then gives $(M, \in) \models \mathsf{ZFC} + \mathsf{Con}(\mathsf{ZFC})^{\in}$, and soundness again gives consistency of $\mathsf{ZFC} + \mathsf{Con}(\mathsf{ZFC})^{\in}$. That consistency sentence is unprovable in $\mathsf{ZFC} + \mathsf{Con}(\mathsf{ZFC})^{\in}$ itself — not by `thm:godel-second` verbatim, which covers computably enumerable $T \supseteq \mathsf{Q}$ in the language of arithmetic and so does not apply to a membership-language theory, but by that theorem transported across the canonical interpretation, under which the fixed provability predicate still satisfies the derivability conditions (this bookkeeping is the third omitted verification); so the hypothesis strictly exceeds $\mathsf{Con}(\mathsf{ZFC})$; other section results stay unchanged. **(3) 9062–9096.** Delete the theorem and sketch (9065–9096); insert: (a) *Lemma (Relativization of derivations)* [*complete proof*]: for membership-language formulas $\varphi_1, \ldots, \varphi_k, \psi$ with free variables among $v_1, \ldots, v_m$, if $\{\varphi_1, \ldots, \varphi_k\} \vdash \psi$ in the fixed natural-deduction calculus, then $\mathsf{ZF} \vdash \forall v_1 \cdots \forall v_m \, (\bigwedge_{j \le m} v_j \in L \wedge \bigwedge_{i \le k} \varphi_i^L \to \psi^L)$. Route (induction on the derivation): $(\cdot)^L$ commutes with connectives and $\bot^L = \bot$, so propositional and classical rules lift verbatim; terms are variables, so $\forall$-elimination and $\exists$-introduction consume a guard $u \in L$ from the antecedent, while $\forall$-introduction and $\exists$-elimination form the guarded clauses of 8718–8723 (eigenvariable condition unchanged; $x \in L$ is parameter-free); a free variable vanishing from the conclusion is discharged via $\mathsf{ZF} \vdash \exists u \, (u \in L)$, witness $\varnothing \in L_1$ (justify inside ZF by the comprehension reading of Def: $\{x \in L_0 : \ldots\} = \varnothing$ by separation whatever the satisfaction clause returns on the empty domain excluded at v1.0 3028; do not route through the proof-sketch result that L contains every ordinal); equality rules act on unchanged atoms; the explicit α rule (v1.0 3475–3487) maps to an α step, since relativization inserts the same parameter-free guard under alpha-variant binders. For $m = 0$ or $k = 0$ the corresponding conjunction is absent from the displayed conclusion. (b) *Theorem (Relative consistency of* CH*)* [*complete proof*]: if ZF is consistent, so is $\mathsf{ZFC} + \mathsf{GCH} + (V{=}L)$; hence if ZFC is consistent, so is $\mathsf{ZFC} + \mathsf{CH}$. Proof: a derivation of $\bot$ from axioms $\sigma_1, \ldots, \sigma_k$, the Lemma ($m = 0$, $\psi = \bot$), and the item-1 scheme give $\mathsf{ZF} \vdash \bot$; ZF-derivations are ZFC-derivations, and $\mathsf{ZFC} + \mathsf{GCH} + (V{=}L)$ derives CH (the $\alpha = 0$ instance of 8633), so inconsistency of $\mathsf{ZFC} + \mathsf{CH}$ lifts to it. (c) *Cited theorem (Relative consistency of* $\neg\mathsf{CH}$*)* [*cited theorem*; Jech, Ch. 14, pp. 201–224]: if ZFC is consistent, so is $\mathsf{ZFC} + \neg\mathsf{CH}$. Imported as ZFC-theorem schemes about the definable class V^B, B the completion of the poset of finite partial functions $\omega_2 \times \omega \to 2$: (i) $\|\sigma\|_B = 1$ for each ZFC axiom; (ii) Boolean soundness: $\{\sigma_1, \ldots, \sigma_k\} \vdash \psi$ implies $\mathsf{ZFC} \vdash \|\sigma_1\| \wedge \cdots \wedge \|\sigma_k\| \le \|\psi\|$, with $\|\bot\| = 0$ and B provably nontrivial; (iii) B is ccc and ccc forcing preserves cardinals: $\|\check{\aleph}_1 = \aleph_1 \wedge \check{\aleph}_2 = \aleph_2\| = 1$; (iv) the

distinctness density argument (9048–9055, per ω_2-section) gives $\|2^{\aleph_0} \geq \check{\aleph}_2\| = 1$, hence $\|\neg\mathsf{CH}\|_B = 1$. A $\bot$-derivation from $\mathsf{ZFC} \cup \{\neg\mathsf{CH}\}$ then yields $\mathsf{ZFC} \vdash 1 = 0$ in B: contradiction. This is the "Boolean-valued" discharge Warning 9107 demands — no countable transitive model; consistency is calculus-invariant via soundness and strong completeness. (d) *Remark (Arithmetized form)* [*proof sketch*]: the two derivation transformations are primitive recursive, so $\mathsf{Con}(\mathsf{ZFC}) \to \mathsf{Con}(\mathsf{ZFC}+\mathsf{CH})$ and $\mathsf{Con}(\mathsf{ZFC}) \to \mathsf{Con}(\mathsf{ZFC}+\neg\mathsf{CH})$ are PA-theorems; omitted: the arithmetization, including a primitive recursive translation between the fixed natural-deduction calculus and the source calculus of (c) — the semantic calculus-invariance argument in (c) does not supply this; the deduction-theorem translation does. (e) *Theorem (Nonderivability)* [*complete proof*]: if ZFC is consistent then $\mathsf{ZFC} \nvdash \mathsf{CH}$ and $\mathsf{ZFC} \nvdash \neg\mathsf{CH}$. Proof: $\mathsf{ZFC} \vdash \mathsf{CH}$ with monotonicity and the assumption rule gives $\mathsf{ZFC} + \neg\mathsf{CH} \vdash \mathsf{CH}, \neg\mathsf{CH}$, negation elimination gives $\vdash \bot$, and the contrapositive of (c) refutes consistency of ZFC; symmetrically $\mathsf{ZFC} \vdash \neg\mathsf{CH}$ makes $\mathsf{ZFC} + \mathsf{CH} \vdash \bot$, contradicting (b). **(4) 9098–9117.** Retain both Warnings, appending to the second that (c) supplies the Boolean-valued discharge; Exercise 9126 and the sketch at 13941–13952 remain correct.

Acceptance. (1) The section replacing 9065–9096 contains exactly the five results of item (3) with the stated statuses; the phrases "the constructions using L and forcing establish" and "the two structures satisfying the displayed theories" and the completeness sentence at 8759–8760 no longer occur in 8665–9133. (2) No proof in the exact-status section uses the countable-transitive-model convention, the generic-filter lemma, or $L \models$; nonderivability uses only monotonicity, assumption, negation elimination, and contrapositives of (b), (c). (3) Jech citations carry pages 175–200, 201–224, 155–174; the retitled document builds cleanly.

R-017 — Split the five-clause finite-coding theorem by status (v1.0 lines 678–702)

Defect. The five-clause theorem `thm:infinite-cardinal-finite-coding` carries the single label "Status: cited theorem," but the source is invoked only "for complete proofs of the cardinal products and union bounds" (clauses 1–4); clause 5 rests on the unlabeled inline sentence "The final item follows by assigning each generated expression its finite syntax string and applying the preceding items" (699–700). A result the manuscript itself proves is hidden inside a cited status, violating the policy at 313–319 that each result carries exactly one of the three status terms and that in-document dependencies be stated separately.

Repair.

1. Restate the theorem with clauses 1–4 only, statements unchanged, label kept, "*Status: cited theorem*"; cite Jech 2003, Theorem 3.5 ($\aleph_\alpha \cdot \aleph_\alpha = \aleph_\alpha$; with the already-cited well-ordering theorem every infinite cardinal is an aleph) and Ch. 5 (cardinal sums and the union bound $|\bigcup_{i\in I} A_i| \leq |I| \cdot \sup_{i\in I}|A_i|$ under AC). The postamble 698–702 shrinks to this citation sentence plus the retained no-infinite-products caveat; the inline clause-5 sentence is deleted.
2. Immediately after (landing at v1.0 line 697) add `cor:syntax-counting`, "*Status: complete proof*": *Assume the Axiom of Choice and let κ be an infinite cardinal. (i) Every set A with $|A| \leq \kappa$ satisfies $|A^{<\omega}| \leq \kappa$, where $A^{<\omega}$ is the set of finite sequences from A. (ii) If Σ is an alphabet with $|\Sigma| \leq \kappa$, then the finite strings over Σ, the finite tuples of such strings, and the formulas generated by any fixed finitary grammar whose alphabet extends Σ by at most countably many further symbols each form a set of cardinality at most κ.* Proof (promoted from the inline argument): (i) fix an injection $f\colon A \to \kappa$ (the meaning of $|A| \leq \kappa$); then $(a_1, \ldots, a_n) \mapsto (f(a_1), \ldots, f(a_n))$ is a length-preserving injection of $A^{<\omega}$ into $\kappa^{<\omega}$, and clause 2 gives $|\kappa^{<\omega}| = \kappa$. (ii) Strings over Σ lie in $\Sigma^{<\omega}$: apply (i) with $A = \Sigma$. Tuples of strings lie in $(\Sigma^{<\omega})^{<\omega}$: apply (i) with $A = \Sigma^{<\omega}$, of cardinality $\leq \kappa$ by the previous step. Every generated expression is a finite string over the grammar's full alphabet (the manuscript's own fact, v1.0 line 1126); writing X for the at most countable set of adjoined symbols

hypothesized in the statement (logical symbols, punctuation, and variables), clause 4 (applied to $\Sigma, X, \varnothing, \varnothing, \ldots$) gives $|\Sigma \cup X| \leq \kappa$, so the formula set injects into $(\Sigma \cup X)^{<\omega}$ and (i) applies.

3. Retarget the clause-5 uses: v1.0 lines 3949 (Henkin–Lindenbaum, "at most κ formulas") and 4347 (downward Löwenheim–Skolem, "at most μ function symbols, formulas, and finite tuples") now cite `cor:syntax-counting`; line 3969 (clause 3) keeps its explicit reference to `thm:infinite-cardinal-finite-coding`; at 4350, the anaphor "The same theorem gives $|N| \leq \mu$" (clause 4) currently resolves through the citation at 4347 and would silently retarget with it, so it must be rewritten as an explicit cross-reference to `thm:infinite-cardinal-finite-coding`.

Acceptance. (1) The region replacing 678–702 contains exactly two labeled results: the four-clause theorem ("Status: cited theorem," Jech citation naming the product and union-bound facts) and `cor:syntax-counting` ("Status: complete proof" with a proof environment containing the injection argument); the sentence beginning "The final item follows" no longer occurs. (2) The corollary's proof appeals only to clauses 2 and 4, the fixed injection, and the formulas-are-strings fact, with no citation to Jech inside the proof. (3) References formerly at 3949 and 4347 resolve to `cor:syntax-counting`, those at 3969 and 4350 to `thm:infinite-cardinal-finite-coding`; the document compiles with no undefined references.

R-002 — Bounded Comp versus the Cantor-pair numbering (v1.0 lines 7004–7031, 6564–6593)

Defect. The theorem at 7015–7028 asserts a bounded $\mathsf{Comp}(e, x, s)$ recognizing accepting computations in the fixed machine numbering (6567–6585), while 7011–7013 concedes that numbering's iterated Cantor pairing "comes with no bounded extraction formula and is not the coding used here." Comp must still decode e and x from iterated Cantor codes; the citations (7030–7031) arithmetize other numberings.

Repair. Option (ii): keep the numbering; make the history coding carry β-extractable *decoding certificates*, so Comp becomes bounded. Replace 7004–7013 by the material below and keep 7015–7027 plus one new clause; 7030–7031 stays.

Lemma A (pairing graph; lands before 7015). $\mathsf{P}(x, y, z) :\equiv (x + y) \times S(x + y) + (y + y) = z + z$ is quantifier-free, defines the graph of $\mathrm{pair}_{\mathbb{N}}$, and $a, b \leq \mathrm{pair}_{\mathbb{N}}(a, b)$ ($b \leq z$; $a \leq a+b \leq (a+b)(a+b+1)/2 \leq z$). Status: complete proof. Lemma B (Gödel β). With the bounded formula

$$\mathsf{B}(w, i, v) :\equiv \exists t < Sw\, \exists m < Sw\, \exists q < St\, \big(\mathsf{P}(t, m, w) \wedge t = d \times q + v \wedge \exists u < d\, (v + Su = d)\big),$$

writing d for the term $S((Si) \times m)$ (u fresh, $u \notin \mathrm{Var}(d)$). The final conjunct is the *marked-bounded* form of $v < d$, $\mathcal{N}$-equivalent to the plain abbreviation of 6924–6926 because any witness of $v + Su = d$ satisfies $Su \leq d$, hence $u < d$; the marked form is required because the book counts only marked bounded quantifiers as bounded (6936–6937, 6939–6951), while a bare comparison $r < t$ expands to an unbounded existential. Claims: (a) for all w, i exactly one $v \leq w$ satisfies B in $\mathcal{N}$; write $\beta(w, i) = v$; (b) every finite sequence $a_0, \ldots, a_{k-1}$ is $\beta(w, \cdot) \restriction k$ for some w. Proof of (a): $\mathrm{pair}_{\mathbb{N}}$ is a bijection, so w determines t, m uniquely, with $t, m \leq w$ by Lemma A, matching the displayed bounds; Euclidean division of t by $d \geq 1$ determines q, v uniquely, with $q \leq t < St$ and $v \leq t \leq w$. Proof of (b): put $m := N!$ with $N := \max(k, a_0, \ldots, a_{k-1}) + 1$; the moduli $1 + (i + 1)m$ are pairwise coprime (a common prime divides $(j - i)m$ but not m, hence divides $j - i$; yet $0 < |j - i| < k \leq N$ forces it to divide m); each $a_i < m$; Chinese remaindering gives $t \equiv a_i \pmod{1 + (i + 1)m}$; put $w := \mathrm{pair}_{\mathbb{N}}(t, m)$. Gödel 1931 and Boolos–Burgess–Jeffrey 2007, ch. 16, are parallel reading. Status: complete proof.

Definition (history coding; replaces 7004–7013). s is a *history code* for (e, x) iff, under $\beta(s, \cdot)$ with a fixed index scheme (families at $\mathrm{pair}_{\mathbb{N}}(t, j)$), it contains: (h1) the binary digits of x with certified expansion $v_0 = d_0$, $v_{j+1} = 2v_j + d_{j+1}$, $v_{L-1} = x$, $v_j \leq x$, $d_j \leq 1$; (h2) the table of e: a chain $\mathsf{P}(k, c_0, e)$, $\mathsf{P}(r_i, c_{i+1}, c_i)$ ($i < k - 1$), $r_{k-1} = c_{k-1}$, rows unpacked by fixed-depth P-steps, plus the validity checks of 6567–6576

under the designated-slot normalization of the R-005 repair; (h3) configurations $C_0, \dots, C_T$, tapes coded in the tape-alphabet base with certified expansions, the first from (h1), each step matching the decoded table by bounded digit comparisons; (h4) final state accepting, tape prefix before the first blank a valid binary numeral (6327–6338). Every auxiliary value is an entry of s (Lemma B(a)) or a Cantor-code component (Lemma A), so all quantifiers are bounded by terms in e, x, s; and every comparison conjunct $r < t$ or $r \le t$ occurring in Comp is written in its marked-bounded form $\exists u < t\,(r + Su = t)$, respectively $\exists u \le t\,(u + r = t)$ with $\le$ as registered by the R-003 repair (u fresh), each $\mathcal{N}$-equivalent to the plain abbreviation, so the matrix is bounded under the counting rule of 6936. Repaired theorem: $\mathsf{Comp}(e, x, s)$ is bounded; $\mathcal{N} \models \mathsf{Comp}(\overline{e}, \overline{x}, \overline{s})$ iff s is a history code for (e, x); and $\varphi_e(x)$ is defined iff $\mathcal{N} \models \exists s\, \mathsf{Comp}(\overline{e}, \overline{x}, s)$ (existence via Lemma B(b); invalid e fail (h2), matching divergence). Status: theorem proof sketch with plan (h1)–(h4); Lemmas A, B complete proof.

Finally, add at 7449–7454 one sentence noting that the coded-computation-sequence apparatus behind `thm:representability`'s Σ^0_1 clause (the construction paragraph at 7351–7357) is the β coding of Lemma B, so a single in-document coding serves machine tables, histories, and proof sequences; this discharges the register's one-coding acceptance for Prov_T.

Acceptance. (1) 7011–7013 deleted; history coding in-document. (2) Every displayed quantifier bounded by a term in e, x, s; every comparison conjunct in marked-bounded form; monotonicity proved. (3) 7040–7051, 7068–7074, 13735–13760 type-check unchanged. (4) The Σ^0_1 presentations of Comp and of Prov_T (7440–7454) both point at Lemma B's apparatus.

R-003 — The two Q-lemmas for Rosser's bounded comparison (v1.0 lines 7663–7758, 6924–6937, 7210–7222)

Defect. The Rosser proof plan (7682–7700) rests on "evaluating the bounded quantifier in Q" and on the exhibited proof witnessing "the bounded existential for every larger candidate." Neither is backed: $\le$ is never introduced (only $<$, 6924–6926), and the two needed Q-facts — bounded-case enumeration and dichotomy — are neither stated nor cited.

Repair. (1) Register, next to $<$ at 6924–6937: $r \le t :\equiv \exists z\,(z + r = t)$, z least fresh; $\exists x \le t\,\varphi :\equiv \exists x(x \le t \wedge \varphi)$, $\forall x \le t\,\varphi :\equiv \forall x(x \le t \to \varphi)$ for $x \notin \mathrm{Var}(t)$, counted as bounded, like $<$. Record: $\mathcal{N} \models x \le t \leftrightarrow x < St$, *not* claimed in Q; the witness sits *left* of $+$ so the lemmas need only Q's right-recursion axioms.

(2) Two numbered lemmas after Q's definition (7222), proved by metainduction on n; each instance is a Q-derivation from $Sx \neq 0$, S-injectivity, $x \neq 0 \to \exists y\, x = Sy$, $x + 0 = x$, $x + Sy = S(x + y)$.

Lemma Q1 (bounded-case enumeration) and Lemma Q2 (dichotomy). For every $n \in \mathbb{N}$:

$$\mathsf{Q} \vdash \forall y\,(y \le \overline{n} \to (y = \overline{0} \vee y = \overline{1} \vee \cdots \vee y = \overline{n})), \qquad \mathsf{Q} \vdash \forall y\,(y \le \overline{n} \vee \overline{n} \le y).$$

Proof of Q1: $n = 0$: from $z + y = 0$, if $y = Sw$ then $S(z + w) = z + Sw = 0$ contradicts $Sx \neq 0$; so $y = 0$. Step: from $z + y = S\overline{n}$: $y = 0$ gives the first disjunct; $y = Sw$ gives $S(z + w) = S\overline{n}$, S-injectivity gives $z + w = \overline{n}$, so $w \le \overline{n}$; the induction hypothesis gives $w = \overline{0} \vee \cdots \vee w = \overline{n}$; apply S. Proof of Q2: $n = 0$: $\overline{0} \le y$, witness $z := y$, by $y + 0 = y$. Step: fix y; if $y = 0$, $y \le S\overline{n}$ (witness $S\overline{n}$); if $y = Sw$, the hypothesis on w gives either $z + w = \overline{n}$, whence $z + Sw = S(z + w) = S\overline{n}$ and $y \le S\overline{n}$, or $u + \overline{n} = w$, whence $u + S\overline{n} = S(u + \overline{n}) = Sw = y$ and $S\overline{n} \le y$. Status of both: complete proof.

(3) Rewrite 7682–7700 to invoke the lemmas by label, with fixed point $\mathsf{Q} \vdash R_T \leftrightarrow \forall y\,(\mathsf{Proof}_T(y, \overline{\ulcorner R_T \urcorner}) \to \exists z \le y\, \mathsf{Proof}_T(z, \overline{\ulcorner \neg R_T \urcorner}))$. Branch $T \vdash R_T$, least proof code p: representability gives $T \vdash \mathsf{Proof}_T(\overline{p}, \overline{\ulcorner R_T \urcorner})$, hence $T \vdash \exists z \le \overline{p}\, \mathsf{Proof}_T(z, \overline{\ulcorner \neg R_T \urcorner})$; consistency gives $T \nvdash \neg R_T$, so numeralwise $T \vdash \neg \mathsf{Proof}_T(\overline{q}, \overline{\ulcorner \neg R_T \urcorner})$ for $q \le p$; Lemma Q1 turns the bounded existential into a finite disjunction: inconsistency. Branch $T \vdash \neg R_T$, least code q: reason in T from $\mathsf{Proof}_T(y, \overline{\ulcorner R_T \urcorner})$; Lemma Q2 gives $y \le \overline{q} \vee \overline{q} \le y$; first case:

Lemma Q1 plus $T \vdash \neg\mathsf{Proof}_T(\overline{k}, \overline{\ulcorner R_T \urcorner})$ for $k \leq q$ (from $T \nvdash R_T$) yields a contradiction; second case: witness $z := \overline{q}$. Hence $T \vdash R_T$, contradicting consistency. Status: Rosser stays proof sketch, now citing Q1/Q2 by label; the solution at 13830–13838 names both lemmas.

Acceptance. (1) $\leq$ is registered with its expansion, the bounded-counting rule, and the non-claim of Q-equivalence with $x < St$; no arithmetic object-formula uses $\leq$ earlier. (2) Q1 and Q2 appear exactly as displayed, proved by metainduction from the Q-axioms, labeled complete proof. (3) Each Rosser branch names which lemma discharges which step.

R-004 — Jump definition and the partiality convention (v1.0 lines 7134–7189, 6576–6582)

Defect. The jump (7154–7161) uses "the e-th oracle machine with oracle B *halts* on input e," while K (6678–6681) uses "$\varphi_e(e)$ is defined," which demands an accepting halt with valid numerical output (6576–6582); halting also covers rejection and invalid output. So the claim at 7163–7165 that the K diagonal "relativizes" fails literally and is unlabeled prose; 7142–7144 also never fixes the oracle I/O convention.

Repair. (1) Replace 7154–7161. Fix an effective numbering of oracle machines analogous to 6567–6585 (same normalization, query rows included, invalid codes diverge everywhere). For valid e and $B \subseteq \mathbb{N}$, let $\varphi_e^B(x)$ be the numerical output when the decoded machine with oracle B, on $\mathrm{bin}(x)$, halts accepting with a valid numerical output, and undefined when it rejects, diverges, or accepts without one — 6576–6582 verbatim, relativized. Define $B' := \{e \in \mathbb{N} : \varphi_e^B(e) \text{ is defined}\}$, $B^{(0)} := B$, $B^{(n+1)} := (B^{(n)})'$. Amend 7142–7145: oracle machines "compute" by this discipline; "computably enumerable relative to B" means: domain of some φ_e^B.

(2) New lemma (relativized adequacy). Every partial function computed with oracle B occurs as φ_e^B; a relativized universal machine gives $U^B(e,x) \simeq \varphi_e^B(x)$; and a relativized parameter theorem holds: for every $m, n \geq 1$ there is a total computable par_n^m with $\varphi^B_{\mathsf{par}_n^m(e,\bar{a})}(\bar{x}) \simeq \varphi_e^B(\bar{a},\bar{x})$ for every oracle B simultaneously. Route: R-005's adequacy proof (renaming commutes with the step relation, queries included), the universal simulation of 6608–6618 with queries passed through, and the compiler construction of 6632–6641 with query rows copied unchanged. Status: proof sketch.

(3) Promote 7163–7165 to a numbered theorem. Jump theorem: for every $B \subseteq \mathbb{N}$, (a) B' is computably enumerable relative to B; (b) $B' \not\leq_T B$. Proof (relativized diagonal): (a) B' is the domain of $e \mapsto U^B(e,e)$. (b) If an oracle machine with oracle B decides B' via total h^B, define $d^B(e) := 0$ if $h^B(e) = 0$, undefined otherwise; d^B is oracle-computable (run the decider; write 0 and accept, or loop); fix b with $\varphi_b^B \simeq d^B$ by the adequacy lemma; then $b \in B'$ iff $\varphi_b^B(b)$ is defined iff $h^B(b) = 0$ iff $b \notin B'$. Status: proof sketch — the displayed diagonal calculation is complete, but clause (a) rests on the relativized universal machine and clause (b) on the occurrence clause of the relativized adequacy lemma, both proof sketches; the status policy (313–319) and the release-checklist rule (14346) forbid complete-proof status over a sketch dependency.

(4) New remark after 7189, keeping citations sound. (i) $K \equiv_m \varnothing'$: an ordinary table embeds as a query-free oracle table; an oracle-$\varnothing$ machine is simulated by answering queries negatively. (ii) The halting-based jump of the cited sources is $\equiv_m$ this B': $e \mapsto e^*$ simulating e and, on any halt, writing 0 and accepting (halting $\leq_m$ defined); $e \mapsto e^{**}$ diverging unless the halt is accepting with valid output (defined $\leq_m$ halting). All four maps are computable table transformations, uniform via the parameter theorem (`thm:parameter`). Jump monotonicity — $A \leq_T B$ implies $A' \leq_m B'$ — is stated and used explicitly: replace each A-query of an oracle-A machine by a run of a fixed oracle-B decider for A, then align self-application indices by the relativized parameter theorem of the adequacy lemma with padding. Hence Post's theorem (7167–7175) is insensitive to the convention; the citations stand. Status: proof

sketch — the four table transformations are elementary, but the monotonicity step runs through the relativized parameter theorem, part of the sketch-status adequacy lemma.

Acceptance. (1) B' is defined via "$\varphi_e^B(e)$ is defined," three-way undefinedness relativized verbatim; 7142–7145 fixes the oracle I/O convention. (2) The jump theorem is a numbered statused result argued by the relativized diagonal invoking the labeled adequacy lemma, with a status label matching its sketch dependencies. (3) The remark states $K \equiv_m \varnothing'$, the $\equiv_m$-bridge to the halting convention, and the jump-monotonicity step it uses, each with honest status.

R-005 — The enumeration claim for the fixed numbering (v1.0 lines 6564–6593, decisive at 6582–6585)

Defect. The claim that "every unary partial computable function occurs in this numbering" and, via the tuple convention, every k-ary one (6582–6585), is unlabeled prose with no proof, status, or citation. Every construction that picks an index for an informally described machine (6692–6699, 6724, 6795–6816, 13735–13760) depends on it, and a bare citation cannot discharge it: the numbering is bespoke (normalization, validity constraint $\{0,1\} \subseteq \mathcal{A}_{\text{in}}$, wrapper).

Repair. (1) Amend 6567–6569 to fix the normalization completely: a description is *normalized* iff its tape symbols are the numbers $0, \ldots, |\mathcal{A}_{\text{tape}}| - 1$ with the designated assignments $\sqcup \mapsto 0$, digit $0 \mapsto 1$, digit $1 \mapsto 2$, and its states are the numbers $0, \ldots, |Q| - 1$ with $q_0 \mapsto 0$, $q_{\text{acc}} \mapsto 1$, $q_{\text{rej}} \mapsto 2$; the remaining numbers are assigned arbitrarily, and the row list is the lexicographic listing by numbered key (q, a) over the non-halting domain. (An "order of first appearance" clause would be circular: the lexicographic row order presupposes the numbering.) No canonical machine-to-code map is needed, since indices are not unique (6587); valid codes thus have pairwise distinct designated states, and machines whose start state is halting (permitted by 6233–6234) are captured up to behavior by the surrogate-start step in the theorem's proof below. Without the designated slots "valid code" is not well defined, and the validity clause $\{0,1\} \subseteq \mathcal{A}_{\text{in}}$ together with the input and output conventions (6296–6338) is to be read at the designated digit numbers $1, 2$.

(2) New lemma (renaming invariance; lands after 6593). If $\iota = (\iota_Q, \iota_A)$ bijectively renames states and tape symbols preserving the roles $q_0, q_{\text{acc}}, q_{\text{rej}}, \sqcup, 0, 1$, then $C \to_{\mathcal{T}} C'$ iff $C^\iota \to_{\mathcal{T}^\iota} (C')^\iota$; hence $\mathcal{T}$ and $\mathcal{T}^\iota$ accept, reject, or diverge together on every input, with corresponding tapes. Proof: one-step induction from 6248–6294. Status: complete proof.

(3) Replace 6582–6585 by a numbered theorem. Adequacy of the numbering (enumeration theorem): (a) every φ_e is partial computable; (b) for every $k \geq 1$ and partial computable $f : \mathbb{N}^k \rightharpoonup \mathbb{N}$ there is e with $\varphi_e(\bar{x}) \simeq f(\bar{x})$ for all $\bar{x} \in \mathbb{N}^k$, under the arity-k convention of 6588–6593. Proof route: (a) is the wrapper argument at 6579–6582, folded in. (b) Let $\mathcal{T}$ witness f per 6340–6349. First, $\{0,1\} \subseteq \mathcal{A}_{\text{in}}$ automatically: code_k is surjective for every $k \geq 1$ (iterated pairing bijections), so the inputs $\text{bin}(\text{code}_k(\bar{x}))$ exhaust all binary numerals; initial configurations exist only for words in $\mathcal{A}_{\text{in}}^*$ (6280–6288), and 6340–6349 demands an accepting or infinite computation on each such input; so both digits lie in $\mathcal{A}_{\text{in}}$, and no alphabet extension is needed. Second, since 6233–6234 requires only $q_{\text{acc}} \neq q_{\text{rej}}$, the start state may coincide with a halting state; in that case first replace it by a fresh start state q_0' with $\delta(q_0', a) = (q_0, a, R)$ for every tape symbol a — the one extra step leaves the tape unchanged, and the output convention ignores the final head position and off-prefix cells (6331–6338), so the accepted outputs and divergence behavior are unchanged. Then the three designated states and three designated symbols are pairwise distinct; choose any role-preserving bijective renaming onto the designated-slot numbering, obtaining a normalized $\mathcal{T}^*$ that behaves identically by the renaming lemma; list its rows lexicographically by numbered key (the row list is the full graph of δ, total on non-halting pairs by 6220–6246), encode as $\text{pair}_{\mathbb{N}}(k, \text{code}_k(\text{rows}))$; the code e is valid by construction, and by the lemma e's behavior on every $\text{bin}(\text{code}_k(\bar{x}))$ is $\mathcal{T}$'s, i.e. $\varphi_e(\bar{x}) \simeq f(\bar{x})$. Retain the Kleene 1952 citation (enumeration and normal form over standard numberings) explicitly as parallel

reading; the in-document proof is primary because the numbering is bespoke. Status afterwards: complete proof.

Acceptance. (1) No unlabeled adequacy prose remains at 6582–6585; the claim exists only as the numbered theorem, clause (b) quantifying over all $k \geq 1$ with $\varphi_e(\bar{x}) \simeq f(\bar{x})$. (2) The normalization convention (designated numbers for $\sqcup, 0, 1, q_0, q_{\text{acc}}, q_{\text{rej}}$) is fixed in the numbering paragraph and used by the proof. (3) The diagonal construction (6692–6699) and the Rice proof (6803–6816) cite the theorem by label when picking indices.

R-006 — Closure-class versus prefix hierarchy (v1.0 lines 6939–7002, 7030–7189)

Defect. The hierarchy definition (6952–6971) builds Σ^0_{n+1}, Π^0_{n+1} as *closure classes* (under $\wedge, \vee$, both bounded quantifiers, one unbounded quantifier), while every hierarchy fact discharged by citation (7030–7031, 7078–7080, 7082–7097, 7167–7189) concerns the literature's *prefix* hierarchy. Nothing identifies the two, so the citations do not apply; the missing ingredient is bounded-quantifier exchange.

Repair. Insert after 7002:

Lemma C (contraction). Let $\mathsf{P}(x, y, z)$ be the quantifier-free pairing-graph formula of the R-002 repair's Lemma A, with its monotonicity bound $a, b \leq \text{pair}_{\mathbb{N}}(a, b)$; the two designs land at adjacent sites (before 7015, after 7002), so the manuscript states P and its lemma exactly once and Lemma C cites it. Then, for z fresh (z not free in θ, $z \notin \{x, y\}$): $\mathcal{N} \models \exists x \exists y\, \theta \leftrightarrow \exists z\, \exists x < Sz\, \exists y < Sz\, (\mathsf{P}(x, y, z) \wedge \theta)$, dually for $\forall\forall$ with $\rightarrow$. Status: complete proof (via Lemma A).

Lemma D (bounded-quantifier exchange). For every formula θ and term t with $y, w \notin \text{Var}(t)$, w fresh:

$$\mathcal{N} \models \forall y < t\, \exists x\, \theta \leftrightarrow \exists w\, \forall y < t\, \exists x < w\, \theta, \quad \mathcal{N} \models \exists y < t\, \forall x\, \theta \leftrightarrow \forall w\, \exists y < t\, \forall x < w\, \theta.$$

Proof: right-to-left immediate; left-to-right: finitely many $y < \text{val}(t)$, choose witnesses x_y, put $w := \max_y x_y + 1$; the second display is dual. Equivalences *in* $\mathcal{N}$, for boldface classes only; no Q- or PA-provability is claimed. Status: complete proof.

Theorem E (prefix normal form). For every $n \geq 0$, relations defined by prefix-form Σ^0_n formulas (n alternating unbounded blocks starting $\exists$, bounded matrix) are closed, up to $\mathcal{N}$-equivalence of the defining formulas, under $\wedge$, $\vee$, bounded $\exists$, bounded $\forall$, unbounded $\exists$; dually for Π^0_n. Hence, by induction on the clauses of 6952–6971, every closure-class formula is $\mathcal{N}$-equivalent to a prefix-form formula of the same class; since conversely every prefix-form formula is generated by those clauses, the boldface classes of 6974–7002 coincide with the cited sources'. Proof route (execute literally): induction on n, both classes at once, Π dual. $n = 0$: all operations except unbounded $\exists$ stay bounded syntactically. $n \geq 1$, relation $\exists x\, B$, B prefix-Π^0_{n-1}: (i) $\wedge, \vee$: rename, pull existentials out, contract by Lemma C; the residual bounded $\exists$'s land on prefix-Π^0_{n-1} formulas, absorbed at level $n-1$. (ii) bounded $\exists$: $\exists y < t\, \exists x\, B \equiv \exists x\, \exists y < t\, B$ by pure logic; absorb at level $n-1$. (iii) bounded $\forall$ (collection): by Lemma D, $\forall y < t\, \exists x\, B$ is $\mathcal{N}$-equivalent to $\exists w\, \forall y < t\, \exists x < w\, B$; absorb both bounded quantifiers at level $n-1$, prepend $\exists w$. (iv) unbounded $\exists$: contract by Lemma C. Base: a level-$(n-1)$ formula converts by hypothesis and is padded with dummy quantifiers. Status afterwards: complete proof. Cite Theorem E at 7078–7080 for the TOT bound and at 7030–7031, 7097, 7178–7189 as license for the prefix-based citations.

Acceptance. (1) Lemma D appears exactly as displayed, proved by the finite-witness maximum, $\mathcal{N}$-equivalence only. (2) Theorem E's induction covers all five operations for both classes at every level, bounded-$\forall$-over-Σ via Lemma D, contractions via Lemma C. (3) All four citation sites reference Theorem E.

R-007 — Prenex normal form: missing lemma chain (v1.0 lines 4396–4446)

Defect. The prenex theorem's proof (4424–4446) opens with unproved global operations: "move negations inward using De Morgan equivalences and quantifier duality, and alpha-rename bound variables so that the formula is rectified" (4425–4427). No first-order negation-normal-form lemma exists (the propositional theorem, 1899–1917, covers no quantifiers and its termination measure counts only connectives below a negation); no rectified-variant existence result exists (no renaming recursion is ever defined); and "quantifier duality" resolves to `lem:quantifier-duality` (3814), a derivability lemma of $\mathsf{ND}_{\mathrm{FOL}}$, where a semantic equivalence is required. The extraction induction then rests on "Rectification guarantees the side condition" (4441) without any stated invariant.

Repair. Insert Lemmas A–C between the definitions (4399–4418) and the theorem (4420), rewrite the theorem's proof as their composition, and attach the missing label `cor:alpha-invariance` at 3203; c is the complexity of 2875 (counting quantifiers, so the measure covers quantifier prefixes) and $\equiv$ is logical equivalence (3256).

Lemma A (`lem:fol-nnf`, negation normal form). Define $\mathrm{nnf}, \mathrm{nnf}' : \mathrm{Fm}(\mathcal{L}) \to \mathrm{Fm}(\mathcal{L})$ by simultaneous recursion: nnf fixes atomic formulas and $\bot$, commutes with $\wedge, \vee, \forall, \exists$, and satisfies $\mathrm{nnf}(\neg\varphi) = \mathrm{nnf}'(\varphi)$, $\mathrm{nnf}(\varphi \to \psi) = \mathrm{nnf}'(\varphi) \vee \mathrm{nnf}(\psi)$; and, for α atomic, $\mathrm{nnf}'(\alpha) = \neg\alpha$, $\mathrm{nnf}'(\bot) = \neg\bot$, $\mathrm{nnf}'(\neg\varphi) = \mathrm{nnf}(\varphi)$, $\mathrm{nnf}'(\varphi \wedge \psi) = \mathrm{nnf}'(\varphi) \vee \mathrm{nnf}'(\psi)$, $\mathrm{nnf}'(\varphi \vee \psi) = \mathrm{nnf}'(\varphi) \wedge \mathrm{nnf}'(\psi)$, $\mathrm{nnf}'(\varphi \to \psi) = \mathrm{nnf}(\varphi) \wedge \mathrm{nnf}'(\psi)$, $\mathrm{nnf}'(\forall x\, \varphi) = \exists x\, \mathrm{nnf}'(\varphi)$, $\mathrm{nnf}'(\exists x\, \varphi) = \forall x\, \mathrm{nnf}'(\varphi)$ (every call is on an immediate subformula, so c strictly decreases and the recursion is legitimate). Claims, by simultaneous strong induction on $c(\varphi)$: (i) both values are in negation normal form (definition 4399–4408; $\neg\bot$ is admitted there); (ii) $\mathrm{nnf}(\varphi) \equiv \varphi$ and $\mathrm{nnf}'(\varphi) \equiv \neg\varphi$, each case checked against the satisfaction clauses (3111–3145): the nnf' quantifier cases are the pointwise computations $\mathcal{M}, s \models \neg\forall x\, \varphi$ iff some a has $\mathcal{M}, s[x \mapsto a] \models \neg\varphi$ iff $\mathcal{M}, s \models \exists x\, \neg\varphi$, dually for $\exists$ — not citations of `lem:quantifier-duality`; (iii) FV and Var of both values equal those of φ.

Lemma B (`lem:rectified-variant`, rectified variant). For φ and finite $V \supseteq \mathrm{Var}(\varphi)$ define $r(\varphi, V)$: atomic and $\bot$ unchanged; $r(\neg\varphi, V) = \neg r(\varphi, V)$; $r(\varphi \circ \psi, V) = r(\varphi, V) \circ r(\psi,\, V \cup \mathrm{Var}(r(\varphi, V)))$ for $\circ \in \{\wedge, \vee, \to\}$; $r(Qx\, \varphi, V) = Qz\, r(\varphi[z/x],\, V \cup \{z\})$ with z least outside V. Well-foundedness is the 2951–2958 pattern: $z \notin \mathrm{Var}(\varphi) \cup \{x\}$, so no binder of φ equals z, the substitution runs on the first two quantifier clauses only (textual replacement, so $\mathrm{Var}(\varphi[z/x]) \subseteq \mathrm{Var}(\varphi) \cup \{z\}$), and $c(\varphi[z/x]) = c(\varphi) < c(Qx\, \varphi)$ by `lem:substitution-complexity` (2965). Claims, by strong induction on $c(\varphi)$ under the invariant $\mathrm{Var}(\varphi) \subseteq V$: (i) $r(\varphi, V) \equiv_\alpha \varphi$ — connective cases by congruence, quantifier case by the generating clause $Qx\, \varphi \equiv_\alpha Qz\, (\varphi[z/x])$ (2984, licensed by $z \notin \mathrm{Var}(\varphi)$), the Q-congruence, and transitivity — hence $r(\varphi, V) \equiv \varphi$ by `cor:alpha-invariance`; (ii) $r(\varphi, V)$ is rectified: its binders are pairwise distinct and outside V (the binary clause forbids the left output's variables to the right recursion), while $\mathrm{FV}(r(\varphi, V)) = \mathrm{FV}(\varphi) \subseteq V$ by (iii); (iii) alpha-equivalent formulas have equal free-variable sets: the equal-FV relation is a congruence equivalence containing each generating pair, since both cases of clause (b) of `lem:substitution-fv` (added by R-010) give $\mathrm{FV}(Qy\, (\varphi[y/x])) = \mathrm{FV}(Qx\, \varphi)$, so leastness of $\equiv_\alpha$ applies; (iv) r preserves the constructor tree, hence negation normal form.

Lemma C (`lem:prenex-extraction`, extraction with invariant). For every rectified φ in negation normal form there is $p(\varphi)$ with: (i) $p(\varphi) \equiv \varphi$; (ii) $p(\varphi)$ rectified and prenex with quantifier-free negation-normal matrix; (iii) *invariant:* FV and Var unchanged, and the prefix consists of exactly the quantifier occurrences of φ (same quantifier, same variable) — no new free or bound variables. Route: structural induction on the negation-normal generation (subformulas of rectified formulas are rectified); p is the identity on base formulas and commutes with Qx. Binary case $\alpha \circ \beta$, $\circ \in \{\wedge, \vee\}$: with $p(\alpha) = Q_1 y_1 \cdots Q_k y_k\, \alpha_0$, $p(\beta) = R_1 z_1 \cdots R_m z_m\, \beta_0$, set $p(\alpha \circ \beta) = Q_1 y_1 \cdots Q_k y_k\, R_1 z_1 \cdots R_m z_m\, (\alpha_0 \circ \beta_0)$, proved $\equiv \alpha \circ \beta$ by an inner induction on $k + m$ from two recorded facts: (C1) $(Qy\, \pi) \circ \chi \equiv Qy\, (\pi \circ \chi)$ and $\chi \circ (Qy\, \pi) \equiv Qy\, (\chi \circ \pi)$ whenever $y \notin \mathrm{FV}(\chi)$, proved from the satisfaction clauses and `lem:fol-coincidence` (3148), displaying the $\forall/\vee$ case (the classical distribution of $\forall$ over a disjunct constant in the quantified variable); (C2) $\equiv$ is a congruence for $\circ$ and Qx (immediate from the satisfaction clauses). The side conditions hold because $y_i \notin \mathrm{FV}(p(\beta)) = \mathrm{FV}(\beta)$ (y_i is bound in the rectified $\alpha \circ \beta$) and $z_j \notin \mathrm{FV}(\alpha_0) \subseteq \mathrm{FV}(\alpha) \cup \{y_1, \ldots, y_k\}$ (rectifiedness, binder distinctness); the output is rectified and satisfies the invariant since nothing is renamed or introduced.

Rewritten theorem proof (replaces 4424–4446; the statement gains label `thm:prenex-normal-form`): by Lemmas A–C, $p\big(r(\mathrm{nnf}(\varphi), \mathrm{Var}(\mathrm{nnf}(\varphi)))\big) \equiv \varphi$ is rectified, prenex, and in the same signature; all four results then carry *Status: complete proof.*

Acceptance. (1) The rewritten proof cites only Lemmas A–C and `cor:alpha-invariance`; "quantifier duality" no longer occurs in 4396–4446. (2) Each lemma names its induction measure; Lemma B's renaming recursion cites `lem:substitution-complexity` for its non-structural call on $\varphi[z/x]$. (3) Lemma C states the no-new-variables

invariant and every extraction step's side condition is discharged from it plus rectifiedness.

R-008 — Skolemization: explicit prefix induction replacing the "quantifier game" (v1.0 lines 4480–4512)

Defect. Both directions of `thm:skolem-equisatisfiable` (4492–4507) are prose: "Induction along the prefix then shows…" states no induction hypothesis, and the converse appeals to "the original quantifier game" (4501) — a notion defined nowhere in the document. The Axiom of Choice is hypothesized for the whole theorem (4482) although only the expansion direction uses it.

Repair. Fix $\sigma = Q_1 x_1 \cdots Q_n x_n\, \theta$ rectified prenex. For $0 \le i \le n$ let $\sigma_i = Q_{i+1} x_{i+1} \cdots Q_n x_n\, \theta$ (so $\sigma_0 = \sigma$, $\sigma_n = \theta$, and $\mathrm{FV}(\sigma_i) \subseteq \{x_1, \dots, x_i\}$); let $q(i)$ count the universal positions among the first i, let $\bar{u}(j)$ list the universal variables before position j, and for existential j let $t_j = f_j(\bar{u}(j))$ be its Skolem term, so $\mathrm{Sk}(\sigma) = \forall u_1 \cdots \forall u_p\, \theta^*$ with $\theta^* = \theta[t_j/x_j]$ simultaneously for all existential j (θ is quantifier-free, so this is plain replacement; since no t_j contains an existential variable, the simultaneous substitution equals the composition of the single substitutions in any order). Given a structure and a tuple $\bar{a}$ of values for the universal variables among $x_1, \dots, x_i$, define the *Skolem-term assignment* $s_{\bar{a}}$: universal x_j gets its entry of $\bar{a}$; existential x_j gets the value v_j of t_j computed from the entries for $\bar{u}(j)$; values of other variables are irrelevant by `lem:fol-coincidence`.

Restated theorem (replaces 4480–4491): let σ be a rectified prenex $\mathcal{L}$-sentence. (1) *(Axiom of Choice.)* Every $\mathcal{L}$-structure satisfying σ has an expansion to the Skolem signature satisfying $\mathrm{Sk}(\sigma)$. (2) *(No choice.)* The $\mathcal{L}$-reduct of every model of $\mathrm{Sk}(\sigma)$ satisfies σ. Hence, assuming the Axiom of Choice, σ has a model iff $\mathrm{Sk}(\sigma)$ has one; the right-to-left implication holds without choice.

Proof of (1). Let $\mathcal{M} \models \sigma$. By finite recursion on i choose set-functions F_j (for existential $j \le i$) while proving

$$\mathrm{Inv}(i): \quad \mathcal{M}, s_{\bar{a}} \models \sigma_i \quad \text{for every } \bar{a} \in |\mathcal{M}|^{\,q(i)},$$

where $s_{\bar{a}}$ computes $v_j = F_j(\text{entries of } \bar{a} \text{ for } \bar{u}(j))$. $\mathrm{Inv}(0)$ is the hypothesis. Universal step: from $\mathcal{M}, s_{\bar{a}} \models \forall x_{i+1}\, \sigma_{i+1}$, the satisfaction clause gives $\mathcal{M}, s_{\bar{a}}[x_{i+1} \mapsto b] \models \sigma_{i+1}$ for every b, which is $\mathrm{Inv}(i+1)$ for $(\bar{a}, b)$. Existential step: by $\mathrm{Inv}(i)$ each witness set $W(\bar{a}) = \{b : \mathcal{M}, s_{\bar{a}}[x_{i+1} \mapsto b] \models \sigma_{i+1}\}$ is nonempty; apply the Axiom of Choice (definition at 651) to $\{\{\bar{a}\} \times W(\bar{a}) : \bar{a} \in |\mathcal{M}|^{\,q(i)}\}$ to obtain F_{i+1} with $F_{i+1}(\bar{a}) \in W(\bar{a})$ — one application of choice per existential quantifier, and nowhere else. Define the expansion $\mathcal{M}^*$ by $f_j^{\mathcal{M}^*} = F_j$. From $\mathrm{Inv}(n)$: for every $\bar{a} \in |\mathcal{M}|^p$, the term-value clauses give t_j the value v_j under $[\bar{u} \mapsto \bar{a}]$ in $\mathcal{M}^*$, so iterating `thm:fol-substitution` once per existential position turns $\mathcal{M}, s_{\bar{a}} \models \theta$ (equal to $\mathcal{M}^*, s_{\bar{a}} \models \theta$ by `lem:reduct-invariance`) into $\mathcal{M}^*, [\bar{u} \mapsto \bar{a}] \models \theta^*$; the p-fold universal clause yields $\mathcal{M}^* \models \mathrm{Sk}(\sigma)$.

Proof of (2). Let $\mathcal{N} \models \mathrm{Sk}(\sigma)$ in the expanded signature; now v_j is the value of the term t_j in $\mathcal{N}$. By downward induction from $i = n$ to $i = 0$ prove $\mathrm{Inv}'(i)$: $\mathcal{N}, s_{\bar{a}} \models \sigma_i$ for every $\bar{a} \in |\mathcal{N}|^{\,q(i)}$. Base $i = n$: instantiate the universal prefix of $\mathrm{Sk}(\sigma)$ at $\bar{a}$ and reverse the same iterated substitution-lemma computation as in (1). Universal step ($Q_{i+1} = \forall$): for every b, $\mathrm{Inv}'(i+1)$ at $(\bar{a}, b)$ gives $\mathcal{N}, s_{\bar{a}}[x_{i+1} \mapsto b] \models \sigma_{i+1}$ (the v_j, $j \le i$, depend only on $\bar{a}$); apply the $\forall$-clause. Existential step: $\mathrm{Inv}'(i+1)$ at $\bar{a}$ exhibits the witness v_{i+1}, and the $\exists$-clause gives $\mathrm{Inv}'(i)$. Then $\mathrm{Inv}'(0)$ and `lem:reduct-invariance` yield $\mathcal{N} \restriction \mathcal{L} \models \sigma$. No choice principle is used. The remark at 4509–4512 stays. The theorem carries *Status: complete proof* afterwards.

Acceptance. (1) The string "quantifier game" occurs nowhere in the manuscript. (2) Each direction displays an induction statement indexed by prefix position with the Skolem-term assignment defined, and every step is discharged by a named satisfaction clause, `thm:fol-substitution`, `lem:fol-coincidence`, or `lem:reduct-invariance`. (3) The Axiom of Choice appears only in clause (1), invoked exactly once per existential quantifier on a displayed family of witness sets; clause (2)'s proof invokes no choice principle.

R-009 — Transfinite Lindenbaum lemma for the Henkin construction (v1.0 lines 3934–3985)

Defect. The second paragraph of the Henkin–Lindenbaum proof (3978–3985) says "Well-order all formulas of $\mathcal{L}^H$ and repeat the Lindenbaum construction" and claims maximal consistency "by the same finite-derivation argument as in `lem:prop-lindenbaum`." That lemma (2381–2405) is an ω-indexed recursion over a countable enumeration; for κ possibly uncountable the construction needs transfinite recursion with limit stages, which the propositional argument does not contain, and the family whose union is taken is never defined. The witness stage two paragraphs

earlier (3961–3976) does all of this explicitly via `thm:transfinite-recursion`; the two halves of one proof sit at different rigor levels.

Repair. Insert between 3932 and 3934:

Lemma (`lem: transfinite-lindenbaum`). Let $\mathcal{L}'$ be a signature, λ an ordinal, and $\langle \varphi_\alpha : \alpha < \lambda \rangle$ an enumeration of $\mathrm{Fm}(\mathcal{L}')$ in which every formula occurs at least once. Then every set $\Sigma \subseteq \mathrm{Fm}(\mathcal{L}')$ consistent for $\mathsf{ND}_{\mathrm{FOL}}$ is contained in a maximal consistent $\Delta \subseteq \mathrm{Fm}(\mathcal{L}')$. (Choice-free given the enumeration; the Axiom of Choice enters only when producing one.)

Proof route. By `thm:transfinite-recursion` (632) with domain $\lambda + 1$ define $\langle \Gamma_\alpha : \alpha \leq \lambda \rangle$:

$$\Gamma_0 = \Sigma, \qquad \Gamma_{\alpha+1} = \begin{cases} \Gamma_\alpha \cup \{\varphi_\alpha\}, & \text{if this set is consistent,} \\ \Gamma_\alpha, & \text{otherwise,} \end{cases} \qquad \Gamma_\beta = \bigcup_{\alpha<\beta} \Gamma_\alpha \ \text{ for limit } \beta.$$

Chain claim: $\alpha \leq \beta \leq \lambda$ implies $\Gamma_\alpha \subseteq \Gamma_\beta$ (transfinite induction on β). *Consistency:* transfinite induction on α; zero by hypothesis; successor by the case split; at a limit β, a derivation of $\bot$ from Γ_β is finite, so its open assumptions form a finite $F \subseteq \Gamma_\beta$; each $\psi \in F$ lies in some Γ_{α_ψ} with $\alpha_\psi < \beta$, and with $\alpha^* = \max_{\psi \in F} \alpha_\psi < \beta$ if $F \neq \varnothing$ (a finite nonempty set of ordinals has a maximum) and $\alpha^* = 0 < \beta$ if $F = \varnothing$, the chain claim locates $F \subseteq \Gamma_{\alpha^*}$, contradicting the induction hypothesis at α^*. *Maximality of* $\Delta = \Gamma_\lambda$: if $\varphi \notin \Delta$, pick α with $\varphi_\alpha = \varphi$; rejection at stage $\alpha+1$ means $\Gamma_\alpha \cup \{\varphi\}$ derives $\bot$, and since a derivation from a set is a derivation from any superset (monotonicity, immediate from the definition of derivability from assumptions), $\Delta \cup \{\varphi\}$ derives $\bot$.

Rewrite of 3978–3985: "The set $\mathrm{Fm}(\mathcal{L}^H)$ has cardinality exactly κ, as established above from `thm:infinite-cardinal-finite-coding` and the formulas $\exists x_0\,(x_0 = d_\xi)$ (the count at 3949–3952 is retained); by the cardinal-assignment corollary (662, under the Axiom of Choice already assumed) fix a bijective enumeration $\langle \varphi_\alpha : \alpha < \kappa \rangle$ of it. Apply `lem:transfinite-lindenbaum` to Σ^H to obtain a maximal consistent $\Delta \supseteq \Sigma^H \supseteq \Sigma$." The existing witness-property sentences (3981–3984) follow unchanged. Status labels afterwards: the new lemma and `lem:henkin-lindenbaum` both *Status: complete proof.*

Acceptance. (1) A lemma between `lem:new-constant-elimination` and `lem:henkin-lindenbaum` displays the recursion with zero, successor, and limit clauses and cites `thm:transfinite-recursion`. (2) Its limit case contains the chain claim and locates an arbitrary finite subset of a limit stage inside a single earlier stage via the maximum of finitely many ordinals. (3) Lines 3978–3981 no longer contain "repeat the Lindenbaum construction" or the appeal to `lem:prop-lindenbaum`; the paragraph applies the new lemma to Σ^H under an explicitly fixed enumeration of order type κ.

R-010 — Vacuous substitution is alpha-invariant: strong induction on complexity (v1.0 lines 3006–3016)

Defect. The unnamed lemma at 3006 ("If $x \notin \mathrm{FV}(\varphi)$, then $\varphi[t/x] \equiv_\alpha \varphi$") is proved "by structural induction," but in the renaming case the recursion visits $\psi[z/y]$, which is not a subformula of $Qy\,\psi$, so plain structural induction does not license the step; the case is discharged by the bare assertion that renaming "changes the formula only within its alpha-equivalence class." The book's own template for this situation is the simultaneous strong induction on complexity at 2951–2958 and its use at 3193–3199 of `lem:substitution-complexity`.

Repair. First insert, after the example ending at 3004 and before the lemma at 3006:

Lemma (`lem: substitution-fv`, free variables under substitution). For every term r: $x \notin \mathrm{Var}(r)$ implies $r[t/x] = r$, and $\mathrm{Var}(r[t/x]) \subseteq (\mathrm{Var}(r) \setminus \{x\}) \cup \mathrm{Var}(t)$ (induction on r). For every formula φ, term t, and variables x, y:

(a) $\mathrm{FV}(\varphi[t/x]) \subseteq (\mathrm{FV}(\varphi) \setminus \{x\}) \cup \mathrm{Var}(t)$ (the book defines no FV on terms; a term's variables are its Var, as in the FV clauses for atomic formulas at 2862–2863);

(b) if $y \notin \mathrm{Var}(\varphi)$, then $\mathrm{FV}(\varphi[y/x]) = (\mathrm{FV}(\varphi) \setminus \{x\}) \cup \{y\}$ when $x \in \mathrm{FV}(\varphi)$, and $\mathrm{FV}(\varphi[y/x]) = \mathrm{FV}(\varphi)$ when $x \notin \mathrm{FV}(\varphi)$.

Route: strong induction on $c(\varphi)$. For (b) the renaming clause of the substitution definition never fires (a binder w of a subformula lies in $\mathrm{Var}(\varphi)$, so $w \neq y$), and the substitution is textual replacement. For (a)'s renaming case $(Qy\,\psi)[t/x] = Qz\,((\psi[z/y])[t/x])$: apply the induction hypothesis to ψ with $[z/y]$ and to $\psi[z/y]$ with $[t/x]$ — the latter at equal complexity $c(\psi[z/y]) = c(\psi) < c(Qy\,\psi)$, licensed by `lem:substitution-complexity` (2965) — then compute, using $z \notin \mathrm{Var}(t) \cup \{x\}$, that $\mathrm{FV}(Qz\,((\psi[z/y])[t/x])) \subseteq (\mathrm{FV}(Qy\,\psi) \setminus \{x\}) \cup \mathrm{Var}(t)$.

Then replace the proof at 3010–3016 by: *Proceed by strong induction on formula complexity c (2875).* Atomic case: x occurs in no term, so every term is unchanged by the term clause of `lem:substitution-fv` and $\varphi[t/x] = \varphi$; $\bot$ likewise. Connective cases: $x \notin \mathrm{FV}$ of each immediate subformula; induction hypotheses and the congruence property of $\equiv_\alpha$. Quantifier $Qy\,\psi$ with $y = x$: substitution stops, reflexivity. With $y \neq x$, $y \notin \mathrm{Var}(t)$: then $x \notin \mathrm{FV}(\psi)$; the induction hypothesis gives $\psi[t/x] \equiv_\alpha \psi$, and the Q-congruence concludes. With $y \neq x$, $y \in \mathrm{Var}(t)$: the definition returns $Qz\,((\psi[z/y])[t/x])$ with z least outside $\mathrm{Var}(\psi) \cup \mathrm{Var}(t) \cup \{x, y\}$. By clause (b) of `lem:substitution-fv`, $x \notin \mathrm{FV}(\psi[z/y])$ (as $x \notin \mathrm{FV}(\psi)$ and $x \neq z$). By `lem:substitution-complexity`, $c(\psi[z/y]) = c(\psi) < c(Qy\,\psi)$, so the *strong* induction hypothesis applies to $\psi[z/y]$ and gives $(\psi[z/y])[t/x] \equiv_\alpha \psi[z/y]$; the Q-congruence gives $Qz\,((\psi[z/y])[t/x]) \equiv_\alpha Qz\,(\psi[z/y])$; the generating clause (licensed since $z \notin \mathrm{Var}(\psi)$) gives $Qz\,(\psi[z/y]) \equiv_\alpha Qy\,\psi$; transitivity concludes. Landing: new lemma at approximately 3005; rewritten proof in place at 3010–3016. Both results carry *Status: complete proof* afterwards. (Clause (b) is also consumed by R-007's Lemma B.)

Acceptance. (1) The proof at 3010–3016 opens "by strong induction on formula complexity" and cites `lem:substitution-complexity` exactly where the induction hypothesis is applied to $\psi[z/y]$. (2) A stated and proved `lem:substitution-fv` precedes the lemma and is cited for $x \notin \mathrm{FV}(\psi[z/y])$. (3) No case applies an induction hypothesis to a formula that is neither of strictly smaller complexity nor of equal complexity justified by `lem:substitution-complexity`.

R-014 — LK completeness for arbitrary sequents (v1.0 lines 4165–4185, 2488–2507; soundness 3674–3687, 2295–2318)

Defect. Soundness (3674) covers every derivable sequent: the calculus is multi-succedent (rules at 2178–2234 and 3583–3618, satisfaction convention at 3579–3581), and the theorem quantifies over sequents $\Gamma \Rightarrow \Delta$. Completeness (4165) is stated only "for every finite multiset Γ and formula φ" — single-succedent. No completeness statement covers sequents with empty or multi-formula succedents, so the two theorems do not compose to a derivability-iff-validity equivalence for the objects the calculus manipulates. The propositional pair exhibits the identical asymmetry — `thm:prop-lk-soundness` (2295–2318) covers every derivable $\mathsf{LK}_{\mathrm{CPL}}$ sequent while completeness (2488–2491) is again single-succedent — and register row R-014 spans both pairs (2488 vs 2295; 4165 vs 3674), so this design must repair both.

Repair. Attach `thm:fol-lk-soundness` to the theorem at 3674. Replace the statement at 4165–4170 by:

Theorem (`thm: fol-lk-completeness`). Assume the Axiom of Choice in the ambient set theory. For all finite multisets Γ, Δ of $\mathcal{L}$-formulas, the sequent $\Gamma \Rightarrow \Delta$ is derivable in $\mathsf{LK}_{\mathrm{FOL}}$ iff for every $\mathcal{L}$-structure $\mathcal{M}$ and assignment s: if $\mathcal{M}, s \models \gamma$ for every $\gamma \in \Gamma$, then $\mathcal{M}, s \models \delta$ for some $\delta \in \Delta$.

(The roadmap's propositional phrasing "every valuation" is this schema instantiated for $\mathsf{LK}_{\mathrm{CPL}}$; at first order the semantic data are structure plus assignment, matching 3579–3581.) Left to right is `thm:fol-lk-soundness`. Right to left, by reduction to `thm:fol-completeness` (4128) plus the existing translation: (1) let $\Delta = \delta_1, \ldots, \delta_m$ ($m \geq 0$) and let S be the set of members of Γ together with $\neg\delta_1, \ldots, \neg\delta_m$; validity of the sequent makes S unsatisfiable, i.e. $S \models \bot$, since a structure and assignment satisfying all of S would satisfy some δ_j and its negation. (2) `thm:fol-completeness` gives $S \vdash_{\mathsf{ND}_{\mathrm{FOL}}} \bot$; the derivation's open assumptions form a finite $S_0 \subseteq S$, say $\gamma_1, \ldots, \gamma_k \in \Gamma$ and $\neg\delta_{j_1}, \ldots, \neg\delta_{j_r}$; the translation of the existing sketch (4171–4185) turns it into an $\mathsf{LK}_{\mathrm{FOL}}$-derivation of $\gamma_1, \ldots, \gamma_k, \neg\delta_{j_1}, \ldots, \neg\delta_{j_r} \Rightarrow \bot$. (3) Cut with the initial $\bot$-sequent $\bot \Rightarrow$ (instance of $\Gamma, \bot \Rightarrow \Delta$ with empty contexts, 2180) empties the succedent. (4) For each j, the sequent $\Rightarrow \delta_j, \neg\delta_j$ is derivable (identity sequent $\delta_j \Rightarrow \delta_j$, then $\neg R$); cutting it against the current sequent, with cut formula $\neg\delta_j$, moves δ_j into the succedent:

$$\frac{\Rightarrow \delta_j, \neg\delta_j \qquad \Gamma', \neg\delta_j \Rightarrow \Pi}{\Gamma' \Rightarrow \delta_j, \Pi} \qquad \text{yielding, after } r \text{ cuts,} \qquad \gamma_1, \ldots, \gamma_k \Rightarrow \delta_{j_1}, \ldots, \delta_{j_r}.$$

(5) Weakening (W_L, W_R) adds the unused members of Γ, Δ and contraction (C_L, C_R) merges duplicates, giving exactly $\Gamma \Rightarrow \Delta$ as multisets. The case $m = 0$ runs with no step-(4) cuts and asserts that no structure and assignment satisfies all of Γ; empty Γ is likewise covered. The route uses cut, which is primitive in the displayed calculus; no cut-free claim is made, and extending the sketch's translation induction to multi-succedent targets is unnecessary since the reduction reuses it verbatim. Landing: statement replaced at 4165–4170; steps (1)–(5) appended to the proof at 4171–4185, whose translation subroutine is retained. Status afterwards: *Status: proof sketch* while the rule-by-rule translation trees remain omitted; the theorem inherits *Status: complete proof* only if a separate work order supplies them.

Propositional half. Attach `thm:prop-lk-completeness` to the theorem at 2488 and replace its statement (2488–2491) by: for all finite multisets Γ, Δ of propositional formulas, the sequent $\Gamma \Rightarrow \Delta$ is derivable in $\mathsf{LK}_{\mathrm{CPL}}$ iff every valuation making every formula of $\mathrm{supp}(\Gamma)$ true makes some formula of $\mathrm{supp}(\Delta)$ true — word-for-word the sequent-satisfaction convention of 2156–2161. Left to right is `thm:prop-lk-soundness` (2295). Right to left is the reduction (1)–(5) verbatim with `thm:prop-completeness` (2433) in place of `thm:fol-completeness`, the translation sketch at 2493–2507 as the retained subroutine, and the same initial sequents and structural rules (2178–2234); no choice principle enters (the propositional Lindenbaum lemma runs on a countable enumeration). Landing: statement replaced at 2488–2491; steps (1)–(5) appended to the proof at 2493–2507. Status afterwards: likewise *Status: proof sketch* while its translation trees remain omitted.

Acceptance. (1) The completeness theorem at 4165 quantifies over sequents $\Gamma \Rightarrow \Delta$ with both contexts arbitrary finite multisets, and its semantic condition is word-for-word the satisfaction convention of 3579–3581, matching the scope of `thm:fol-lk-soundness`. (2) The proof displays the reduction chain of steps (1)–(5), including the $\Rightarrow \delta_j, \neg\delta_j$ construction and the empty-succedent case. (3) The status label matches the evidence actually present: sketch while the translation trees are omitted, complete only if they are supplied. (4) The propositional theorem at 2488 likewise quantifies over sequents with both contexts arbitrary finite multisets, matches the scope of `thm:prop-lk-soundness`, is proved by the same reduction without any choice principle, and carries the same status discipline; after both restatements, soundness and completeness are paired over the same sequent class in each calculus, discharging both line pairs of the register row.

R-011 — CNF half of the normal-form theorem (v1.0 lines 1919–1966; defective half 1958–1965)

Defect. The CNF half of the theorem at 1919–1923 reads "Apply the disjunctive construction to $\neg\varphi$, negate the result, and use De Morgan equivalences" (1958–1959). Against the string-valued CNF definition at 1878–1897 (a CNF formula is $\top$, or the right-associated conjunction N_1 of clause interpretations $\mathrm{form}(C_i)$, each clause listed in strictly increasing code order), this fails: for valid φ the DNF of $\neg\varphi$ is $\bot$, and $\neg\bot$ is not a CNF string; for variable-free false φ it yields $\neg\top$ instead of $\bot = \mathrm{form}(\varnothing)$; and "use De Morgan" silently performs double-negation elimination and re-association that the definition does not supply.

Repair. Replace 1958–1965 by a dual direct construction mirroring 1924–1956, plus one auxiliary lemma inserted after 1897. *Auxiliary lemma* (complete proof; two inductions on the definitional recursions at 1855–1870 and 1883–1890): (i) a valuation w satisfies $\mathrm{form}(C)$ iff w satisfies some literal in C (induction on $|C|$ along $D_k = L_k$, $D_i = (L_i \vee D_{i+1})$; the case $C = \varnothing$ is the falsity of $\bot$); (ii) w satisfies N_1 iff w satisfies $\mathrm{form}(C_j)$ for every $j \le m$ (induction on m along the N_i recursion). *Construction.* Let $p_1, \ldots, p_n$ be the variables of φ in increasing index order; reuse the DNF half's valuations on this finite set and the coincidence lemma (1925–1930). For a valuation v on $\{p_1, \ldots, p_n\}$, let $\bar{\ell}_i^v$ be $\neg p_i$ if $v(p_i) = \mathsf{T}$ and p_i if $v(p_i) = \mathsf{F}$, and put $K_v := \{\bar{\ell}_1^v, \ldots, \bar{\ell}_n^v\}$. The coding at 1849–1854 gives the variable of index i the code $2i$ and its negation the code $2i+1$ — both strictly increasing in i — and the enumeration $p_1, \ldots, p_n$ is in increasing index order (the indices need not form an initial segment), so any literal on p_j has smaller code than any literal on p_k for $j < k$: if their indices are $i_j < i_k$, then $2i_j + 1 < 2i_k$. The definitional listing of K_v in increasing code order is therefore exactly $\bar{\ell}_1^v, \ldots, \bar{\ell}_n^v$: $\gamma_v := \mathrm{form}(K_v)$ needs no re-ordering. By lemma (i), w falsifies γ_v iff $w(p_i) = v(p_i)$ for all $i \le n$. Let W_φ be the set of valuations on $\{p_1, \ldots, p_n\}$ falsifying φ, enumerated lexicographically ($\mathsf{F} < \mathsf{T}$, as at 1945–1948) as $v_1, \ldots, v_m$. Define $C_\varphi := \top$ if $m = 0$, and otherwise $C_\varphi := N_1$ built from the clauses $C_j := K_{v_j}$ by the recursion at 1883–1890. Then C_φ is literally a CNF string of 1878–1897, including the degenerate cases: valid φ gives $m = 0$, the canonical $\top$; variable-free true φ gives $n = m = 0$, again $\top$; variable-free false φ gives $n = 0$, $m = 1$, $K_{v_1} = \varnothing$, so $C_\varphi = \mathrm{form}(\varnothing) = \bot$, the CNF whose single clause is empty. *Equivalence:* for a total valuation w with restriction w_0 to $\{p_1, \ldots, p_n\}$ and $m > 0$: by lemma (ii) and (i), w satisfies C_φ iff for every $j \le m$ it satisfies some literal of K_{v_j}, iff $w_0 \ne v_j$ for all j, iff $w_0 \notin W_\varphi$, iff w_0 satisfies φ, iff (coincidence) w satisfies

φ. If $m = 0$, every valuation on the variable set satisfies φ, so by coincidence every w satisfies both φ and $\top$. The theorem keeps its plain proof environment; the DNF half is unchanged except that it may cite lemma (i).

Acceptance. (1) For the three degenerate inputs (valid; variable-free true; variable-free false) the constructed strings are $\top$, $\top$, form($\varnothing$), each satisfying the definition at 1878–1897 verbatim. (2) The CNF half invokes only the definitional recursions, the auxiliary lemma, and the coincidence lemma — no De Morgan, double negation, idempotence, or associativity. (3) The auxiliary lemma carries a complete proof by induction on both recursions.

R-012 — Mix rule, measure, and honest corollary statuses (v1.0 lines 9505–9548, 9555–9565, 9575–9586)

Defect. Step 3 of the cut-elimination sketch (9528–9532) appeals to "the multicut (mix) rule" and to elimination "by the same lexicographic measure," but no mix rule is displayed anywhere, the measure at 9509–9511 is defined only for cut, and the equipollence of the cut and mix calculi is unstated. The corollaries at 9555 and 9575 carry unlabeled proof environments, implying complete proofs, yet both consume the sketched theorem, contravening the status policy at 313–319.

Repair. (1) *Display mix* after 9511, in the notation of 9200–9224, where A^k abbreviates k comma-separated occurrences of A. For LK, with $m, n \geq 1$ and Δ, Σ permitted to contain further occurrences of A:

$$\frac{\Gamma \Rightarrow \Delta, A^m \qquad A^n, \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma}\ \textsc{Mix}.$$

For LJ (single succedent, so the left premise's succedent multiplicity m is fixed at one): from $\Gamma \Rightarrow A$ and $A^n, \Pi \Rightarrow C$ with $n \geq 1$, conclude $\Gamma, \Pi \Rightarrow C$. (2) *Equipollence lemma* (new; status: complete proof). Cut is the instance $m = n = 1$. Conversely, mix is derived from the displayed primitives: $m - 1$ applications of CR to the left premise, $n - 1$ of CL to the right, then one cut. Hence the cut and mix formulations derive the same sequents (LJ: CL only), so eliminating mix eliminates cut. (3) *Restate the measure* (replace 9505–9511): a mix instance with mix formula A and immediate subderivations D_1, D_2 has rank $r(A)$ and height $h(D_1) + h(D_2)$; the reduction is controlled by the lexicographic order on $\langle r(A),\ h(D_1) + h(D_2)\rangle$, with cut the case $m = n = 1$. (4) *Preface the sketch* (after 9519) with: "By the equipollence lemma, first replace every cut in the given derivation by the mix instance $m = n = 1$; the reduction below eliminates mixes, and steps 1, 2, 5, and 6 apply verbatim with 'mix' for 'cut' and the mix formula for the cut formula." *Rewrite sketch step 3* (9528–9532): if a premise of a mix ends in a contraction of the mix formula (CL above the right premise, CR above the left), replace that premise by the contraction's premise and raise the corresponding multiplicity by one: rank unchanged, height decreased by one. Add to step 4: when the mix formula is principal on both sides with multiplicities (m, n), the principal reduction yields mixes on proper subformulas (smaller rank) plus residual mixes on remaining occurrences at equal rank but smaller height; the full case table remains the omitted verification (citation at 9548 unchanged). (5) *Honest statuses.* Append *Status: proof sketch.* to the theorem statement (9513–9517). Split the corollary at 9555: a new lemma "every cut-free propositional LK- or LJ-derivation has the subformula property" takes the existing induction (9560–9565) as a complete proof; the corollary (existence of a cut-free derivation with the property) is restated with *Status: proof sketch.*, its proof converted to the sketch environment with the dependency sentence modeled on 7728–7729: "This derivation uses the preceding proof sketch of the cut-elimination theorem; a complete proof supplies its reduction case table." The corollary at 9575 keeps its argument verbatim, converted to the sketch environment with the same dependency sentence and *Status: proof sketch.* on its statement. When cut elimination is proved in full, exactly these two status lines and two sentences flip to complete proof.

Acceptance. (1) Mix is displayed for both calculi with explicit multiplicities $m, n \geq 1$, and the equipollence lemma is proved using only CL, CR, and Cut as displayed at 9204–9224 and 9307–9375. (2) The sketch opens by converting every cut to a mix via the equipollence lemma, so no step manipulates an object the measure does not cover; step 3 names only displayed rules and exhibits the measure decrease (rank equal, height -1). (3) The theorem and both corollaries carry *Status: proof sketch.*; the cut-free-subformula lemma and the equipollence lemma carry *Status: complete proof.*; no complete-proof item in the chapter cites the cut-elimination theorem.

R-013 — Repetition and vacuous discharge in $\mathsf{ND}_{\mathsf{CPL}}$ (v1.0 lines 2006–2021, 2351–2379, 2443–2455; first-order model 3799–3812)

Defect. Line 2020–2021 licenses, in prose only, "Repetition of an already available formula is permitted," though the calculus is defined (2006–2018) as a tree generated by the displayed rules and no repetition rule is displayed; and the definition never says whether a discharging rule may discharge zero occurrences. Clause 2 of the closure lemma (2360–2362) and the completeness theorem (2446–2449) apply $\neg I/\mathsf{RAA}$ to derivations that may contain no open occurrence of the discharged assumption, and clause 5 (2374–2375) says "assume φ, repeat ψ, and apply $\to I$."

Repair. Eliminate the uses, matching 3799–3812. (a) Delete the repetition sentence (2020–2021). Append to the derivation definition (after 2011): "An application of $\to I$, $\neg I$, or RAA carries a label i and discharges exactly the open occurrences marked i of the bracketed assumption above its premise; an application of $\vee E$ carries two labels i, j, as displayed, and discharges exactly the open occurrences of the first disjunct marked i above its second premise and the open occurrences of the second disjunct marked j above its third premise. Every label carried by such an application discharges at least one occurrence. Marking any set of open occurrences of one formula with a common label fresh from every label in the derivation preserves the derivation and the occurrences discharged by each of its rule applications, since no rule application carries the fresh label; this merging relabeling is distinct from the injective renamings above, and is the operation used when a newly added root inference is to discharge several open occurrences at once. There is no repetition step: where an informal presentation repeats a formula, the tree reuses the node already present." (b) Closure clause 2 (2358–2362), mirroring 3803–3807: if $\varphi \notin \Delta$, maximality yields a derivation D of $\bot$ with open assumptions in $\Delta \cup \{\varphi\}$; because Δ is consistent, at least one open occurrence of φ remains in D; mark all open occurrences of φ in D with a common fresh label i (the merging relabeling of (a)), and apply $\neg I$ with label i at the root, discharging every such occurrence, to obtain $\Delta \vdash \neg\varphi$; clause 1 gives $\neg\varphi \in \Delta$. (c) Closure clause 5 (2374–2375), replacing repetition by a conjunction detour: if $\psi \in \Delta$, assume φ with label i and take ψ as an open assumption; $\wedge I$ gives $\varphi \wedge \psi$; $\wedge E_2$ gives ψ; $\to I$ discharges the single open occurrence marked i, concluding $\varphi \to \psi$ with open assumptions in Δ. The discharge is of an occurrence genuinely present, and only displayed rules occur. (d) Completeness (2446–2449): suppose a derivation D of $\bot$ has open assumptions in $\Gamma \cup \{\neg\varphi\}$. If no open occurrence of $\neg\varphi$ appeared in D, then D would witness $\Gamma \vdash \bot$, and $\bot E$ would give $\Gamma \vdash \varphi$, contrary to the supposition $\Gamma \nvdash \varphi$. So at least one open occurrence remains; mark all open occurrences of $\neg\varphi$ with a common fresh label i (the merging relabeling of (a)) and apply RAA with label i at the root, discharging every such occurrence, to obtain $\Gamma \vdash \varphi$ — again contrary to the supposition. Hence $\Gamma \cup \{\neg\varphi\}$ is consistent. (e) Sweep every displayed $\vee E/\to I/\neg I/\mathsf{RAA}$ application in the ND chapters, examples, exercises, and solutions for compliance (the examples at 2077–2110 already comply; the intuitionistic calculus, defined at 2106–2109 by omitting RAA, inherits the convention; 3799–3812 stands unchanged and now matches, its "discharging every such occurrence" being an instance of the merging relabeling followed by a single labeled application).

Acceptance. (1) "Repetition" no longer occurs in the $\mathsf{ND}_{\mathsf{CPL}}$ section, and the derivation definition states explicitly both the at-least-one-occurrence discharge convention (with $\vee E$ carrying two labels) and the

fresh-label merging relabeling used by the closure and completeness proofs. (2) The proofs of closure clauses 2 and 5 and of completeness contain, respectively, the non-vacuity argument and the conjunction detour, and invoke no rule outside the display at 2022–2065. (3) An audit of all discharging-rule applications in displayed derivations finds none that discharges zero occurrences.

R-015 — Reduction of arbitrary embeddings to substructure inclusions (v1.0 lines 5989–5992, 6009–6012; supporting definitions 5042–5057, 5093–5103)

Defect. Model completeness is defined at 5989 as "every embedding between its models is elementary," but the DLO corollary's sketch verifies only inclusions: "If $\mathcal{M} \subseteq \mathcal{N}$ are models of DLO ... This is model completeness" (6009–6012). The passage from inclusions to arbitrary embeddings is missing, and the manuscript nowhere proves that satisfaction is isomorphism-invariant or that an embedding is an isomorphism onto its image.

Repair. Insert Lemmas 1–2 after 5125 (end of the Substructures section) and Lemma 3 after 5992; patch 6011–6012. *Lemma 1 (Isomorphism invariance; complete proof).* If $h : \mathcal{M} \to \mathcal{N}$ is an isomorphism, then for every formula $\varphi(x_1, \ldots, x_k)$ and all $a_1, \ldots, a_k \in M$: $\mathcal{M} \models \varphi(\bar{a}) \iff \mathcal{N} \models \varphi(h\bar{a})$; in particular isomorphic structures satisfy the same sentences. *Route:* induction on φ; atomic case by the term-preservation lemma (5060–5076), injectivity for equality atoms, and preservation with reflection (5042–5052) for relation atoms; Boolean cases by induction hypothesis; for $\exists y$, a witness $a' \in M$ maps to $h(a')$, and conversely a witness $b \in N$ is $h(a')$ by surjectivity; $\forall$ dually. *Lemma 2 (Image of an embedding; complete proof).* If $h : \mathcal{M} \to \mathcal{N}$ is an embedding, then $h(M)$ contains $c^{\mathcal{N}}$ for every constant and is closed under every $f^{\mathcal{N}}$; the structure $h[\mathcal{M}]$ with domain $h(M)$, constants $c^{\mathcal{N}}$, the restrictions of the $f^{\mathcal{N}}$, and relations $R^{\mathcal{N}}$ restricted to $h(M)$ is a substructure of $\mathcal{N}$ per 5093–5103; and h with codomain $h(M)$ is an isomorphism $\mathcal{M} \to h[\mathcal{M}]$. *Route:* $c^{\mathcal{N}} = h(c^{\mathcal{M}}) \in h(M)$ and $f^{\mathcal{N}}(h\bar{a}) = h(f^{\mathcal{M}}(\bar{a})) \in h(M)$ by the homomorphism clauses; the inclusion preserves and reflects atomic formulas because interpretations are restrictions, so $h[\mathcal{M}] \subseteq \mathcal{N}$; the corestricted h is injective (given), surjective by construction, preserves and reflects relations since $R^{h[\mathcal{M}]} = R^{\mathcal{N}} \cap h(M)^k$ while h reflects $R^{\mathcal{N}}$; hence a surjective embedding, an isomorphism per 5055–5057. *Lemma 3 (Reduction to inclusions; complete proof).* If $\mathcal{M} \preccurlyeq \mathcal{N}$ whenever $\mathcal{M} \subseteq \mathcal{N}$ are both models of T, then T is model complete. *Route:* an embedding $h : \mathcal{M} \to \mathcal{N}$ between models of T factors (Lemma 2) as the isomorphism $h_0 : \mathcal{M} \to h[\mathcal{M}]$ followed by the inclusion $h[\mathcal{M}] \subseteq \mathcal{N}$; Lemma 1 gives $h[\mathcal{M}] \models T$; the hypothesis gives $h[\mathcal{M}] \preccurlyeq \mathcal{N}$; then for all $\varphi, \bar{a}$: $\mathcal{M} \models \varphi(\bar{a}) \iff h[\mathcal{M}] \models \varphi(h\bar{a})$ (Lemma 1) $\iff \mathcal{N} \models \varphi(h\bar{a})$ (elementary inclusion). Hence h is elementary. *Patch 6011–6012:* replace "This is model completeness." by "By the reduction lemma, every embedding between models of DLO is elementary; this is model completeness." The corollary keeps *Status: proof sketch.* (its remaining omission is the literal-case verification named at 6013–6014, not the embedding reduction).

Acceptance. (1) The three lemmas appear with complete proofs using only 5022–5057, 5093–5103, and the term-preservation lemma; Lemma 1's quantifier case invokes surjectivity explicitly, and Lemma 2 verifies closure of $h(M)$ before forming $h[\mathcal{M}]$. (2) The DLO sketch derives model completeness by citing Lemma 3. (3) No passage concludes model completeness from inclusions alone.

R-016 — Extension clause in the back-and-forth test (v1.0 lines 6068–6093; corollary 6097–6110)

Defect. The back-and-forth theorem (6068–6081) concludes only "$\mathcal{M} \cong \mathcal{N}$," but the corollary's proof asserts "The resulting isomorphism extends the original map" (6108–6109) — a property of the proof's construction, which begins at an arbitrary $f_0 \in \mathcal{F}$, not of the stated theorem. The corollary's use is therefore unlicensed.

Repair. *Restated theorem (6068–6081).* Let $\mathcal{M}, \mathcal{N}$ be countable structures and $\mathcal{F}$ a nonempty set of finite partial isomorphisms from $\mathcal{M}$ to $\mathcal{N}$ such that for every $f \in \mathcal{F}$: (1) for every $a \in M$, some $g \in \mathcal{F}$ extends f with a in its domain; (2) for every $b \in N$, some $g \in \mathcal{F}$ extends f with b in its range. Then for every $f_0 \in \mathcal{F}$ there is an isomorphism $h : \mathcal{M} \to \mathcal{N}$ with $f_0 \subseteq h$ (extension as sets of ordered pairs); in particular $\mathcal{M} \cong \mathcal{N}$. *Proof edits (6083–6093):* replace "begin with any $f_0 \in \mathcal{F}$" by "let f_0 be the given member of $\mathcal{F}$"; the recursion and union argument are unchanged; append "Since $f_0 \subseteq f_1 \subseteq \cdots$ and $h = \bigcup_k f_k$, the isomorphism extends f_0." *Status: complete proof* retained. *Corollary patch (6104–6110).* Replace the final sentence by an explicit application: the set $\mathcal{F}$ of finite elementary maps from $\mathcal{M}$ to itself extending the given map f is nonempty ($f \in \mathcal{F}$); every member is a finite partial isomorphism, since elementary maps preserve and reflect atomic formulas and reflect equality, hence are injective; clause (1) holds within $\mathcal{F}$ because the saturation-homogeneity theorem (5567–5571) extends any $g \in \mathcal{F}$ by any prescribed domain element to an elementary map still extending f; clause (2) holds by applying the same theorem to g^{-1} (the inverse of a finite elementary self-map is elementary) and inverting the result, which extends $g \supseteq f$ and has the prescribed element in its range. The restated theorem applied with $f_0 := f$ yields an isomorphism $h : \mathcal{M} \to \mathcal{M}$ with $f \subseteq h$: an automorphism extending f. *Status: complete proof* retained. The forward reference at 5563–5564 remains accurate.

Acceptance. (1) The theorem statement contains the quantified extension clause "for every $f_0 \in \mathcal{F}$ there is an isomorphism $h \supseteq f_0$." (2) Its proof starts the recursion at the given f_0 and derives $f_0 \subseteq h$ from the chain union. (3) The corollary invokes the extension clause with $f_0 :=$ the given map, and both results retain *Status: complete proof.*

4 Depth-gap inventory

Gaps *within* the eleven existing parts, each assigned a target home in the new architecture (§6) and a **treatment level**, which is a ceiling, not a floor to exceed:

E	full reference entry (definitions, adequacy theorems with status, tables)
T	table row(s) only, in a master or per-chapter table
S	statement with exact hypotheses + status + citation; no development
P	pointer: one sentence and a source

ID	Item	Target	Tr	Prio	Notes
D-P1-001	Zorn's lemma: statement, proof from AC, equivalence noted	P6	E	must	absent book-wide; felt at 3978 where it is the standard tool for Henkin–Lindenbaum; cross-cutting list (Zorn/AC-equivalents)
D-P1-002	AC-equivalents theorem and master table (AC ⇔ well-ordering ⇔ Zorn ⇔ Tukey; BPI strictly weaker)	P6	T	must	cross-cutting list; AC-equivalents table required by target architecture sec. 6
D-P1-003	Cardinal trichotomy: statement and AC-dependence	P6	S	must	used at 672 but never stated
D-P1-004	König's lemma and fan theorem, with the choice-free route to propositional compactness	P6	E	must	absence blocks choice-free propositional compactness; ties to the broken forward promise at 546–548 (countable-language completeness)
D-P1-005	BPI/Ultrafilter Lemma ⇔ propositional compactness	P2.1	E	should	Ultrafilter Lemma present at 704–724 but never linked to compactness
D-P1-006	Countable choice: statement, status, and where the book uses it	P6	S	should	raised in an exercise, resolved nowhere
D-P2-001	Hilbert-style calculus for CPL with deduction theorem and ND/LK/Hilbert equivalence and comparison	P2.1	E	must	no Hilbert system book-wide; cross-cutting list

continued

ID	Item	Target	Tr	Prio	Notes
D-P2-002	Post's functional-completeness classification (five maximal clones + negative results + Sheffer stroke/NOR)	P2.1	E	must	one positive basis only in v1.0; cross-cutting list; Post's lattice figure in apparatus register
D-P2-003	Duality theorem for propositional logic	P2.1	E	should	cross-cutting list
D-P2-004	Decidability of CPL stated as a theorem	P2.1	S	must	never stated despite the truth-table method being developed
D-P2-005	SAT NP-completeness statement in the propositional chapter with forward reference to P4	P2.1	S	must	Cook cited only at 12118 with no forward ref from Part II
D-P2-006	Propositional interpolation theorem	P2.1	E	should	
D-P2-007	Tseitin CNF transformation: statement and cross-reference from the CNF section	P2.1	S	should	exists at 11945 but unreferenced from the CNF section
D-P2-008	Horn-SAT and 2-SAT tractable classes	P2.1	E	should	also feeds P9/P12 SAT material
D-P2-009	Lindenbaum–Tarski algebra and Boolean-valued semantics for CPL	P10	E	must	quotient machinery from Ch. 1 never exploited; also listed in B3 algebraic/categorical and A(d) – coordinate with new-content register
D-P2-010	Substitution and replacement congruence theorems promoted from exercise to stated results	P2.1	S	should	
D-P2-011	Propositional logic over uncountable languages (compactness/completeness beyond the fixed countable alphabet)	P2.1	S	could	alphabet fixed countable at 1565
D-P3-001	Hilbert-style FOL system with deduction theorem	P2.2	E	must	cross-cutting list
D-P3-002	Classical analytic tableaux for propositional and first-order logic with countermodel extraction and completeness	P2.2	E	must	absent book-wide (only the many-valued matrix tableau at 11255); also listed in B3 automation and cross-cutting list
D-P3-003	Herbrand's theorem proper: Herbrand universe/base/structure/expansion and the validity form	P2.2	E	must	only a cited clause corollary at 4799; cross-cutting list; link to resolution
D-P3-004	Craig interpolation and Beth definability: statements and forward references in the first-order chapter	P2.2	S	must	currently only cited in Part VIII (9627/9645) with no forward reference; also flagged in B2 model theory
D-P3-005	Robinson joint consistency theorem	P3	E	should	
D-P3-006	Decidable fragments of FOL: catalog table (monadic; FO^2; guarded; Bernays–Schönfinkel)	P2.2	T	must	monadic FOL one aside at 6875; master decidability table in P14
D-P3-007	Many-sorted first-order logic: syntax, semantics, reduction to one-sorted	P2.4	E	should	only a second-order translation target at 10204
D-P3-008	Empty domains and the inclusive/free-logic convention with pointer to the free-logic entry card	P8.4	S	should	nonempty domain imposed at 3028; full free-logic card is new-content N-b; also listed in B3 nonmonotonic/philosophical
D-P3-009	Infinitary logic $L_{\omega_1\omega}$ overview and pointer	P2.4	S	could	one mention at 9602
D-P3-010	Definitional extensions: conservativity theorem stated and proved	P2.2	E	must	unproved in example 3415–3418; Beth definability leans on it; coordinate with defect register
D-P3-011	Equality metatheory consolidated: free-standing congruence lemma and equality-axiom theorems	P2.2	E	must	symmetry an unlabeled example; transitivity an exercise
D-P3-012	Clausal-form theorem composing prenex + Skolem + CNF	P2.2	S	should	currently left to an exercise
D-P3-013	Compactness applications suite: nonstandard models, Skolem paradox discussion, upward LS via elementary diagrams	P2.2	E	should	one corollary only in v1.0
D-P3-014	First-order subformula definition and subformula property	P2.2	S	must	Sub is propositional-only; subformula property is a stated learning outcome
D-P4-001	Ehrenfeucht–Fraïssé games with inexpressibility applications	P3	E	must	absent book-wide; listed in both B (first-order) and B2 (model theory); also A(a)
D-P4-002	Preservation theorems (Łoś–Tarski; Chang–Łoś–Suszko)	P3	E	should	absent despite a full ultraproduct section; also A(a)
D-P4-003	Keisler–Shelah isomorphism theorem (statement)	P3	S	should	
D-P4-004	Compactness via ultraproducts drawn as a corollary of Łoś	P3	S	should	
D-P4-005	Prime and atomic models with the omitting-types connection	P3	E	must	undefined though Ryll-Nardzewski is stated
D-P4-006	Vaught two-cardinal theorem (statement)	P3	S	could	
D-P4-007	Indiscernibles and Ehrenfeucht–Mostowski models (statement level)	P3	S	should	

continued

ID	Item	Target	Tr	Prio	Notes
D-P4-008	Stability, forking, and simplicity survey	P3	S	should	research-level; survey ceiling per treatment policy; also A(a)
D-P4-009	NIP and o-minimality survey	P3	S	should	research-level survey; also A(a)
D-P4-010	Finite model theory: 0–1 laws and Fagin's theorem	P3	S	should	only Trakhtenbrot in v1.0; descriptive-complexity link to P9; also A(a)
D-P4-011	Computable model theory pointer	P3	P	could	also A(a)
D-P4-012	Lindström's theorem repositioned to model theory with abstract model theory context	P3	S	should	currently placed in Part IX, not IV
D-P5-001	μ-recursion and partial recursive functions with the TM $\equiv$ μ-recursive equivalence theorem	P4	E	must	partial computable is machine-only in v1.0 so the standard equivalence cannot be stated; also A(a)
D-P5-002	Church–Turing thesis discussion	P4	S	must	
D-P5-003	Turing degrees as a structure (poset, join, jump on degrees)	P4	E	should	only $\leq_T$ and the jump in v1.0; also A(a)
D-P5-004	Post's problem, finite-injury priority method, Friedberg–Muchnik (statement + method sketch)	P4	S	should	also A(a)
D-P5-005	C.e.-set structure theory: simple and creative sets, Myhill isomorphism, Rice–Shapiro	P4	S	should	also A(a)
D-P5-006	Kolmogorov complexity: invariance, incompressibility, halting-probability pointer	P4	S	could	also A(a)
D-P5-007	Complexity theory expansion: PSPACE, time/space hierarchy theorems, PH, complexity-class ladder table	P4	T	must	only a short section at 12105; descriptive complexity via P9; also A(a)
D-P6-001	Presburger arithmetic: decidability by quantifier elimination (statement + proof route)	P5	S	must	absent book-wide; flagged in both B (first-order) and B2 (arithmetic); also A(a)
D-P6-002	Tennenbaum's theorem (statement)	P5	S	should	also A(a)
D-P6-003	Nonstandard models of PA: order type $\mathbb{N} + \mathbb{Z} \cdot \mathbb{Q}$, overspill lemma	P5	E	should	
D-P6-004	Paris–Harrington and Goodstein independence results (statements)	P5	S	should	also A(a)
D-P6-005	Interpretability: relative interpretation with examples	P5	S	could	also A(a)
D-P6-006	Provable Σ_1-completeness stated and proved	P5	E	must	used implicitly in the Gödel II machinery, never stated
D-P6-007	Provability logic GL: axioms, Kripke semantics, fixed-point theorem; Solovay arithmetical completeness (S)	P5	E	must	natural companion to the proved Löb theorem; also flagged in B3 modal and A(b); entry card in P8.2
D-P6-008	Reverse mathematics: Z_2 subsystems and calibration table (RCA_0, WKL_0, ACA_0, ATR_0, Π^1_1-CA_0)	P5	T	should	also A(a)
D-P7-001	Ordinal arithmetic: sums, products, exponentiation, Cantor normal form, transfinite induction vs recursion	P6	E	must	ε_0/Cantor normal form used at 9854–9876 but never developed; listed in both B (foundations) and B2 (set theory); also A(a)
D-P7-002	Cardinal arithmetic beyond 2^κ: sums, products, exponentiation, $\aleph/\beth$ hierarchies, regular vs singular; CH/GCH statements and pointer	P6	E	must	listed in both B (foundations) and B2 (set theory)
D-P7-003	Cofinality and König's theorem	P6	E	must	
D-P7-004	Easton's theorem (statement)	P6	S	should	
D-P7-005	Martin's axiom: statement and sample consequences	P6	S	should	also A(a)
D-P7-006	Large-cardinal ladder: inaccessible through measurable and beyond, with consistency-strength chart	P6	T	must	0 occurrences of inaccessible in v1.0 – needed at the ctm hypothesis 8767; research-level so table + statements ceiling; also A(a)
D-P7-007	Lévy hierarchy, reflection principle, absoluteness (Shoenfield statement)	P6	E	must	needed for forcing and L discipline; also A(a)
D-P7-008	Condensation and the structure of L (condensation statement; fine-structure pointer)	P6	S	should	also A(a)
D-P7-009	Clubs, stationary sets, Fodor's lemma	P6	E	should	also A(a)
D-P7-010	Δ-system lemma and Ramsey's theorem	P6	E	must	Δ-system lemma needed for ccc of Cohen forcing (D-P7-012); also A(a)
D-P7-011	Boolean-valued models (statement level) and product/iterated/proper forcing pointers	P6	S	should	iterated/proper forcing NOT-IN-SCOPE beyond pointer per roadmap; also A(a)
D-P7-012	ccc preservation of cardinals and ccc of the Cohen poset, proved	P6	E	must	currently cited only; feeds the honest Con(ZFC+$\neg$CH) treatment

continued

ID	Item	Target	Tr	Prio	Notes
D-P7-013	Descriptive set theory: Borel and projective hierarchies, regularity properties (survey)	P6	S	should	research-level survey; also A(a)
D-P7-014	Determinacy: Borel determinacy statement, AD vs AC (survey)	P6	S	should	research-level survey; also A(a)
D-P8-001	Completeness of LK for arbitrary sequents in the proof-theory part	P7	E	must	Part VIII has only a soundness sketch
D-P8-002	LK $\leftrightarrow$ ND equivalence theorem	P7	E	must	
D-P8-003	Midsequent theorem and Herbrand's theorem via cut elimination	P7	S	should	also A(a)
D-P8-004	Craig interpolation proved by Maehara's method	P7	E	must	Maehara named but not carried out; cross-cutting list: Craig/Beth proved rather than cited
D-P8-005	Gentzen consistency-proof components: derivation coding, ordinal assignment, decrease lemma	P7	S	should	current treatment is an honest survey with no coding/assignment/decrease lemma
D-P8-006	Ordinal-analysis calibration table beyond PA with proof-theoretic ordinal defined	P7	T	should	v1.0 declines to define proof-theoretic ordinal; ordinal arithmetic must land before the ordinal-analysis table
D-P8-007	Proof complexity: Frege systems, resolution lower bounds (pigeonhole), extended resolution	P7	S	should	flagged in both B (propositional) and B2 (proof theory); also A(a)
D-P9-001	Second-order/HOL proof calculus displayed (comprehension and choice schemas as a calculus) with Henkin completeness route	P2.3	E	must	explicit decision in v1.0 to omit; also A(a)
D-P9-002	Analytical hierarchy (Σ^1_n): table and basic results	P4	T	should	also A(a)
D-P9-003	Untyped λ-calculus with Church–Rosser/confluence	P7	E	must	absent book-wide while β-reduction, subject reduction, and normalization are present; also A(a)
D-P9-004	System F: definition and strong normalization (statement)	P7	S	should	also A(a)
D-P10-001	Canonical models constructed with completeness proofs for K, T, S4, S5	P8.2	E	must	named in sketches, never constructed; cross-cutting list
D-P10-002	Filtration, finite model property, and decidability for the basic modal systems	P8.2	E	must	filtration only an omitted obligation; no FMP/decidability result for any modal logic in v1.0; cross-cutting list
D-P10-003	Frame-correspondence table with the standard translation; Sahlqvist theorem as statement	P8.2	E	must	no correspondence theory in v1.0; axiom $\leftrightarrow$ frame-condition $\leftrightarrow$ algebra master table in P14
D-P10-004	Bisimulation and the van Benthem characterization theorem	P8.2	E	should	
D-P10-005	Modal systems lattice completion: entry cards for D, B, K4, KD45, Grz	P8.2	E	must	GL handled at D-P6-007
D-P10-006	Tense logic and multimodal fusions	P8.3	S	should	
D-P10-007	Neighborhood semantics for non-normal modal logics	P8.2	S	could	
D-P10-008	Quantified modal logic: Barcan formulas, constant vs varying domains	P8.2	E	must	absent book-wide; also A(b) – coordinate with new-content register
D-P10-009	CTL, CTL*, and μ-calculus entry cards	P8.3	E	must	also A(c); complexity fields via D-P11-010
D-P10-010	Büchi automata and automata-theoretic LTL model checking	P9	S	must	also A(c) model checking – coordinate with new-content register
D-P10-011	Dynamic epistemic logic, public announcement, distributed knowledge	P8.3	S	should	
D-P10-012	Common-knowledge axiomatization: induction rule, completeness and non-compactness caveat	P8.3	E	must	C_G has language + semantics but no axioms at 11179–11191; an exercise uses it; defect-register overlap (C3 finding 3)
D-P10-013	Heyting algebras with algebraic completeness of IPL	P8.1	E	must	also A(d) – coordinate with new-content register
D-P10-014	Topological semantics for IPL	P8.1	S	should	
D-P10-015	BHK reading and Kleene realizability survey	P8.1	S	should	
D-P10-016	Existence property for intuitionistic logic	P8.1	S	must	cross-cutting list; disjunction property already proved in v1.0
D-P10-017	First-order intuitionistic logic: Kripke semantics with expanding domains, completeness statement, CD axiom	P8.1	E	must	chapter is propositional-only; cross-cutting list
D-P10-018	Intermediate logics KC and LC (Gödel–Dummett as intermediate logic)	P8.1	S	should	Gödel t-norm semantics under D-P10-021
D-P10-019	Markov's principle: statement and status	P8.1	S	could	

continued

ID	Item	Target	Tr	Prio	Notes
D-P10-020	Łukasiewicz logics, finite- and infinite-valued: matrices, axiomatization, completeness statements	P8.5	E	must	
D-P10-021	t-norm fuzzy logics: BL, Gödel, product	P8.5	S	should	
D-P10-022	MV-algebras and Chang completeness (statement)	P10	S	should	also A(d)
D-P10-023	Post logics (statement level)	P8.5	S	could	
D-P10-024	FDE/Belnap four-valued logic entry	P8.5	E	should	
D-P10-025	Weak Kleene and Bochvar logics	P8.5	S	could	
D-P10-026	Supervaluations	P8.5	S	should	also A(b) vagueness – coordinate with new-content register
D-P10-027	Registered facts: K3 has no tautologies; LP tautologies = classical; K3–LP duality	P8.5	S	must	never registered in v1.0; cheap entry-card fields
D-P10-028	Full classical linear logic: additives, multiplicatives, units, exponentials, linear negation, one-sided sequents, cut-elimination statement	P8.6	E	must	cross-cutting list
D-P10-029	Phase semantics: soundness and completeness statement	P8.6	S	should	
D-P10-030	Coherence spaces and proof nets pointers	P8.6	P	could	proof nets pointer-only per scope ruling
D-P10-031	Display, deep-inference, hypersequent, and labelled calculi survey pointer	P7	P	could	
D-P10-032	Relevance systems R and E as named axiom systems with the variable-sharing theorem	P8.6	S	should	omission was deliberate in v1.0 but leaves no variable-sharing theorem
D-P10-033	Lambek calculus entry	P8.6	S	should	only named in v1.0
D-P10-034	Residuated lattices as the unifying substructural semantics	P10	S	should	one definition only in v1.0; also A(d)
D-P10-035	Nonmonotonic logics: default logic, circumscription, autoepistemic logic	P8.7	S	should	ASP under D-P11-007; also A(b) – coordinate with new-content register
D-P10-036	AGM revision postulates table	P8.7	T	should	also A(b)
D-P10-037	Deontic logic: SDL and its paradoxes	P8.4	S	should	also A(b) – coordinate with new-content register
D-P10-038	Counterfactuals: Stalnaker–Lewis sphere semantics	P8.4	S	should	also A(b) – coordinate with new-content register
D-P10-039	Boolean algebras chapter: axioms, filters and ultrafilters, atomic/atomless, representation	P10	E	must	also A(d) – coordinate with new-content register
D-P10-040	Stone duality	P10	E	should	also A(d)
D-P10-041	Cylindric and relation algebras pointer	P10	P	could	pointer-level per A(d)
D-P10-042	CCC ↔ STLC categorical semantics	P10	S	should	also A(d) – coordinate with new-content register
D-P10-043	Hyperdoctrines and topos logic pointer	P10	P	could	NOT-IN-SCOPE beyond CCC/topos overview per roadmap
D-P11-001	CDCL: clause learning, watched literals, restarts, DRAT proofs	P12	E	must	chapter presents 1962 DPLL only; also A(c) – coordinate with new-content register
D-P11-002	Unification: Martelli–Montanari algorithm with the unification theorem	P12	E	must	mgu defined, occurs check in prose, no algorithm or theorem; also A(c)
D-P11-003	Superposition calculus: rules, term orderings (KBO/LPO), redundancy criterion, completeness statement	P12	S	must	named but undefined at the Part III resolution section (paramodulation/superposition); also A(c)
D-P11-004	Resolution refinements: ordered/hyper/semantic resolution, set-of-support, subsumption	P12	S	should	
D-P11-005	SMT: Nelson–Oppen combination and DPLL(T)	P12	S	should	
D-P11-006	Logic programming: SLD resolution with soundness/completeness, negation as failure, Prolog	P9	E	must	also A(c) – coordinate with new-content register
D-P11-007	Answer set programming and Datalog	P9	S	should	also A(c)
D-P11-008	Hoare logic rules table and weakest preconditions; separation logic pointer	P9	E	must	also A(c) – coordinate with new-content register
D-P11-009	BDDs, QBF, #SAT, MaxSAT	P12	S	should	BDD absence also flagged in B (propositional)
D-P11-010	Decidability and complexity results for every presented system: CPL (validity coNP-complete / SAT NP-complete), FOL (undecidable, c.e.), K/T/S4 and IPL PSPACE-complete, S5 NP-complete, LTL PSPACE-complete, QBF PSPACE-complete, plus decidable fragments	P14	T	must	cross-cutting list; B3 automation: complexity beyond two facts absent and the coNP macro defined but never used; B3 temporal: LTL PSPACE-completeness lands here; master table harvested from entry-card fields – depends on the P8 per-system entry cards
D-P11-011	Prover and proof-assistant landscape table (Agda, HOL Light, HOL4, PVS, Mizar, ACL2, Metamath)	P12	T	should	only Lean 4, Rocq, Isabelle named in v1.0

continued

ID	Item	Target	Tr	Prio	Notes
D-P11-012	MLTT/CIC foundations: universes, inductive types, W-types, definitional vs propositional equality, extraction	P12	S	should	dependent types one passing mention in Part IX (B2 higher-order); also A(c)
D-P11-013	HoTT and univalence pointer	P12	P	could	NOT-IN-SCOPE beyond pointer per roadmap
D-P11-014	LCF architecture and the de Bruijn criterion	P12	S	should	
D-P11-015	Landmark formalizations table (Four Color, Odd Order, Flyspeck, CompCert, seL4)	P12	T	could	cross-link to P13 chronology

5 New-domain specification

One register row per deliverable, keyed to the four domains of decision D1. The *sources* column seeds the bibliography intake (§7); rows with domain **x** are the **NOT-IN-SCOPE register** — exclusions recorded as decisions so that absence is never again ambiguous between "omitted deliberately" and "overlooked." The spec-first rule (§9) applies: writing beyond a row's treatment ceiling requires editing the row first.

ID	Target	Item	Tr	Prio	Sources
N-a-001	P3 3.5 countable-models	ω-categoricity: Ryll-Nardzewski/Engeler/Svenonius characterization via types, DLO and random-graph examples	E	must	Ryll-Nardzewski 1959; Engeler 1959; Svenonius 1959; Hodges 1993
N-a-002	P3 3.5 countable-models	Fraisse constructions: hereditary/joint-embedding/amalgamation properties, existence and uniqueness of the Fraisse limit, random graph and extension axioms	E	must	Fraisse 1954; Rado 1964; Hodges 1993
N-a-003	P3 3.5 countable-models	prime, atomic, countably saturated, and homogeneous models: existence and uniqueness theorems over countable complete theories	E	must	Vaught 1961; Chang-Keisler 1990; Marker 2002
N-a-004	P3 3.5 countable-models	Vaught's never-two theorem (T); Vaught conjecture status report (P)	T	should	Vaught 1961; Morley 1970
N-a-005	P3 3.5 countable-models	two-cardinal theorems: Vaught's two-cardinal theorem statement and method sketch	S	could	Vaught 1965; Chang-Keisler 1990
N-a-006	P3 3.6 stability-classification	Morley rank and totally transcendental theories; outline of the proof of Morley's categoricity theorem; Baldwin-Lachlan refinement statement	S	should	Morley 1965; Baldwin-Lachlan 1971; Marker 2002
N-a-007	P3 3.6 stability-classification	stability spectrum and forking: stable/superstable/ω-stable, nonforking extensions, symmetry and transitivity survey	S	should	Shelah 1978; Pillay 1996; Tent-Ziegler 2012
N-a-008	P3 3.6 stability-classification	indiscernible sequences and Ehrenfeucht-Mostowski models: existence via Ramsey's theorem, applications to non-isomorphic models	T	should	Ehrenfeucht-Mostowski 1956; Ramsey 1930; Hodges 1993
N-a-009	P3 3.7 tame-topology	o-minimality: definition, monotonicity and cell-decomposition statements, RCF as an o-minimal structure	S	should	Pillay-Steinhorn 1986; Knight-Pillay-Steinhorn 1986; van den Dries 1998
N-a-010	P3 3.7 tame-topology	NIP and simple theories: definitions, examples, map of the neostability landscape	S	could	Shelah 1971; Simon 2015; Tent-Ziegler 2012
N-a-011	P3 3.8 preservation-model-completeness	preservation theorems: Los-Tarski (universal), Lyndon (positivity), Chang-Los-Suszko (unions of chains)	T	should	Los 1955; Lyndon 1959; Chang-Keisler 1990; Hodges 1993
N-a-012	P3 3.8 preservation-model-completeness	model completeness and model companions: Robinson's test, ACF and RCF worked examples, existentially closed structures	E	must	Robinson 1956; Chang-Keisler 1990; Marker 2002
N-a-013	P3 3.4 interpolation-definability	Robinson joint consistency theorem and Beth definability with model-theoretic proofs; equivalence with Craig interpolation	T	should	Robinson 1956; Beth 1953; Hodges 1993
N-a-014	P3 3.2 types-saturation	ultrafilter existence from BPI, BPI versus AC status, Keisler-Shelah ultrapower isomorphism theorem statement	S	could	Keisler 1961; Shelah 1971; Bell-Slomson 1969
N-a-015	P7 7.4 lambda-calculi	untyped lambda calculus and combinatory logic as computation models: Church-Rosser theorem statement, lambda-definability equals partial recursiveness	T	must	Church 1936; Church-Rosser 1936; Kleene 1936; Curry-Feys 1958; Barendregt 1984

continued

ID	Target	Item	Tr	Prio	Sources
N-a-016	P4 4.4 degrees-and-priority	Turing degrees: partial order, join, jump operator, Kleene-Post construction of incomparable degrees below $\mathbf{0}'$	E	must	Kleene-Post 1954; Rogers 1967; Soare 2016
N-a-017	P4 4.4 degrees-and-priority	Post's problem and the Friedberg-Muchnik theorem: finite-injury priority construction in full	T	must	Post 1944; Friedberg 1957; Muchnik 1956; Soare 2016
N-a-018	P4 4.4 degrees-and-priority	simple and creative sets, many-one and one-one reducibility, Myhill isomorphism theorem	T	should	Post 1944; Myhill 1955; Rogers 1967
N-a-019	P4 4.5 frontiers-of-computability	hyperarithmetical and analytical hierarchies: $\Delta^1_1 = \mathrm{HYP}$ statement, ordinal notations $\mathcal{O}$ sketch	S	should	Kleene 1955; Rogers 1967; Sacks 1990
N-a-020	P4 4.5 frontiers-of-computability	Kolmogorov complexity: invariance theorem, incompressibility, Chaitin-style incompleteness	T	should	Kolmogorov 1965; Chaitin 1974; Li-Vitanyi 2008; Downey-Hirschfeldt 2010
N-a-021	P4 4.5 frontiers-of-computability	Hilbert's tenth problem: MRDP theorem statement, Diophantine sets equal c.e. sets, proof survey with Pell-equation core	S	must	Davis-Putnam-Robinson 1961; Matiyasevich 1970; Davis 1973
N-a-022	P4 4.5 frontiers-of-computability	undecidable word problems: Post-Markov theorem for semigroups (T), Novikov-Boone theorem for groups (S)	T	could	Post 1947; Markov 1947; Novikov 1955; Boone 1958
N-a-023	P4 4.5 frontiers-of-computability	busy beaver function: definition and Rado's non-computability theorem	T	could	Rado 1962
N-a-024	P4 4.5 frontiers-of-computability	computable structure theory pointer: computable presentations, indices of structures	P	could	Ash-Knight 2000
N-a-025	P5 5.6 provability-logic	GL: axioms, Kripke semantics for transitive conversely-well-founded frames, completeness and FMP, fixed-point theorem, Solovay arithmetical completeness (S)	E	must	Solovay 1976; Sambin 1976; Segerberg 1971; Boolos 1993
N-a-026	P5 5.6 provability-logic	Rosser provability and witness comparison; Guaspari-Solovay R pointer	P	could	Guaspari-Solovay 1979; Lindstrom 2003
N-a-027	P5 5.5 independence-and-strength	Paris-Harrington theorem: statement, unprovability in PA via indicator/model-theoretic argument (S), provability in ACA context	S	must	Paris-Harrington 1977; Kanamori-McAloon 1987; Kaye 1991
N-a-028	P5 5.5 independence-and-strength	Goodstein sequences and the Kirby-Paris hydra: termination via ε_0 (T), unprovability in PA statement	T	must	Goodstein 1944; Kirby-Paris 1982
N-a-029	P5 5.5 independence-and-strength	interpretability and relative interpretation: Tarski-Mostowski-Robinson method, essential undecidability transfer, interpretability logic pointer	S	should	Tarski-Mostowski-Robinson 1953; Visser 1997; Lindstrom 2003
N-a-030	P5 5.5 independence-and-strength	subsystems of second-order arithmetic and reverse mathematics: RCA_0, WKL_0, ACA_0, ATR_0, Π^1_1-CA_0; Big Five phenomenon with sample equivalences	S	must	Friedman 1975; Simpson 2009
N-a-031	P5 5.5 independence-and-strength	intensionality of consistency statements: Feferman's analysis, dependence on the provability predicate	S	should	Feferman 1960
N-a-032	P5 5.4 models-of-pa	nonstandard models of PA: existence and order type $\mathbb{N}+\mathbb{Z}\cdot\mathbb{Q}$, overspill and its uses, Tennenbaum's theorem	E	must	Skolem 1934; Tennenbaum 1959; Kaye 1991
N-a-033	P5 5.7 decidable-arithmetic	Presburger arithmetic: quantifier elimination with congruence predicates, decidability, Fischer-Rabin lower bound statement	T	must	Presburger 1929; Fischer-Rabin 1974; Enderton 2001
N-a-034	P5 5.7 decidable-arithmetic	Skolem arithmetic of multiplication: decidability via Feferman-Vaught statement	S	could	Skolem 1930; Feferman-Vaught 1959
N-a-035	P6 6.5 cardinal-arithmetic	cofinality, regular and singular cardinals, Konig's theorem, Hausdorff recursion, Easton's theorem statement	E	must	Konig 1905; Easton 1970; Jech 2003; Kunen 2011
N-a-036	P6 6.5 cardinal-arithmetic	singular cardinal behavior: Silver's theorem statement, PCF pointer	S	could	Silver 1975; Shelah 1994
N-a-037	P6 6.4 infinitary-combinatorics	closed unbounded and stationary sets, Fodor's pressing-down lemma, diagonal intersection	E	must	Fodor 1956; Jech 2003; Kunen 2011
N-a-038	P6 6.4 infinitary-combinatorics	Delta-system lemma with proof and the ccc verification pattern for Cohen forcing	T	must	Shanin 1946; Kunen 2011
N-a-039	P6 6.4 infinitary-combinatorics	infinite Ramsey theorem (T); Erdos-Rado theorem and partition calculus notation (S)	T	must	Ramsey 1930; Erdos-Rado 1956; Jech 2003
N-a-040	P6 6.4 infinitary-combinatorics	trees: Konig infinity lemma, Aronszajn tree construction, Suslin hypothesis and its independence status survey	S	should	Konig 1927; Kurepa 1935; Todorcevic 1984; Kunen 2011

continued

ID	Target	Item	Tr	Prio	Sources
N-a-041	P6 6.4 infinitary-combinatorics	combinatorial principles: diamond and square statements with one application each; Martin's axiom with ccc applications	S	should	Jensen 1972; Martin-Solovay 1970; Kunen 2011
N-a-042	P6 6.3 constructibility-forcing	Cohen forcing of ¬CH: full ccc cardinal-preservation verification, nice names counting	T	must	Cohen 1963; Kunen 2011
N-a-043	P6 6.3 constructibility-forcing	independence of AC: symmetric extensions method survey; permutation models of ZFA	S	should	Fraenkel 1922; Cohen 1963; Jech 1973; Halbeisen 2017
N-a-044	P6 6.6 large-cardinals	large cardinals: inaccessible, Mahlo, weakly compact, measurable via ultrapowers and elementary embeddings, Scott's theorem $V \neq L$, consistency-strength hierarchy chart to supercompact	S	must	Ulam 1930; Scott 1961; Keisler 1962; Kanamori 2003
N-a-045	P6 6.6 large-cardinals	Levy hierarchy, absoluteness for transitive models, Shoenfield absoluteness statement with applications	T	should	Levy 1965; Shoenfield 1961; Jech 2003
N-a-046	P6 6.7 descriptive-set-theory	descriptive set theory: Polish spaces, Borel hierarchy, analytic sets, Suslin's theorem, perfect-set property, regularity properties	E	must	Suslin 1917; Kechris 1995; Moschovakis 2009
N-a-047	P6 6.7 descriptive-set-theory	effective descriptive set theory bridge to computability: lightface pointclasses pointer	P	could	Moschovakis 2009; Sacks 1990
N-a-048	P6 6.8 determinacy	infinite games: Gale-Stewart theorem (T), Borel determinacy statement, AD versus AC, projective determinacy and Woodin cardinals pointer	S	should	Gale-Stewart 1953; Mycielski-Steinhaus 1962; Martin 1975; Martin-Steel 1989
N-a-049	P7 7.5 arithmetic-proof-theory	Gentzen's consistency proof for PA: reduction procedure on derivations, transfinite induction up to ε_0, ordinal assignment in full for the propositional core	T	must	Gentzen 1936; Gentzen 1938; Takeuti 1987; Schwichtenberg 1977
N-a-050	P7 7.5 arithmetic-proof-theory	ordinal notation systems: Cantor normal form arithmetic, Veblen functions, Γ_0 and Feferman-Schutte statement	S	should	Veblen 1908; Feferman 1964; Schutte 1977; Pohlers 2009
N-a-051	P7 7.5 arithmetic-proof-theory	PRA and finitism: Skolem's arithmetic, Tait's identification of finitism with PRA	S	should	Skolem 1923; Tait 1981
N-a-052	P7 7.5 arithmetic-proof-theory	omega-logic: the ω-rule, cut elimination for PA with the ω-rule, bound ε_0	S	could	Schutte 1951; Schwichtenberg 1977; Pohlers 2009
N-a-053	P7 7.6 functional-interpretations	Godel's Dialectica interpretation: System T, interpretation of HA, consistency corollary	S	should	Godel 1958; Troelstra 1973; Avigad-Feferman 1998
N-a-054	P7 7.6 functional-interpretations	Kleene number realizability: soundness for HA, ECT and Markov's principle independence applications	T	must	Kleene 1945; Troelstra 1998; van Oosten 2008
N-a-055	P7 7.6 functional-interpretations	disjunction and numerical existence properties for IPC and HA with proofs	T	must	Godel 1932; Gentzen 1935; Kleene 1945; Troelstra-van Dalen 1988
N-a-056	P7 7.7 structural-extensions	epsilon calculus: epsilon terms, first and second epsilon theorems statements, Herbrand's theorem in general form	S	should	Hilbert-Bernays 1939; Herbrand 1930; Buss 1998
N-a-057	P7 7.7 structural-extensions	generalized sequent frameworks survey: hypersequents, labelled calculi, display calculi, deep inference, with one worked modal example each	S	should	Avron 1996; Negri 2005; Belnap 1982; Guglielmi 2007; Negri-von Plato 2001
N-a-058	P7 7.7 structural-extensions	proof mining pointer: extraction of bounds from classical proofs	P	could	Kohlenbach 2008
N-a-059	P3 3.9 abstract-model-theory	infinitary logic $L_{\omega_1\omega}$: syntax, Scott isomorphism theorem, Scott sentences, Karp completeness statement, compactness failure	S	should	Karp 1964; Scott 1965; Keisler 1971; Marker 2016
N-a-060	P3 3.9 abstract-model-theory	generalized quantifiers: Mostowski and Lindstrom quantifiers, completeness for $L(Q_1)$ statement	S	could	Mostowski 1957; Lindstrom 1966; Keisler 1970
N-a-061	P3 3.9 abstract-model-theory	abstract logics and the full proof of Lindstrom's first theorem	T	should	Lindstrom 1969; Barwise 1974; Ebbinghaus-Flum-Thomas 1994
N-a-062	P3 3.9 abstract-model-theory	independence-friendly and dependence logic pointer	P	could	Hintikka-Sandu 1989; Vaananen 2007
N-a-063	P4 4.1 machines-and-models	μ-recursion and partial recursive functions: normal form and s-m-n recap, equivalence theorem Turing-computable $\equiv$ μ-recursive	E	must	Kleene 1936; Turing 1936; Kleene 1952; Odifreddi 1989

continued

ID	Target	Item	Tr	Prio	Sources
N-a-064	P4 4.6 complexity	complexity theory as a subject: class ladder P, NP, coNP, PH, PSPACE, EXPTIME, time and space hierarchy theorems, logic-relevant completeness results table	T	must	Hartmanis-Stearns 1965; Cook 1971; Karp 1972; Stockmeyer 1976; Papadimitriou 1994; Arora-Barak 2009
N-a-065	P3 3.1 elementary-maps-ef-games	Ehrenfeucht-Fraisse games: game characterization of elementary equivalence, quantifier rank, inexpressibility applications	E	must	Fraisse 1954; Ehrenfeucht 1961; Hodges 1993; Ebbinghaus-Flum-Thomas 1994
N-a-066	P6 6.1 ordinal-arithmetic	ordinal arithmetic developed as theory: sums, products, exponentiation, Cantor normal form, transfinite induction and recursion theorems	E	must	Cantor 1897; Sierpinski 1965; Jech 2003; Kunen 2011
N-a-067	P6 6.3 constructibility-forcing	constructible hierarchy L: condensation lemma statement, GCH in L, reflection principle, fine-structure pointer	S	should	Godel 1938; Levy 1960; Devlin 1984; Jech 2003
N-a-068	P6 6.3 constructibility-forcing	Boolean-valued models at statement level; product and iterated forcing pointers	S	should	Scott-Solovay 1967; Bell 2005; Kunen 2011
N-a-069	P7 7.2 cut-elimination	midsequent theorem via cut elimination and Herbrand's theorem as corollary	S	should	Gentzen 1935; Herbrand 1930; Takeuti 1987; Buss 1998
N-a-070	P7 7.2 cut-elimination	Craig interpolation proved by Maehara's method with Beth definability corollary	E	must	Craig 1957; Maehara 1960; Takeuti 1987
N-a-071	P2 2.3 second-higher-order	second-order and HOL proof calculi displayed: comprehension and choice schemas as a calculus, Henkin completeness route	E	must	Church 1940; Henkin 1950; Andrews 2002; Shapiro 1991
N-b-001	P8 8.2 modal	canonical models and completeness for K, T, D, B, K4, S4, S5: Lindenbaum construction for normal modal logics, canonicity discussion	E	must	Lemmon-Scott 1977; Chellas 1980; Blackburn-de Rijke-Venema 2001
N-b-002	P8 8.2 modal	frame correspondence: definability of the T/4/5/B/D axioms, Sahlqvist theorem statement with algorithmic examples (proof S)	T	must	Sahlqvist 1975; van Benthem 1983; Blackburn-de Rijke-Venema 2001
N-b-003	P8 8.2 modal	bisimulation: definition, invariance, van Benthem characterization theorem	T	must	van Benthem 1976; Blackburn-de Rijke-Venema 2001
N-b-004	P8 8.2 modal	filtration, finite model property, and decidability for K, T, S4, S5	T	must	Segerberg 1971; Chagrov-Zakharyaschev 1997; Blackburn-de Rijke-Venema 2001
N-b-005	P8 8.2 modal	Kripke frame incompleteness phenomena: general frames, Thomason and Fine counterexamples survey	S	should	Thomason 1974; Fine 1974; Chagrov-Zakharyaschev 1997
N-b-006	P8 8.2 modal	non-normal modal logics and neighborhood semantics: systems E, M, C, N, regular logics, completeness	E	should	Scott 1970; Montague 1970; Chellas 1980; Pacuit 2017
N-b-007	P8 8.2 modal	per-system entry cards for the modal cube: K, D, T, B, K4, S4, S4.2, S5, GL, Grz – axioms, frame class, FMP, decidability, characteristic theorems	E	must	Chellas 1980; Hughes-Cresswell 1996; Chagrov-Zakharyaschev 1997
N-b-008	P8 8.2 modal	two-dimensional semantics: actuality operator, fixedly, Kaplan character/content distinction	S	should	Crossley-Humberstone 1977; Davies-Humberstone 1980; Kaplan 1989
N-b-009	P8 8.2 modal	quantified modal logic: constant versus varying domains, Barcan and converse Barcan formulas, rigid designators, existence predicate, actualism/possibilism	E	must	Barcan 1946; Kripke 1963; Fitting-Mendelsohn 1998; Garson 2001; Williamson 2013
N-b-010	P8 8.4 conditionals-deontic-free	free logic: positive/negative/neutral systems, inner and outer domains, definite descriptions (Russell versus free-logic treatments)	E	should	Russell 1905; Lambert 1967; Bencivenga 1986; Lehmann 2002
N-b-011	P8 8.3 temporal-dynamic-epistemic	multi-agent epistemic logic: $S5_n$/$KD45_n$, common knowledge as fixed point, muddy children analysis	E	must	Hintikka 1962; Aumann 1976; Fagin-Halpern-Moses-Vardi 1995
N-b-012	P8 8.3 temporal-dynamic-epistemic	dynamic epistemic logic: public announcement logic axioms, reduction laws, unsuccessful updates	S	should	Plaza 1989; Baltag-Moss-Solecki 1998; van Ditmarsch-van der Hoek-Kooi 2008
N-b-013	P8 8.4 conditionals-deontic-free	deontic logic: SDL as KD, Ross and Good Samaritan and Chisholm contrary-to-duty paradoxes, dyadic conditional obligation	E	must	von Wright 1951; Chisholm 1963; Hansson 1969; McNamara 2006
N-b-014	P8 8.3 temporal-dynamic-epistemic	philosophical tense logic: Prior's systems, since/until and Kamp's expressive completeness statement, Ockhamist versus Peircean branching time	S	should	Prior 1967; Kamp 1968; Burgess 1984
N-b-015	P8 8.3 temporal-dynamic-epistemic	hybrid logic: nominals, satisfaction operators, axiomatization survey	S	could	Prior 1967; Blackburn 2000; Areces-ten Cate 2007

continued

ID	Target	Item	Tr	Prio	Sources
N-b-016	P8 8.4 conditionals-deontic-free	counterfactuals: Stalnaker selection functions versus Lewis sphere systems, conditional excluded middle debate, limit assumption, completeness statements for C1 and VC	E	must	Stalnaker 1968; Lewis 1973; Nute 1980
N-b-017	P8 8.4 conditionals-deontic-free	indicative conditionals: paradoxes of material implication, Adams's thesis, Lewis triviality results, suppositional theories	S	must	Adams 1975; Lewis 1976; Jackson 1987; Edgington 1995; Bennett 2003
N-b-018	P8 8.4 conditionals-deontic-free	conditional logics as a family: CK and neighbors, selection-function and premise semantics, correspondence table	T	should	Chellas 1975; Nute 1980
N-b-019	P11 11.2 truth-theories	Tarskian truth as a philosophical unit: convention T, object/metalanguage discipline, hierarchy of languages	S	should	Tarski 1935; Tarski 1944
N-b-020	P11 11.2 truth-theories	Kripke's fixed-point theory of truth: partial Kleene valuation schemes, least and maximal fixed points, groundedness, ghost of the Tarski hierarchy	T	must	Kripke 1975; Martin 1984; Halbach 2011
N-b-021	P11 11.2 truth-theories	revision theory of truth survey	S	could	Herzberger 1982; Gupta-Belnap 1993
N-b-022	P11 11.2 truth-theories	axiomatic theories of truth: TB/CT conservativity, KF, FS, comparative survey	S	should	Friedman-Sheard 1987; Feferman 1991; Halbach 2011; Horsten 2011
N-b-023	P11 11.1 paradoxes	paradox catalog with formal diagnoses: strengthened liar, Curry, Yablo, Grelling, Berry	E	must	Curry 1942; Yablo 1993; Priest 2006; Beall-Glanzberg-Ripley 2018
N-b-024	P11 11.4 vagueness	vagueness: sorites structure, supervaluationism with global versus local validity, degree-theoretic approaches, epistemicism, contextualism pointer	S	must	Fine 1975; Williamson 1994; Keefe 2000; Smith 2008
N-b-025	P8 8.5 many-valued-fuzzy	first-degree entailment: Belnap-Dunn four-valued semantics, bilattice reading, relation to K3 and LP	E	must	Dunn 1976; Belnap 1977; Omori-Wansing 2017
N-b-026	P8 8.7 paraconsistent-nonmonotonic	dialetheism and the choice of paraconsistent consequence: motivation from paradox, inclosure schema, replies	S	should	Priest 1979; Priest 2006
N-b-027	P8 8.6 substructural-linear	relevance logic landscape: E and R axiomatic systems, variable-sharing proof, ternary-frame conditions per axiom, admissibility of gamma	S	should	Anderson-Belnap 1975; Meyer-Dunn 1969; Routley-Meyer 1973; Dunn-Restall 2002; Mares 2004
N-b-028	P8 8.5 many-valued-fuzzy	Lukasiewicz logics: L3, finite Ln, infinite-valued L, axiomatization, McNaughton theorem statement, real-valued semantics	E	must	Lukasiewicz 1920; Lukasiewicz-Tarski 1930; McNaughton 1951; Cignoli-D'Ottaviano-Mundici 2000
N-b-029	P8 8.5 many-valued-fuzzy	t-norm fuzzy logics: continuous t-norms, BL and MTL, Godel and product logics, standard completeness survey	S	must	Hajek 1998; Esteva-Godo 2001; Cintula-Hajek-Noguera 2011
N-b-030	P8 8.1 intuitionistic-intermediate	intuitionism beyond the calculus: BHK reading, Brouwerian counterexamples, choice sequences, bar induction, continuity principles, Markov's principle status	S	should	Brouwer 1908; Heyting 1956; Dummett 1977; Troelstra-van Dalen 1988
N-b-031	P8 8.1 intuitionistic-intermediate	intermediate logics: Godel-Dummett LC, KC, the lattice of superintuitionistic logics survey	S	should	Godel 1932; Dummett 1959; Jankov 1968; Chagrov-Zakharyaschev 1997
N-b-032	P8 8.7 paraconsistent-nonmonotonic	nonmonotonic consequence: preferential models, KLM system P, rational monotony; AGM belief revision postulates, Levi and Harper identities	E	must	Alchourron-Gardenfors-Makinson 1985; Kraus-Lehmann-Magidor 1990; Lehmann-Magidor 1992; Makinson 2005
N-b-033	P8 8.7 paraconsistent-nonmonotonic	default logic and circumscription entry cards	S	should	Reiter 1980; McCarthy 1980
N-b-034	P11 11.3 consequence-and-constants	philosophy of logical consequence: Tarski's analysis, model-theoretic versus proof-theoretic accounts, demarcation of logical constants (invariance), harmony	S	should	Tarski 1936; Prawitz 1974; Etchemendy 1990; Sher 1991; Dummett 1991
N-b-035	P11 11.5 ontological-commitment	plural quantification and the status of second-order logic: Boolos's plural reading, ontological commitment, Quine's 'set theory in disguise' charge	S	should	Quine 1970; Boolos 1984; Boolos 1985; Shapiro 1991
N-c-001	P9 9.1 finite-model-theory	finite model theory foundations: failure of compactness and completeness in the finite, Craig/Beth failure, EF-game method on finite structures, Trakhtenbrot recap	E	must	Ebbinghaus-Flum 1995; Libkin 2004

continued

ID	Target	Item	Tr	Prio	Sources
N-c-002	P9 9.1 finite-model-theory	zero-one laws for first-order logic: Glebskii and Fagin theorems, extension axioms, the random-graph link	T	must	Glebskii et al. 1969; Fagin 1976; Libkin 2004
N-c-003	P9 9.1 finite-model-theory	locality: Hanf and Gaifman theorems with inexpressibility applications (connectivity, acyclicity)	T	should	Hanf 1965; Gaifman 1982; Libkin 2004
N-c-004	P9 9.1 finite-model-theory	descriptive complexity: Fagin's theorem (existential SO captures NP) with proof sketch, Immerman-Vardi (LFP captures P on ordered structures) statement	T	must	Fagin 1974; Vardi 1982; Immerman 1986; Immerman 1999
N-c-005	P9 9.2 decidable-fragments	the classical decision problem: prefix-class classification, Bernays-Schonfinkel-Ramsey, Ackermann, Godel-Kalmar-Schutte classes, reduction classes	S	should	Lewis 1979; Borger-Gradel-Gurevich 1997
N-c-006	P9 9.2 decidable-fragments	two-variable fragment: finite model property and NEXPTIME decidability statement	S	should	Mortimer 1975; Gradel-Kolaitis-Vardi 1997
N-c-007	P9 9.2 decidable-fragments	guarded fragment: decidability and the modal connection	S	should	Andreka-van Benthem-Nemeti 1998; Gradel 1999
N-c-008	P9 9.2 decidable-fragments	monadic first-order logic: Lowenheim-Behmann decidability	T	should	Lowenheim 1915; Behmann 1922; Borger-Gradel-Gurevich 1997
N-c-009	P9 9.3 description-logics	ALC: syntax, model semantics, TBox/ABox, tableau decision procedure with blocking, ExpTime complexity statement	E	must	Schmidt-Schauss-Smolka 1991; Baader et al. 2003; Baader-Horrocks-Lutz-Sattler 2017
N-c-010	P9 9.3 description-logics	lightweight and expressive DLs: EL family with polynomial classification, DL-Lite and query answering, SHOIQ/OWL profiles map	S	should	Baader-Brandt-Lutz 2005; Calvanese et al. 2007; Baader-Horrocks-Lutz-Sattler 2017
N-c-011	P9 9.3 description-logics	ALC as notational variant of multimodal K (Schild correspondence)	T	should	Schild 1991
N-c-012	P9 9.4 temporal-logics-automata	LTL deepened: past operators, Kamp and Gabbay-Pnueli-Shelah-Stavi expressive completeness statements, safety/liveness, PSPACE satisfiability	T	must	Kamp 1968; Pnueli 1977; Gabbay-Pnueli-Shelah-Stavi 1980; Sistla-Clarke 1985
N-c-013	P9 9.4 temporal-logics-automata	omega-automata: Buchi/Rabin/parity acceptance, closure properties, complementation statement, Vardi-Wolper LTL-to-automata translation	S	must	Buchi 1962; McNaughton 1966; Vardi-Wolper 1986; Safra 1988; Thomas 1997
N-c-014	P9 9.4 temporal-logics-automata	MSO and automata: Buchi-Elgot-Trakhtenbrot theorem on finite words (T), S1S decidability, Rabin's S2S statement	T	should	Buchi 1960; Elgot 1961; Trakhtenbrot 1961; Rabin 1969; Thomas 1997
N-c-015	P9 9.4 temporal-logics-automata	tree automata pointer and applications to decidability transfer	P	could	Thatcher-Wright 1968; Comon et al. 2007
N-c-016	P9 9.5 model-checking	CTL and CTL*: syntax, semantics, fixpoint characterizations of EG/EU, labeling algorithm with complexity, LTL versus CTL expressiveness	E	must	Clarke-Emerson 1981; Queille-Sifakis 1982; Emerson-Halpern 1986; Clarke-Grumberg-Peled 1999; Baier-Katoen 2008
N-c-017	P9 9.5 model-checking	symbolic model checking: BDDs and Bryant canonicity, fixpoint computation, bounded model checking via SAT, CEGAR pointer	S	must	Bryant 1986; Burch et al. 1992; McMillan 1993; Biere-Cimatti-Clarke-Zhu 1999; Clarke et al. 2003
N-c-018	P9 9.5 model-checking	modal mu-calculus: syntax, Knaster-Tarski semantics, relation to parity games, alternation hierarchy statement	S	should	Kozen 1983; Emerson-Jutla 1991; Bradfield-Stirling 2007
N-c-019	P9 9.6 program-logics	Hoare logic: partial and total correctness, axioms and rules, soundness proof, relative completeness (Cook), weakest preconditions (Dijkstra)	E	must	Floyd 1967; Hoare 1969; Dijkstra 1975; Cook 1978; Apt 1981
N-c-020	P9 9.6 program-logics	separation logic: heap assertions, frame rule, symbolic execution, bi-abduction pointer	S	should	O'Hearn-Reynolds-Yang 2001; Reynolds 2002; Calcagno et al. 2011; O'Hearn 2019
N-c-021	P9 9.6 program-logics	propositional dynamic logic: syntax, regular programs, Segerberg axioms with Kozen-Parikh completeness statement, Fischer-Ladner closure, ExpTime decidability	E	should	Pratt 1976; Fischer-Ladner 1979; Kozen-Parikh 1981; Harel-Kozen-Tiuryn 2000
N-c-022	P9 9.7 logic-programming-databases	logic programming: definite programs, least Herbrand model, immediate-consequence fixpoint, SLD-resolution soundness and completeness, negation as failure with Clark completion, SLDNF	E	must	Kowalski 1974; van Emden-Kowalski 1976; Clark 1978; Apt-van Emden 1982; Lloyd 1987
N-c-023	P9 9.7 logic-programming-databases	answer set programming: stable-model semantics via Gelfond-Lifschitz reduct, complexity statement	S	should	Gelfond-Lifschitz 1988; Marek-Truszczynski 1999; Brewka-Eiter-Truszczynski 2011

continued

ID	Target	Item	Tr	Prio	Sources
N-c-024	P9 9.7 logic-programming-databases	Datalog and conjunctive queries: Chandra-Merlin homomorphism theorem, containment decidability, data/combined complexity	S	should	Chandra-Merlin 1977; Vardi 1982; Abiteboul-Hull-Vianu 1995
N-c-025	P9 9.7 logic-programming-databases	constraint satisfaction and logic pointer: CSP as homomorphism problem, dichotomy theorem statement	P	could	Feder-Vardi 1998; Bulatov 2017; Zhuk 2017
N-c-026	P9 9.8 complexity-of-logic	complexity of logical tasks: Cook-Levin recap, QBF PSPACE-completeness with proof, Ladner-style table for modal logics (K/S4 PSPACE, S5 NP), FO validity undecidable recap	T	must	Cook 1971; Stockmeyer-Meyer 1973; Ladner 1977; Papadimitriou 1994
N-c-027	P7 7.8 proof-complexity	proof complexity: Cook-Reckhow framework, resolution lower bound for the pigeonhole principle (Haken) proof sketch, Frege systems open-problems map	S	should	Cook-Reckhow 1979; Haken 1985; Beame-Pitassi 1998; Krajicek 2019
N-c-028	P9 9.8 complexity-of-logic	interpolation in verification pointer: McMillan-style interpolant-based model checking	P	could	McMillan 2003
N-c-029	P7 7.4 lambda-calculi	System F: syntax, impredicative encodings, Girard's strong normalization statement, Curry-Howard for second-order, parametricity pointer	S	must	Girard 1972; Reynolds 1974; Reynolds 1983; Girard-Lafont-Taylor 1989
N-c-030	P7 7.4 lambda-calculi	Martin-Lof type theory: judgments, Pi/Sigma/identity types, universes, propositions-as-types with proof relevance, normalization and decidable type checking statements	S	must	Martin-Lof 1984; Nordstrom-Petersson-Smith 1990; Hofmann 1997
N-c-031	P12 12.5 interactive-proof	lambda cube and pure type systems; calculus of (inductive) constructions as the Coq/Lean foundation	S	should	Coquand-Huet 1988; Barendregt 1992; Paulin-Mohring 2015
N-c-032	P12 12.1 sat	CDCL: clause learning with UIP, watched literals, restarts, VSIDS; GRASP/Chaff/MiniSat lineage; DRAT proofs and certified UNSAT	E	must	Marques-Silva-Sakallah 1999; Moskewicz et al. 2001; Een-Sorensson 2003; Marques-Silva-Lynce-Malik 2009; Wetzler-Heule-Hunt 2014
N-c-033	P12 12.1 sat	tractable cases and hardness structure: 2-SAT via implication graphs, Horn-SAT unit propagation, random k-SAT phase transition pointer	T	should	Aspvall-Plass-Tarjan 1979; Dowling-Gallier 1984; Mitchell-Selman-Levesque 1992
N-c-034	P12 12.2 smt	SMT internals: DPLL(T) architecture, congruence closure for EUF, simplex for linear arithmetic, Nelson-Oppen combination with proof for convex stably-infinite theories	E	must	Nelson-Oppen 1979; Dutertre-de Moura 2006; Nieuwenhuis-Oliveras-Tinelli 2006; de Moura-Bjorner 2008; Barrett-Sebastiani-Seshia-Tinelli 2009
N-c-035	P12 12.3 first-order-atp	unification theory deepened: Martelli-Montanari rule-based algorithm with termination proof, mgu theorem, occurs-check, equational and AC-unification survey, undecidability of higher-order unification	E	must	Robinson 1965; Huet 1973; Goldfarb 1981; Martelli-Montanari 1982; Baader-Snyder 2001
N-c-036	P12 12.3 first-order-atp	ordered resolution and the superposition calculus: reduction orders, redundancy and saturation, model-construction completeness proof sketch	S	must	Bachmair-Ganzinger 1994; Nieuwenhuis-Rubio 2001
N-c-037	P12 12.4 rewriting	term rewriting: abstract reduction systems, Newman's lemma, critical pairs and Knuth-Bendix completion, termination orders (LPO/KBO), word-problem applications	E	must	Newman 1942; Knuth-Bendix 1970; Huet 1980; Dershowitz 1982; Baader-Nipkow 1998
N-c-038	P12 12.5 interactive-proof	LCF architecture and kernels: de Bruijn criterion, proof terms versus proof scripts, trusted computing bases of Coq/Lean/Isabelle/HOL, reflection	S	should	de Bruijn 1970; Gordon-Milner-Wadsworth 1979; Pollack 1998; Harrison 2009
N-c-039	P12 12.5 interactive-proof	program extraction from constructive proofs: realizability-based extraction	S	could	Paulin-Mohring 1989; Letouzey 2003
N-c-040	P12 12.5 interactive-proof	ATP-ITP integration pointer: hammers and proof reconstruction	P	could	Paulson-Blanchette 2010; Blanchette et al. 2016
N-d-001	P10 10.1 universal-algebra	universal algebra toolkit: algebras, homomorphisms, congruences and quotients, subdirect representation, Birkhoff HSP variety theorem	T	must	Birkhoff 1935; Birkhoff 1944; Burris-Sankappanavar 1981
N-d-002	P10 10.1 universal-algebra	equational logic: term algebras, free algebras, Birkhoff completeness for equational consequence	T	must	Birkhoff 1935; Burris-Sankappanavar 1981
N-d-003	P10 10.2 boolean-stone	Boolean algebras: filters, ideals, ultrafilters, atomic versus atomless, BPI status relative to AC	E	must	Halmos 1963; Bell-Slomson 1969; Koppelberg 1989

continued

ID	Target	Item	Tr	Prio	Sources
N-d-004	P10 10.2 boolean-stone	Stone representation and duality: proof of the representation theorem, Stone spaces, clopen-algebra inverse, uniqueness of the countable atomless BA	T	must	Stone 1936; Halmos 1963; Johnstone 1982; Givant-Halmos 2009
N-d-005	P10 10.2 boolean-stone	Lindenbaum-Tarski algebras: algebraic completeness proof for CPL, free Boolean algebras	E	must	Tarski 1935; Rasiowa-Sikorski 1963
N-d-006	P10 10.2 boolean-stone	Priestley duality for distributive lattices	S	should	Priestley 1970; Davey-Priestley 2002
N-d-007	P10 10.2 boolean-stone	Stone-duality dividends for logic: compactness as compactness of the Stone space, type spaces of theories as Stone spaces	S	should	Johnstone 1982; Marker 2002
N-d-008	P10 10.3 heyting-algebras	Heyting algebras: definition, opens of a topological space, algebraic completeness of IPC, Rasiowa-Sikorski lemma	T	must	Heyting 1930; McKinsey-Tarski 1946; Rasiowa-Sikorski 1963; Balbes-Dwinger 1974
N-d-009	P10 10.3 heyting-algebras	Esakia duality and the lattice of intermediate logics survey	S	should	Esakia 1974; Chagrov-Zakharyaschev 1997
N-d-010	P10 10.3 heyting-algebras	locales and pointfree topology pointer	P	could	Johnstone 1982; Picado-Pultr 2012
N-d-011	P10 10.4 modal-algebras	Boolean algebras with operators: Jonsson-Tarski representation, canonical extensions, duality with general frames, algebraic completeness of normal modal logics	T	must	Jonsson-Tarski 1951; Goldblatt 1989; Blackburn-de Rijke-Venema 2001; Venema 2007
N-d-012	P10 10.4 modal-algebras	topological semantics: McKinsey-Tarski completeness of S4 for topological interior, completeness for the real line statement	T	should	McKinsey-Tarski 1944; Rasiowa-Sikorski 1963; van Benthem-Bezhanishvili 2007
N-d-013	P10 10.5 residuated-mv	MV-algebras and Chang completeness for infinite-valued Lukasiewicz logic; Mundici's l-group correspondence statement	T	must	Chang 1958; Chang 1959; Mundici 1986; Cignoli-D'Ottaviano-Mundici 2000
N-d-014	P10 10.5 residuated-mv	residuated lattices and FL-algebras: algebraization of substructural logics, subvariety lattice survey, cut elimination correspondences	S	must	Ono-Komori 1985; Galatos-Jipsen-Kowalski-Ono 2007
N-d-015	P10 10.5 residuated-mv	BL- and MTL-algebras and standard completeness statements	S	should	Hajek 1998; Cignoli-Esteva-Godo-Torrens 2000; Jenei-Montagna 2002
N-d-016	P10 10.6 algebraized-fol	cylindric algebras: algebraization of first-order logic, representability problems survey	S	should	Henkin-Monk-Tarski 1971; Henkin-Monk-Tarski 1985; Monk 2000
N-d-017	P10 10.6 algebraized-fol	relation algebras: Tarski's calculus of relations, representable relation algebras, undecidability statement, set theory without variables pointer	S	should	Tarski 1941; Tarski-Givant 1987; Maddux 2006
N-d-018	P10 10.6 algebraized-fol	Blok-Pigozzi algebraizability pointer	P	could	Blok-Pigozzi 1989; Font 2016
N-d-019	P10 10.7 category-toolkit	category theory for logicians: categories, functors, natural transformations, adjunctions, limits and colimits, Yoneda lemma statement	E	must	Mac Lane 1998; Awodey 2010
N-d-020	P10 10.7 category-toolkit	Lawvere algebraic theories and functorial semantics	S	should	Lawvere 1963; Adamek-Rosicky-Vitale 2011
N-d-021	P10 10.8 ccc-lambda	cartesian closed categories and typed lambda calculus: Curry-Howard-Lambek three-way correspondence with the equivalence theorem	T	must	Lambek 1980; Lambek-Scott 1986
N-d-022	P10 10.9 topos-overview	quantifiers as adjoints; hyperdoctrine view of first-order logic overview	S	should	Lawvere 1969; Pitts 2000
N-d-023	P10 10.9 topos-overview	elementary topos overview: subobject classifier, internal logic is intuitionistic HOL, presheaf examples, Kripke-Joyal semantics	S	must	Lawvere 1970; Lambek-Scott 1986; Mac Lane-Moerdijk 1992; Johnstone 2002
N-d-024	P10 10.9 topos-overview	topos-theoretic independence pointer: Tierney's sheaf-theoretic forcing and CH	P	could	Tierney 1972; Mac Lane-Moerdijk 1992
N-x-001	P7 –	homotopy type theory and univalent foundations beyond a one-page pointer	P	–	Univalent Foundations Program 2013
N-x-002	P7 –	higher inductive types and cubical type theory internals	P	–	Cohen-Coquand-Huber-Mortberg 2016
N-x-003	P6 –	proper forcing, iterated forcing, and the forcing-axiom landscape (PFA/MM) beyond a pointer	P	–	Shelah 1982; Foreman-Magidor-Shelah 1988
N-x-004	P6 –	inner model theory, fine structure, and core models	P	–	Jensen 1972; Mitchell-Steel 1994
N-x-005	P6 –	PCF theory and advanced cardinal arithmetic beyond Silver's theorem statement	P	–	Shelah 1994
N-x-006	P8 –	quantum logic (orthomodular lattices as logic)	P	–	Birkhoff-von Neumann 1936
N-x-007	P8 –	adaptive logics	P	–	Batens 2007

continued

ID	Target	Item	Tr	Prio	Sources
N-x-008	P11 –	inductive logic and confirmation theory	P	–	Carnap 1950
N-x-009	P8 –	formal argumentation frameworks	P	–	Dung 1995
N-x-010	P10 – pointer	abstract algebraic logic beyond the Blok-Pigozzi	P	–	Font-Jansana-Pigozzi 2003; Font 2016
N-x-011	P7 –	geometry of interaction	P	–	Girard 1989
N-x-012	P7 –	proof nets beyond a pointer (correctness criteria, MLL net theory)	P	–	Girard 1987; Danos-Regnier 1989
N-x-013	P10 –	categorical logic beyond the CCC and topos overview (fibrations, sketches, infinity-topoi)	P	–	Jacobs 1999; Lurie 2009
N-x-014	P4 –	computable analysis and computability over the reals (TTE, Weihrauch degrees)	P	–	Weihrauch 2000; Pour-El-Richards 1989
N-x-015	P9 –	probabilistic and stochastic model checking, probability logics beyond a pointer	P	–	Halpern 2003; Baier-Katoen 2008

6 Target handbook architecture

6.1 Design rules

1. The handbook has **14 parts plus a proof annex**; the body is lookup material (entries, tables, statements with status and citation) and the annex holds compressed proofs, organized parallel to the body.
2. Per-chapter proof sections and a separate companion volume are both rejected: the former re-interleaves proofs with lookup material; the latter is premature (the annex can be split out later without rework).
3. The v1.0 manuscript is **not printed**. After Phase-2 repairs it lives at `source/` as the verification substrate; its exercises, feedback chapter, and qualifying problems are retired from print but retained as verification assets, and solutions that currently do theorems' work are promoted into annex proofs during migration.
4. Every numbered body item carries a status field with an address: *proved* (Annex P$n.x$) / *sketched* (Annex ...) / *cited* (source). Compression may only *downgrade* status; upgrading requires re-verification.

6.2 The fourteen parts

Part 1 — Notation, Symbols, and Conventions.
Master symbol registry (generated from `registry/symbols.csv`); binding and precedence for *all* systems; notation variants across the literature; the 16 binary truth functions; mathematical preliminaries as definition/theorem tables.

Part 2 — Classical Logic.
2.1 propositional (semantics, equivalences, normal forms incl. Tseitin, functional completeness with Post's lattice, duality, calculi side by side: Hilbert / ND / LK / tableaux / resolution, metatheorems, complexity); 2.2 first-order (syntax, semantics, calculi, metatheorem catalog incl. the countable/choice-free completeness case, prenex–Skolem–clausal pipeline, decidable-fragments table, equality metatheory); 2.3 second- and higher-order (full vs. Henkin, a displayed SOL calculus, simple type theory); 2.4 variants (many-sorted, free, infinitary, generalized quantifiers).

Part 3 — Model Theory.
Elementary maps and diagrams; Ehrenfeucht–Fraïssé games; types, saturation, prime/atomic models; ultraproducts and Łoś; preservation theorems; the QE catalog per theory; categoricity and Morley; stability/NIP/o-minimality overview tables; finite model theory; Lindström.

Part 4 — Computability and Complexity.
Machine models and μ-recursion with the equivalence theorem; Church–Turing; s-m-n, recursion, Rice; undecidability catalog; c.e.-set structure; degrees, jump, Post's problem, priority method;

arithmetical and analytical hierarchies; Kolmogorov complexity; complexity-class ladder and logic-relevant completeness results; descriptive complexity.

Part 5 — Arithmetic and Incompleteness.
Q and PA; representability and coding (repaired); the diagonal lemma; the Gödel I and II, Rosser, Löb, and Tarski entries; provability logic GL with Solovay completeness; decidable arithmetics; nonstandard models and Tennenbaum; independence catalog (Paris–Harrington, Goodstein); interpretability; second-order arithmetic and the reverse-mathematics calibration table.

Part 6 — Set Theory.
ZFC axiom table and system variants; ordinal arithmetic and Cantor normal form; cardinal arithmetic, cofinality, König; AC equivalents; clubs/stationary/Fodor, Δ-system, Ramsey; combinatorial principles ($\diamondsuit$, $\square$, MA); V, L, HOD; absoluteness and reflection; forcing entries with the honest $\neg$CH treatment; independence catalog; the large-cardinal chart; descriptive set theory and determinacy.

Part 7 — Proof Theory and λ-Calculi.
Calculi catalog with identity cards; structural-rules table; cut elimination with mix displayed; subformula property, midsequent, Herbrand; interpolation (Maehara); normalization and strong normalization; Curry–Howard correspondence table; untyped λ-calculus and Church–Rosser; System F; MLTT/CIC overview; ordinal-analysis calibration; Gentzen's consistency proof (survey, honestly labeled); proof complexity.

Part 8 — Nonclassical Logics.
The flagship part: uniform per-system **entry cards** (name/aliases, language, semantics, axioms/rules with a pointer to the rule sheet, adequacy theorems with status, characteristic theorems and failures, decidability and complexity, translations, algebraic counterpart, sources). 8.1 intuitionistic and intermediate; 8.2 modal (systems lattice, correspondence table, canonical models, filtration, decidability, quantified modal logic, GL); 8.3 temporal, dynamic, epistemic (LTL, CTL/CTL*, μ-calculus, PDL, S5$_n$, common knowledge with its induction axiom, public announcement); 8.4 conditionals, deontic, free logic; 8.5 many-valued and fuzzy; 8.6 substructural (family map, relevance, full classical linear logic, Lambek); 8.7 paraconsistent and nonmonotonic (with the AGM postulates table).

Part 9 — Logic in Computer Science.
Description-logic family table with complexity; logic programming (SLD resolution, negation as failure, Prolog, ASP, Datalog); Hoare logic and weakest preconditions, separation-logic pointer; model checking (automata-theoretic LTL, fixpoint CTL, BDDs); database and finite-model connections; type systems.

Part 10 — Algebraic and Categorical Logic.
Boolean algebras and Stone duality; Lindenbaum–Tarski algebras; Heyting, MV, and modal algebras with Jónsson–Tarski; residuated lattices; cylindric/relation-algebra pointers; categorical semantics (CCC $\leftrightarrow$ STLC, hyperdoctrine pointer, topos overview).

Part 11 — Philosophy of Logic.
Paradox catalog with resolution strategies; theories of truth (Tarski hierarchy, Kripke fixed points, revision); consequence relations; vagueness; identity; ontological commitment; schools capsules.

Part 12 — Automation and Mechanization.
SAT: DPLL and CDCL, 2-SAT/Horn; resolution refinements; unification (Martelli–Montanari algorithm and theorem); superposition as a calculus; SMT (theories table, Nelson–Oppen, DPLL(T)); tableau and connection methods; QBF/#SAT/MaxSAT; prover and proof-assistant landscape; kernel foundations (LCF, de Bruijn criterion, MLTT/CIC in kernels); landmark formalizations; typesetting and reproducibility reference.

Part 13 — Chronology and Biography.
Chronology table; landmark-papers table; $\sim$40 capsule biographies; notation-history table.

Part 14 — Apparatus.
Complete rule sheets ($\sim$20, generated from the calculus registry); master cross-reference tables

(decidability and complexity per logic; consistency-strength ladder; expressive power; axiom ↔ frame ↔ algebra correspondence) — *harvested* from entry-card fields, not authored; glossary; per-part further reading; subject, symbol, person, and named-theorem indexes.

Appendix P — Proof Annex.

Parallel to Parts 2–12. Tier 1: method entries (Henkin, diagonalization, Łoś, back-and-forth, canonical models, filtration, cut elimination, normalization, forcing, EF, priority) — compressed proof plus a body method box. Tier 2: results the manuscript proves completely — compressed to 40–60% of source length, `proofboundary` markers retained. Tier 3: cited theorems — statement only, no annex entry.

6.3 Source-to-target mapping

v1.0 part (lines)	Target	Disposition
I Foundations (363–1546)	P1, P6	compress ∼4:1 to registry tables; proof-writing chapter's core into front matter
II Propositional (1547–2695)	P2.1	entries + tables; proofs to annex
III First-order (2696–4994)	P2.2, P2.4	entries + tables; Henkin to annex Tier 1
IV Model theory (4995–6203)	P3	entries; Łoś, back-and-forth to annex Tier 1
V Computability (6204–7190)	P4	entries + hierarchy tables
VI Arithmetic (7191–8068)	P5	entries; audit table institutionalized
VII Set theory (8069–9134)	P6	entries; forcing to annex Tier 1 (repaired)
VIII Proof theory (9135–9948)	P7	entries; cut elimination to annex Tier 1
IX Second/higher-order (9949–10621)	P2.3	entries; many-sorted translation to annex Tier 2
X Nonclassical (10622–11683)	P8	exploded into entry cards
XI Practice/automation (11684–12675)	P12	compressed + heavy additions
Back matter (12676–14780)	P14 / retired	registers and sheets rebuilt from data; exercises retired to verification assets

6.4 Templates

The entry-card and theorem environments carry *mandatory* keys; compilation fails without them (ruling 1). Sketch of the planned interface:

```
\begin{systementry}{name=S5, aliases={KT5},
    language=modal-prop, sheet=sheet:S5}
  \entrysemantics{Kripke frames with equivalence relations}
  \entryadequacy{sound+strongly complete for local consequence,
    status=proved, annex=P8.2.4}
  \entrydecidability{decidable; SAT is NP-complete,
    status=cited, source=Ladner1977}
  \entrytranslations{into FOL via standard translation}
  \entryalgebra{monadic algebras}
  \entrysources{BlackburnDeRijkeVenema2001}
\end{systementry}

\begin{hbtheorem}{title=Compactness, status=proved,
    annex=P2.2.7, sources={Godel1930,Malcev1936}}
  ...statement only; the proof lives at the annex address...
\end{hbtheorem}
```

Label taxonomy for the new tree: `entry:`, `thm:`, `tab:`, `fig:`, `sheet:`, `annex:`, `sec:` — replacing the v1.0 label population of which ∼85% is never referenced.

7 Apparatus program

Indexes. Four indexes, hand-authored at definition sites with inverted headings and sort keys (`Godel@Gödel`); *see*/*see also* entries; target 2,500–4,000 subject entries. The v1.0 auto-index wrapper (theorem titles as entries) is retired.

Registries as data. `registry/symbols.csv` (glyph, macro, role, scope, precedence rank, first use) generates the Part-1 tables and drives the notation lint; `registry/calculi/*.tex` holds every inference rule — body chapters, entry cards, and Part-14 rule sheets all `\input` from it, making body–sheet divergence structurally impossible.

Bibliography. Manual `thebibliography` (64 entries) converts to biblatex in Phase 3; the five missing primaries (Glivenko 1929, Gödel 1933, Gentzen 1936, Kleene 1938, Priest 1979) enter immediately; the per-domain source lists in `new-content.csv` are the intake queue. Projected final size 250–350 entries.

The build list.

ID	Category	Item	Specification	Prio	Ph
A-001	rule-sheet	Rule sheet: ND-CPL	All twelve natural-deduction rules plus the repetition rule and the vacuous-discharge convention, displayed, named, and labeled, generated from the calculus-registry file.	must	6
A-002	rule-sheet	Rule sheet: ND-FOL	Quantifier and equality rules with the generalization-variable and eigenvariable side conditions stated on the sheet itself.	must	6
A-003	rule-sheet	Rule sheet: unified LK	Canonical LK (primitive negation, contexted initial sequents) with the Part VIII presentation documented as a named variant and an equivalence-theorem cross-reference.	must	6
A-004	rule-sheet	Rule sheet: LK with equality	$\mathsf{LK}^{=}$ as a separately named calculus displaying the equality rules and stating that cut elimination for it is a distinct theorem with its own status.	must	6
A-005	rule-sheet	Rule sheet: LJ	Single-succedent intuitionistic sequent calculus, load-bearing for the disjunction-property proof, with the restriction stated explicitly.	must	6
A-006	rule-sheet	Rule sheet: Hilbert CPL	Axiom schemas and modus ponens with the deduction theorem stated as a named result on the sheet.	must	6
A-007	rule-sheet	Rule sheet: Hilbert FOL	Axiom schemas, generalization rule, and the deduction theorem with its free-variable side condition.	must	6
A-008	rule-sheet	Rule sheet: Hilbert K	Axiom K, necessitation, and uniform substitution as the base normal modal system, with the substitution terminology matching the registered term.	must	6
A-009	rule-sheet	Rule sheet: T	System T as K plus the reflexivity axiom, with its frame condition cross-referenced to the correspondence table.	must	6
A-010	rule-sheet	Rule sheet: S4	System S4 as T plus axiom 4, with its frame condition cross-referenced to the correspondence table.	must	6
A-011	rule-sheet	Rule sheet: S5	System S5 with axiom 5 (or B plus 4), its equivalence-relation frame condition, and the NP satisfiability note.	must	6
A-012	rule-sheet	Rule sheet: GL	Löb axiom system with necessitation, the arithmetical interpretation, and the Solovay completeness statement cross-referenced.	must	6
A-013	rule-sheet	Rule sheet: LTL	Axioms including U-unfolding and the induction rule, with operator precedence licensed by the completed precedence table.	must	6
A-014	rule-sheet	Rule sheet: S5n epistemic	Multi-agent $\mathsf{S5}_n$ with K_i, E_G, and C_G axioms, the C_G induction rule, and the common-knowledge weak-completeness caveat.	must	6

continued

ID	Category	Item	Specification	Prio	Ph
A-015	rule-sheet	Rule sheet: classical propositional tableaux	Expansion and closure rules with the systematic-tableau completeness statement and countermodel read-off.	must	6
A-016	rule-sheet	Rule sheet: classical first-order tableaux	Gamma and delta rules with freshness side conditions displayed on the sheet.	must	6
A-017	rule-sheet	Rule sheet: many-valued matrix tableau	Value-labelled expansion rules with the notion of a saturated branch defined on the sheet, curing the undefined-term defect.	must	6
A-018	rule-sheet	Rule sheet: resolution, factoring, superposition	Rules with the soundness-critical side conditions (renaming apart, most general unifier) displayed, plus the term ordering (KBO/LPO) and redundancy criteria for superposition.	must	6
A-019	rule-sheet	Rule sheet: IMLL/ILL/CLL	Sequent rules for intuitionistic and classical linear logic including the four exponential rules, each named, with the one-sided classical presentation.	must	6
A-020	rule-sheet	Rule sheet: Lambek calculus	Directional implication introduction and elimination rules in sequent form with the no-structural-rules statement.	should	6
A-021	rule-sheet	Rule sheet: STLC / Curry-Howard	Typing rules annotated with proof terms beside the corresponding natural-deduction rules.	must	6
A-022	rule-sheet	Rule sheet: Hoare logic	Axiom schemas and inference rules including the consequence rule, cross-referenced to the weakest-precondition table.	must	6
A-023	rule-sheet	Rule sheet: SLD resolution	Definite-clause resolution with the selection rule stated and the negation-as-failure caveat.	should	6
A-024	table	Table: the sixteen binary connectives	All sixteen binary truth functions with standard names, truth tables, and definability notes including Sheffer stroke and NOR.	must	4
A-025	table	Table: notation variants across the literature	Peano/Russell, Polish, Hilbert, and Gentzen notations for connectives and quantifiers keyed to the handbook's registered symbols.	should	4
A-026	table	Table: Greek, Fraktur, and blackboard letter conventions	Registered uses of Greek, Fraktur, and blackboard alphabets by domain with scope column matching the symbol registry.	should	4
A-027	table	Table: standard propositional equivalence laws	De Morgan, distribution, absorption, double negation, material implication, and contraposition laws with labels citable from body chapters.	must	4
A-028	table	Table: duality	Dual connectives and the duality theorem for propositional formulas with the negation-normal-form connection.	should	4
A-029	table	Table: Post functional-completeness criteria	The five maximal clones with membership tests and the completeness criterion for a set of connectives, supporting the Post-lattice figure.	must	4
A-030	table	Table: tractable propositional classes	Horn, dual-Horn, 2-SAT, and XOR-SAT with their satisfiability algorithms and complexity.	should	4
A-031	table	Table: propositional proof systems side-by-side	Hilbert, ND, LK, tableau, and resolution for CPL compared on rules, adequacy theorems, and proof-search behavior.	must	4
A-032	table	Table: natural-language translation guide	Canonical translations of English connective idioms (only, unless, neither, only if) correcting the v1.0 translation-table errors.	should	4
A-033	table	Table: quantifier equivalence laws	Prenexing, distribution, vacuous-quantifier, and duality equivalences with side conditions on free variables.	must	4
A-034	table	Table: normal-form comparison	NNF, prenex, Skolem, CNF, DNF, and clausal form compared on existence, equivalence versus equisatisfiability, and size blowup.	must	4
A-035	table	Table: first-order metatheorem summary	Soundness, completeness, compactness, both Löwenheim–Skolem theorems, Herbrand, interpolation, Beth, and Lindström with exact hypotheses including choice-principle usage.	must	4
A-036	table	Table: decidable fragments of first-order logic	Monadic FOL, FO^2, guarded fragment, and Bernays–Schönfinkel class with decision complexity and undecidable boundary cases.	must	4
A-037	table	Table: first-order proof systems side-by-side	Hilbert, ND, LK, and tableaux for FOL compared on rules, side conditions, and adequacy theorems.	should	4

continued

ID	Category	Item	Specification	Prio	Ph
A-038	table	Table: extensions of first-order logic	Many-sorted, free, infinitary, generalized-quantifier, and second-order extensions with the Lindström-theorem calibration of which metatheorems survive.	should	4
A-039	table	Table: full versus Henkin second-order semantics	Comparison of full and Henkin semantics on completeness, compactness, Löwenheim–Skolem, and categoricity, resolving the D2 gap.	must	4
A-040	table	Table: first-order theories master table	DLO, ACF, RCF, Presburger, PA, true arithmetic, and ZFC classified by completeness, decidability, quantifier elimination, and categoricity spectrum.	must	4
A-041	table	Table: quantifier-elimination catalog	Per-theory QE status with eliminating extensions (definable Skolem functions, model completeness) and the DLO/ACF/RCF witnesses.	must	4
A-042	table	Table: preservation theorems	Łoś–Tarski, Chang–Łoś–Suszko, and Lyndon theorems pairing syntactic classes with preservation properties.	must	4
A-043	table	Table: stability, NIP, and o-minimality overview	Classification-theoretic dividing lines with defining conditions and canonical example theories.	should	4
A-044	table	Table: finite model theory results	Zero-one laws, Fagin's theorem, and Trakhtenbrot's theorem with the finite-versus-unrestricted contrast.	should	4
A-045	table	Table: machine models and equivalences	Turing machines, register machines, mu-recursive functions, and lambda-definability with the pairwise equivalence theorems and the Church–Turing discussion pointer.	must	4
A-046	table	Table: undecidability catalog	Halting problem, Rice, Hilbert's tenth, word problems, tiling, and FOL validity with reduction sources and citations.	must	4
A-047	table	Table: arithmetical hierarchy	$\Sigma^0_n/\Pi^0_n/\Delta^0_n$ levels with closure properties, complete problems, Post's theorem, and the jump correspondence.	must	4
A-048	table	Table: analytical hierarchy	Σ^1_n/Π^1_n levels with complete sets and the relation to second-order arithmetic.	must	4
A-049	table	Table: Turing degrees and jump facts	Degree-structure facts (jump, join, Post's problem, Friedberg–Muchnik) at statement level with citations.	should	4
A-050	table	Table: complexity-class ladder	P, NP, coNP, PH, PSPACE, EXPTIME, and beyond with known inclusions, separations, open problems, and canonical complete problems.	must	4
A-051	table	Table: logic decision-problem complexity (master)	For every logic in the handbook, satisfiability/validity complexity or undecidability with completeness degree, harvested from entry-card fields.	must	6
A-052	table	Table: descriptive-complexity correspondences	FO, existential SO (Fagin), least-fixed-point, and their capture results over ordered finite structures.	should	4
A-053	table	Table: Q and PA axioms	Robinson arithmetic and PA axiom schemas in the fixed signature, with the induction schema's side conditions.	must	4
A-054	table	Table: arithmetic independence catalog	Gödel and Rosser sentences, Con statements, Paris–Harrington, and Goodstein with the theory each is independent of.	must	4
A-055	table	Table: decidable arithmetic theories	Presburger and Skolem arithmetic with decision procedures, complexity bounds, and the contrast with full arithmetic.	should	4
A-056	table	Table: reverse-mathematics calibration	The Big Five subsystems of Z_2 with axioms and representative equivalent theorems.	must	4
A-057	table	Table: provability-logic dictionary	GL modal principles paired with their arithmetical readings and the Solovay arithmetical-completeness statement.	must	4
A-058	table	Table: ZFC axioms	Formal statement of every ZFC axiom and schema with side conditions, replacing the prose-only presentation.	must	4
A-059	table	Table: set-theory system variants	ZF, Z, KP, NBG, MK, and NF compared on classes, foundation, choice, and relative strength.	must	4
A-060	table	Table: ordinal arithmetic laws	Addition, multiplication, and exponentiation laws with continuity and commutativity failures and Cantor normal form.	must	4
A-061	table	Table: cardinal arithmetic laws	Sum, product, and exponentiation rules, cofinality, regular/singular classification, König's theorem, and the $\aleph/\beth$ hierarchies.	must	4

continued

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A-062	table	Table: AC equivalents	Well-ordering theorem, Zorn's lemma, Tychonoff, and further equivalents with ZF-provability status of each implication.	must	4
A-063	table	Table: choice-principle dependency	Which handbook theorems require AC, DC, countable choice, or BPI, including the choice-free countable-language completeness theorem.	must	4
A-064	table	Table: combinatorial principles	Diamond, square, MA, club and stationary sets, Fodor, Delta-system lemma, and Ramsey's theorem with implication structure.	must	4
A-065	table	Table: large-cardinal ladder	Inaccessible through supercompact and beyond in consistency-strength order with defining properties and first failures of $V = L$.	must	4
A-066	table	Table: independence catalog	Statements independent of ZFC (CH, GCH patterns, Suslin hypothesis, AC over ZF) with the method (forcing, inner models) and sources.	must	4
A-067	table	Table: forcing-poset catalog	Cohen, random, and collapse forcings with what each adds and preserves and its chain conditions.	should	4
A-068	table	Table: Borel and projective hierarchies	Pointclass levels with closure properties, regularity properties, and the determinacy calibration (Borel determinacy in ZFC; PD from large cardinals; AD versus AC).	should	4
A-069	table	Table: consistency-strength ladder (master)	PRA through PA, subsystems of Z_2, Z, ZF, ZFC, and large-cardinal extensions in one ordered ladder harvested from entry cards, paired with the ladder figure.	must	6
A-070	table	Table: structural rules per calculus	Weakening, contraction, exchange, and cut across LK, LJ, linear, relevance, affine, and Lambek systems, keying the substructural family map.	must	4
A-071	table	Table: calculi catalog	LK, LJ, NK, NJ with hypersequent, display, and deep-inference pointers, each with its adequacy status.	should	4
A-072	table	Table: Curry-Howard correspondence	Logic to type theory to category theory dictionary (formulas/types/objects, proofs/terms/morphisms, normalization/reduction).	must	4
A-073	table	Table: lambda-calculi comparison	Untyped, simply typed, System F, and MLTT/CIC compared on Church–Rosser, normalization, and typability decidability.	must	4
A-074	table	Table: ordinal-analysis calibration	Proof-theoretic ordinals for Q-range fragments, PA (ε_0), and stronger systems at statement level with citations.	must	6
A-075	table	Table: proof-complexity systems	Resolution, Frege, and extended Frege with known lower bounds (pigeonhole in resolution) and simulation relations.	should	4
A-076	table	Table: modal systems and axioms	K, D, T, B, K4, KD45, S4, S4.2, S4.3, S5, GL, and Grz with axioms and containments, backing the lattice figure.	must	4
A-077	table	Table: frame correspondence (axiom, condition, algebra)	Modal axioms paired with first-order frame conditions and algebraic conditions, with Sahlqvist coverage noted, harvested from entry cards.	must	6
A-078	table	Table: Barcan and domain variation	Barcan and converse Barcan formulas against constant, increasing, and varying domains with validity status in each regime.	must	4
A-079	table	Table: temporal logics comparison	LTL, CTL, CTL*, and the modal μ-calculus compared on expressiveness and satisfiability and model-checking complexity.	must	4
A-080	table	Table: epistemic systems	$\mathsf{S5}_n$, group knowledge $E_G/C_G/D_G$, and public-announcement axioms with the C_G induction rule included.	must	4
A-081	table	Table: intermediate logics	KC and LC (Gödel–Dummett) with axioms, frame classes, disjunction property, and finite-model-property status.	should	4
A-082	table	Table: negative translations	Gödel–Gentzen, Glivenko, Kolmogorov, and Kuroda translations with exact clauses (settling the $\bot$-atomic case) and preservation theorems.	must	4
A-083	table	Table: modal embeddings and standard translation	The FO standard translation, the Gödel–McKinsey–Tarski embedding of IPL into S4, and the GL arithmetical interpretation as one translation catalog.	should	4

continued

ID	Category	Item	Specification	Prio	Ph
A-084	table	Table: many-valued matrices comparison	K3, LP, weak Kleene, FDE, and finite Łukasiewicz matrices with designated values, tautology sets, and the K3/LP duality registered.	must	4
A-085	table	Table: t-norm fuzzy logics	BL, Łukasiewicz, Gödel, and product logics with t-norms, axiomatizations, and MV-algebra completeness statements.	should	4
A-086	table	Table: substructural family map	Which structural rules are dropped to obtain relevance, linear, affine, and Lambek systems, paired with the family-map figure.	must	4
A-087	table	Table: relevance systems	R, E, and RM axiom systems with the variable-sharing property and its status per system.	should	4
A-088	table	Table: linear-logic connectives	Multiplicative, additive, and exponential connectives with units, polarities, and De Morgan duals in one registry-aligned table.	must	4
A-089	table	Table: conditional and counterfactual systems	Stalnaker and Lewis sphere-semantics systems with characteristic validities and failures (antecedent strengthening, transitivity).	must	4
A-090	table	Table: deontic systems	Standard deontic logic with its axioms and the classical paradoxes (Ross, gentle murder) at statement level.	should	4
A-091	table	Table: free-logic variants	Positive, negative, and neutral free logics with existence predicate, quantifier rules, and definite-description treatment.	should	4
A-092	table	Table: nonmonotonic formalisms comparison	Default logic, circumscription, autoepistemic logic, and ASP compared on consequence definition and complexity.	should	4
A-093	table	Table: AGM postulates	Contraction and revision postulates with the Levi and Harper identities.	must	4
A-094	table	Table: paraconsistent systems	LP and da Costa C-systems with explosion failure and designated-value or consistency-operator mechanisms.	should	4
A-095	table	Table: description-logic family complexity	ALC through SROIQ with OWL correspondences and reasoning-problem complexity per logic.	must	4
A-096	table	Table: Hoare logic rules	All Hoare rules with side conditions plus the weakest-precondition calculus and a separation-logic pointer.	must	4
A-097	table	Table: logic-programming semantics	SLD resolution, negation as failure, completion, well-founded, and stable-model semantics compared, with Datalog noted.	should	4
A-098	table	Table: Boolean algebra laws	Axiomatizations of Boolean algebras with the lattice-theoretic and ring-theoretic presentations.	must	4
A-099	table	Table: Stone duality dictionary	The algebra-to-space correspondences (algebras/Stone spaces, homomorphisms/continuous maps, ideals/open sets), paired with the duality figure.	must	4
A-100	table	Table: logic-algebra correspondences	CPL/Boolean, IPL/Heyting, S4/interior, Łukasiewicz/MV, and substructural/residuated-lattice pairs via the Lindenbaum–Tarski construction.	must	4
A-101	table	Table: paradox catalog	Liar, Russell, Curry, Berry, Yablo, sorites, and Skolem paradoxes with formulation and principal resolution strategies.	must	4
A-102	table	Table: truth-theories comparison	Tarski hierarchy, Kripke fixed points, and revision theory compared on object-language truth predicates and paradox handling.	must	4
A-103	table	Table: consequence-relation properties	Tarskian closure conditions (reflexivity, monotonicity, cut) and which substructural and nonmonotonic systems drop each.	should	4
A-104	table	Table: vagueness approaches	Supervaluationist, degree-theoretic, and epistemicist treatments of the sorites at capsule level.	could	4
A-105	table	Table: SAT-solving techniques	DPLL versus CDCL with clause learning, watched literals, restarts, and DRAT proof logging.	must	4
A-106	table	Table: unification algorithm	Martelli–Montanari transformation rules with the occurs check and the mgu existence/uniqueness theorem.	must	4
A-107	table	Table: SMT theory catalog	EUF, arrays, bit-vectors, and arithmetic theories with decidability, complexity, and Nelson–Oppen combinability conditions.	must	4

continued

ID	Category	Item	Specification	Prio	Ph
A-108	table	Table: prover and proof-assistant landscape	Lean, Rocq, Isabelle, Agda, HOL Light/HOL4, PVS, Mizar, and ACL2 classified by foundation, kernel architecture, and de Bruijn criterion.	must	4
A-109	table	Table: chronology of logic	Dated milestone table from Aristotle to the present keyed to handbook entries.	should	6
A-110	table	Table: landmark papers	The landmark-papers register with full citations and one-line significance statements linked to the bibliography.	should	6
A-111	table	Table: capsule biographies	Short biographical capsules for the figures in the person index.	could	6
A-112	table	Table: expressive-power hierarchy (master)	Propositional, modal (via bisimulation invariance), FO, LFP, and SO ordered by expressive power with witnessing separations.	must	6
A-113	table	Table: per-part audit registers	One generated audit table per part on the model of the v1.0 incompleteness audit table (line 8005), emitted by the audit-table generator, never hand-authored.	must	6
A-114	figure	Figure: formula syntax tree	Parse tree of a worked formula supporting the unique-readability material and the syntax-tree exercises.	must	4
A-115	figure	Figure: ND derivation tree example	A complete natural-deduction derivation displayed as a tree with discharge labels.	must	4
A-116	figure	Figure: LK derivation tree example	A cut-free LK derivation displayed as a tree with rule names at each inference.	must	4
A-117	figure	Figure: sequent-versus-ND correspondence	Side-by-side translation of one derivation between ND and LK showing the correspondence of rules.	should	4
A-118	figure	Figure: normal-form conversion pipeline	Flowchart from formula through NNF, prenex, Skolem, and clausal form marking equivalence-preserving versus equisatisfiability-preserving steps.	should	4
A-119	figure	Figure: Post's lattice	The lattice of clones with the five maximal clones highlighted, supporting the functional-completeness criterion table.	must	4
A-120	figure	Figure: closed propositional tableau example	A closed classical tableau tree for a worked unsatisfiable formula.	must	4
A-121	figure	Figure: resolution refutation example	A resolution refutation displayed as a DAG from input clauses to the empty clause.	should	4
A-122	figure	Figure: Ehrenfeucht–Fraïssé game	A game board for two small structures illustrating a winning duplicator strategy over a fixed number of rounds.	should	4
A-123	figure	Figure: bisimulation zigzag	The zigzag diagram for bisimulation between two Kripke models supporting the modal-invariance material.	should	4
A-124	figure	Figure: arithmetical hierarchy	Inclusion diagram of $\Sigma^0_n/\Pi^0_n/\Delta^0_n$ with strictness arrows and complete-set labels.	must	4
A-125	figure	Figure: analytical hierarchy	Inclusion diagram of Σ^1_n/Π^1_n atop the arithmetical hierarchy.	should	4
A-126	figure	Figure: Turing degrees and the jump	Schematic of the degree ordering with $\mathbf{0}$, $\mathbf{0}'$, incomparable c.e. degrees, and the jump operation.	should	4
A-127	figure	Figure: complexity-class inclusion diagram	Hasse-style diagram of P through EXPTIME with known strict inclusions marked, paired with the ladder table.	must	4
A-128	figure	Figure: reverse-mathematics Big Five	Inclusion diagram of the five subsystems of Z_2 with representative theorems attached.	should	4
A-129	figure	Figure: ordinals to epsilon-zero	Number-line schematic of ordinals through ω, ω^ω, and ε_0 with Cantor-normal-form annotations.	should	4
A-130	figure	Figure: cumulative hierarchy	The V_α picture with rank axis and the position of small large cardinals indicated.	must	4
A-131	figure	Figure: constructible hierarchy	The L_α hierarchy inside V illustrating condensation and the $V = L$ contrast.	should	4
A-132	figure	Figure: forcing extension	Ground model, generic filter, and extension $M[G]$ with names and their evaluations schematically indicated.	must	4
A-133	figure	Figure: choice-principle implication diagram	AC, DC, countable choice, and BPI with known non-reversals marked, paired with the dependency table.	should	4
A-134	figure	Figure: consistency-strength ladder	Vertical ladder from PRA to large cardinals matching the master consistency-strength table.	must	6

continued

ID	Category	Item	Specification	Prio	Ph
A-135	figure	Figure: Borel and projective hierarchies	Pointclass diagram with the Borel levels and projective classes and their closure arrows.	should	4
A-136	figure	Figure: modal-systems lattice	Containment lattice of K, D, T, B, K4, S4, S4.2, S4.3, S5, GL, and Grz, paired with the axiom table.	must	4
A-137	figure	Figure: Kripke countermodel for excluded middle	Two-world intuitionistic model refuting $p \vee \neg p$ with forcing annotations at each world.	must	4
A-138	figure	Figure: Kripke countermodel for Peirce's law	Intuitionistic countermodel to $((p \to q) \to p) \to p$ with forcing annotations.	must	4
A-139	figure	Figure: Kripke countermodel refuting T	Non-reflexive frame refuting $\Box p \to p$ demonstrating the correspondence direction.	must	4
A-140	figure	Figure: Kripke countermodel refuting 4	Non-transitive frame refuting $\Box p \to \Box\Box p$.	must	4
A-141	figure	Figure: Kripke countermodel refuting 5	Non-Euclidean frame refuting $\Diamond p \to \Box\Diamond p$.	must	4
A-142	figure	Figure: Kripke countermodel for the Barcan formula	Increasing-domain model refuting the Barcan formula, paired with the domain-variation table.	must	4
A-143	figure	Figure: Routley–Meyer ternary relation	Diagram of the ternary accessibility relation evaluating a relevant conditional at a world.	should	4
A-144	figure	Figure: lattice of intermediate logics	IPL to CPL interval with KC and LC placed, matching the intermediate-logics table.	could	4
A-145	figure	Figure: LTL evaluation on a path	A timeline diagram evaluating X, U, F, and G formulas at positions of one path.	must	4
A-146	figure	Figure: CTL computation tree	Branching-time tree contrasting path quantifiers with the linear LTL picture.	should	4
A-147	figure	Figure: Büchi automaton example	A Büchi automaton for a worked LTL formula with its acceptance condition, supporting the model-checking entry.	should	4
A-148	figure	Figure: SLD tree	An SLD derivation tree for a worked definite program and goal showing success and failure branches.	should	4
A-149	figure	Figure: CDCL implication graph	Implication graph at a conflict with the learned clause and the backjump level indicated.	must	4
A-150	figure	Figure: Stone duality picture	A finite Boolean algebra beside its Stone space with the correspondence arrows drawn.	should	4
A-151	figure	Figure: Curry-Howard triangle	Triangle diagram connecting logic, type theory, and category theory with the translations labeled on the edges.	should	4
A-152	figure	Figure: type-derivation tree	An STLC typing derivation displayed as a tree beside its ND counterpart.	must	4
A-153	figure	Figure: substructural family map	Map from dropped structural rules to the resulting logics, paired with the family table.	must	4
A-154	index	Subject index: inverted headings policy	Handbook-grade subject index of 2500–4000 entries with inverted noun-first headings replacing the sentence-shaped auto-titles, tagged inline from birth during migration.	must	6
A-155	index	Subject index: sort keys and escaping	Every entry carries an explicit sort key; accented names (Gödel, Löb, Löwenheim, Lindström, Schröder) and math-leading heads are escaped so makeindex sorts and renders them correctly.	must	6
A-156	index	Subject index: see and see-also layer	Cross-reference layer of see and see-also entries connecting synonyms and neighboring topics across parts.	must	6
A-157	index	Symbol index	Index of every registered symbol with page of registration and principal uses, generated from the symbol-registry CSV.	must	6
A-158	index	Person index	Index of persons with accented names correctly sorted, linked to the biography capsules where present.	must	6
A-159	index	Named-theorem index	Index of all named theorems and lemmas (including the currently unindexable lemmas such as Lindenbaum and both truth lemmas) pointing to statement and annex proof.	must	6
A-160	symbol-registry	Symbol-registry CSV as single source	One machine-readable registry (symbol, meaning, scope, precedence, first-use anchor) that generates the Part 1 tables and drives the notation lint.	must	3
A-161	symbol-registry	Registry sub-table: classical logic symbols	Connectives, quantifiers, turnstiles, and metavariable conventions for Parts 1–2 with scope column.	must	4
A-162	symbol-registry	Registry sub-table: model-theory symbols	Adds the omitted Part IV notation: $\preceq$, elementary $\equiv$, Diag/ElDiag, $S_n(T)$, ultraproducts $\prod \mathcal{M}_i/\mathcal{U}$, and $\simeq$.	must	4

continued

ID	Category	Item	Specification	Prio	Ph
A-163	symbol-registry	Registry sub-table: computability and arithmetic symbols	Adds $\leq_m$, $\leq_T$, φ_e, $\emptyset^{(n)}$, $\ulcorner\cdot\urcorner$, $\bar{n}$, Prov_T, $\mathsf{Con}(T)$, and the object-language $\leq$.	must	4
A-164	symbol-registry	Registry sub-table: set-theory symbols	Adds V_α, rank, L_α, $\mathbb{P}$, names and evaluations, $M[G]$, and the forcing relation.	must	4
A-165	symbol-registry	Registry sub-table: nonclassical symbols	Adds $\Vdash$, $\Box$, $\Diamond$, $\mathsf{X/U/F/G}$, $K_i/E_G/C_G$, linear connectives, matrix data, and the per-system turnstiles the style chapter promises.	must	4
A-166	symbol-registry	Precedence completion	Precedence and associativity ranks for every Part X operator (modal, temporal, epistemic, linear) extending the seven classical ranks, licensing the U-unfolding axiom.	must	3
A-167	symbol-registry	Collision ruling: absolute-value bars	$\lvert\cdot\rvert$ means cardinality only; structure domains are written $\mathrm{dom}\,\mathcal{M}$, closing the Part III/IV seam and the nested-bars defect.	must	3
A-168	symbol-registry	Collision ruling: equivalence signs	$\equiv$ means elementary equivalence only; formula equivalence is $\Leftrightarrow$; the definitional sign is $:\equiv$.	must	3
A-169	symbol-registry	Collision ruling: forcing macro	One registered forcing macro rendering $\Vdash$ everywhere, ending the body-versus-feedback drift between the two current macros.	must	3
A-170	symbol-registry	Collision ruling: letter overloads	The overloaded letters (formula metavariable versus carrier A, worlds versus valuation v, V, D, M) are separated by scope entries or renamed.	must	3
A-171	symbol-registry	Registry row corrections	Remove the over-scoped R^* common-knowledge claim, fix the valuation-row category error, and clarify the boldface Σ^0_n pointclass convention.	must	3
A-172	symbol-registry	Term-denotation bridging convention	Register once the bridge between the Part III denotation-bracket notation and the Part IV superscript form for term values.	should	3
A-173	environment	Entry-card environment with mandatory status key	Theorem-class entry-card environment whose status key (complete proof / proof sketch / cited theorem) is compile-mandatory, with sources field, erroring on omission.	must	3
A-174	environment	Numbered warning environment	Numbered, labeled, indexable warning environment replacing prose warnings.	must	3
A-175	environment	Numbered proofboundary environment	Numbered internal-proof-boundary environment replacing the unreferenceable paragraph device.	must	3
A-176	environment	Audit-table generator	Script emitting the per-part audit longtables (with proper continuation heads and feet) from entry-card metadata; audit tables are generated, never authored.	must	3
A-177	environment	Calculus-registry rule display macro	Named, labeled rule-display macro used only inside registry files; body, entry cards, and rule sheets all input the registry, making body-sheet divergence impossible.	must	3
A-178	environment	Numbered convention environment book-wide	The convention environment numbered and used from Part 1 onward, promoting the precedence and quantifier-scope conventions out of prose.	must	3
A-179	environment	Uniform solution environment	All exercise solutions use the defined solution environment (replacing the 112 hand-rolled italic headers) with graded solution/sketch variants.	must	3
A-180	bibliography	Biblatex conversion	Convert the 64 manual bibitems to a biblatex database with normalized DOI/URL practice and stub keys for planned content.	must	3
A-181	bibliography	Five missing primaries	Add Glivenko 1929, Gödel 1933, Gentzen 1936, Kleene's K3 source, and Priest 1979, cited at their points of use.	must	3
A-182	bibliography	Per-domain shopping-list intake	Canonical-source lists per new domain (including Martelli–Montanari, CDCL sources, Sahlqvist, Solovay, Paris–Harrington) entered as the domains are written.	must	4
A-183	navigation	tocdepth and numbered rule-sheet sections	Set tocdepth to show subsections and number the rule-sheet sections so every sheet is referenceable from the body.	must	3
A-184	navigation	List of tables	A list of tables covering every register and comparison table in the handbook.	must	6
A-185	navigation	List of figures and listings	Lists of figures and of captioned listings, absent entirely from v1.0.	must	6

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ID	Category	Item	Specification	Prio	Ph
A-186	navigation	Per-part further reading	A further-reading section per part drawn from the per-domain shopping lists.	must	4
A-187	navigation	Glossary	Alphabetical glossary of registered terms with one-sentence definitions and pointers to the defining entries.	must	6
A-188	navigation	Exercise-solution bidirectional links	Every exercise hyperlinks to its solution and every solution back to its exercise, preserving the 112=112 completeness discipline.	must	4
A-189	navigation	Body-to-appendix cross-references	Body chapters cite the rule sheets, precedence table, and notation registry, and forward references connect LK to cut elimination and resolution to automated reasoning.	should	4
A-190	check	Check: status coverage	Every theorem-class environment carries a status key; coverage must be 100 percent and the build fails otherwise.	must	3
A-191	check	Check: unlabeled-environment scan	No numbered result environment without a label; no result-bearing sectioning paragraphs.	must	3
A-192	check	Check: rule-display ban	No inference-rule display outside the calculus-registry files.	must	3
A-193	check	Check: symbol-registry coverage	Notation lint verifying every registered glyph has one role in scope and no body glyph is missing from the registry.	must	3
A-194	check	Check: body-annex bidirectional completeness	Every body entry has its annex proof or citation and every annex item has a body entry, generalizing the exercise-feedback discipline.	must	3
A-195	check	Check: zero-citation chapters	No chapter with zero citations, enforcing the primary-source rule.	must	3
A-196	check	Check: index health	Every index entry has a sort key, no raw-math or sentence-shaped headings, and accented heads escape correctly under makeindex.	must	3
A-197	check	Check: dead labels and dangling references	No dangling cross-reference or citation targets, unreferenced labels reported, and prose-relative pointers (above, below, the cited theorem) flagged for conversion to references.	must	3
A-198	check	Check: bibliography hygiene	Cited keys and bibliography entries match bidirectionally with consistent DOI/URL fields.	must	3
A-199	check	Check: compile warnings as errors	latexmk build treating the declared warning subset (overfull tables, missing characters, undefined references) as errors.	must	3
A-200	check	Check: audit-table freshness	Per-part audit tables regenerate cleanly and match entry-card metadata at every session close.	must	3
A-201	table	Notation-history table (Part 13)	One row per major notation lineage (Peano–Russell dots, Polish, Gentzen sequent, Hilbert school) with period, exemplar glyphs, and modern descendants; feeds the Part-1 notation-variants table	could	6

8 Standards and rulings

Binding for the new tree; enforced by the check suite (`tools/check.py -strict`).

Ruling 1. Status is compile-mandatory. Every theorem-class environment takes a status key (`proved`/`sketched`/`cited`) with an annex address or source; omission is a compile error. The body contains no `proof` environments at all — proofs live only in the annex.

Ruling 2. LK nomenclature. Canonical LK is the Parts II–III presentation (primitive negation rules, contexted initial sequents) — the one the rule sheets and completeness proofs already use. Part VIII's defined-negation variant is documented as a named variant with an equivalence theorem. $\mathsf{LK}^{=}$ (with equality rules) is a separately named calculus whose cut-elimination status is its own row. No theorem cites a calculus without its exact registry name.

Ruling 3. Glyph roles. $|\cdot|$ means cardinality only; structure domains are written $\mathrm{dom}\,\mathcal{M}$ (closing the Part III/IV seam). $\equiv$ means elementary equivalence only; formula equivalence is $\Leftrightarrow$; the definitional sign is $:\equiv$. Forcing is one registered macro ($\Vdash$ with a negated partner), ending the $\models/\Vdash$ drift. Worlds, matrices, and carriers get registered letters distinct from valuations, structures, and formulas.

Ruling 4. Rules live in the registry. Every inference rule is named, labeled, and displayed only in `registry/calculi/`; every display shows premises, conclusion, name, discharge, and side conditions (the book's own requirement at v1.0 lines 12518–12525, mechanized).

Ruling 5. Numbered semantic environments. `warning` and `proofboundary` are numbered, indexed environments; `\paragraph` never again does semantic work.

Ruling 6. Generated audit tables. Every part ends with a generated register of its numbered results with status and dependencies (institutionalizing the v1.0 incompleteness audit table at lines 8002–8035).

Ruling 7. Bidirectional completeness. Every *proved* status resolves to exactly one annex item and every annex item is pointed to exactly once — the Exercise Feedback chapter's 112=112 discipline, mechanized as a check.

Ruling 8. Primary sources. Historical results cite primary sources (with modern expositions alongside); the zero-citation-chapter check enforces the floor.

9 Campaign plan and session protocol

9.1 Phases

Ph	Name	Output	Scale
0	Infrastructure	repo, `v1.0-audit` tag, baseline build, check suite	done
1	Gap report	this document + four registers	done at sign-off
2	Source math repairs	FIX-SOURCE rows executed; tag `v1.1-verified`	∼25–40 proof sites
3	Skeleton + registries	new tree, templates, calculus/symbol registries, biblatex	+5%
4	Migration + depth fill	Parts 1–7 body+annex, Part 8/12 cores; one part = one visit	0.9–1.1×
5	New domains	Parts 8–13 completed; deepenings in 3–7	+0.6–0.9×
6	Master apparatus	Part 14, four indexes, master tables	+0.1×
7	Release audit	the book passes its own checklist; v2.0	—

Projected final size 1.7–2.1× the current volume; a hard ceiling of 2.2× triggers re-scoping review.

9.2 Dependency register

(1) Phase-2 repairs strictly before migration of the affected parts. (2) Rulings before registries before any prose in the new tree. (3) Calculus registry before any calculus-bearing chapter migrates. (4) biblatex before migration. (5) Inline `\index` tagging from each chapter's birth; the index *pass* only in Phase 6. (6) Part 8 entry cards before Parts 9–11 (they cite into them). (7) Part-14 master tables after Parts 2–12 land (harvested from entry fields). (8) Part 13 last.

9.3 Session protocol

Every later session: (1) run `tools/check.py`; (2) pick open register rows for the current phase; (3) **spec-first rule**: any scope deviation edits the register row before prose is written; (4) work; (5) flip row status with the commit hash; (6) regenerate tables (`tools/csv2tex.py`); (7) run checks; (8) commit. The registers alone make any session resumable.

9.4 Standing controls

Treatment-level ceilings and the NOT-IN-SCOPE register against scope explosion; the calculus registry against body–sheet divergence; the compile-mandatory status key against policy drift (base rate: the v1.0

policy collapsed twice); the mechanical check suite every session; `source/` immutable outside Phase 2 so the old book remains intact and compilable throughout.

10 Traceability

Finding-to-ID mapping lives in the registers themselves (each row cites its v1.0 lines); the full audit reports are archived in the campaign plan file. During migration each executed row gains the new tree's label in its status cell, so the specification remains navigable in both directions for the whole campaign.

10.1 Frozen v1.0 table of contents

Parts and chapters of the audited manuscript with their line anchors (generated from the tagged source):

Part	Chapter	Line
Foundations of Formal Work		363
	Sets, Relations, Functions, and Cardinality	365
	Levels, Objects, and Consequence	798
	Finite Strings, Induction, and Recursion	1101
	Writing Formal Definitions and Proofs	1411
Classical Propositional Logic		1649
	Propositional Syntax and Semantics	1658
	Natural Deduction and Sequents	2187
	Soundness, Completeness, Compactness, and Resolution	2451
Classical First-Order Logic		3022
	First-Order Syntax	3032
	Structures, Satisfaction, and Translation	3429
	Theories and Axiomatization	3729
	First-Order Deduction	3927
	Henkin Completeness and Compactness	4306
	Cardinality, Normal Forms, and Resolution	4968
Model Theory		6049
	Maps, Substructures, and Elementarity	6052
	Types, Saturation, and Ultraproducts	6510
	Quantifier Elimination and Categoricity	7150
Computability		7494
	Machines and Effective Procedures	7497
	Universal Computation, Reductions, and Fixed Points	7851
	Logical Decision Problems and the Arithmetical Hierarchy	8248
Formal Arithmetic and Incompleteness		9080
	Arithmetic, Representation, and Coding	9083
	Diagonalization and the First Incompleteness Theorem	9439
	Provability, the Second Incompleteness Theorem, and Truth	9750
Axiomatic Set Theory		10060
	The Axioms of ZF and ZFC	10063
	Hierarchy, Ordinals, and Cardinals	10353
	Constructibility and Forcing	10675
Proof Theory		11354
	Classical and Intuitionistic Calculi	11357
	Cut Elimination and Analytic Derivations	11711
	Normalization and Ordinal Analysis	11961
Second-Order and Higher-Order Logic		12249
	Second-Order Syntax and Two Semantics	12252
	Simple Type Theory	12721
Nonclassical Logic		12970
	Intuitionistic Logic	12973

continued

Part	Chapter	Line
	Modal, Temporal, and Epistemic Logic	13201
	Many-Valued and Paraconsistent Logic	13584
	Relevance and Linear Logic	13789
Formal Practice, Automation, and Verification		14095
	Constructing Derivations, Models, and Countermodels	14097
	Automated Reasoning	14271
	Mechanized Proof Checking	14591
	Writing and Typesetting Formal Logic	14729
	Research Workflow and Reproducibility	15030
	Registered Notation	15095
	Binding and Precedence	15207
	Rule Sheets	15240
	Qualifying-Examination Problems and Selected Solutions	15489
	Exercise Feedback	15638
	Editorial and Formal-Logical Release Checklist	16743